

Limits and Guarantees for Quadratic Intersection Cuts

Abstract

An intersection cut depends on both the corner relaxation and the convex set used to exclude its vertex. We study how this choice affects the objective bound for quadratic constraints. We first give a self-contained account of the unrestricted benchmark: with positive reduced costs, the supremum over all convex free sets equals the nonlinear corner bound. We distinguish this equality from attainment and give a sufficient transversality condition for attainment. For bilinear constraints, we optimize a three-parameter family obtained from determinant-preserving transformations, and its upward completions. A scaled measure of vertex depth gives a positive approximation guarantee whose order is sharp. For a fixed quadratic representation, the point rule used by SCIP admits a sharp guarantee that also depends on ray conditioning. Exact rational examples show that the transformed families can miss the corner bound, even at a unique minimizing point reached on one ray, and that taking every cut from a family need not remove the gap. We give a contact criterion that separates supremum equality from attainment, and extend the orbit analysis to homogeneous 2×2 minors. For successive cutting rounds, a uniform depth condition ensures convergence on a compact relaxation; an explicit sequence of maximal bilinear-free sets shows why maximality and uniformly pointed basis cones alone are insufficient. Archived computational results compare corner-bound selection, reoptimized LP bounds, and solver performance. They show that improving the first criterion does not consistently improve the others.

Keywords: intersection cuts; quadratic optimization; bilinear constraints; free sets; corner relaxations; cutting-plane convergence.

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1 Introduction

Consider an LP vertex that violates a quadratic constraint. A basis gives a cone with that vertex as its apex. A closed convex set whose interior contains the apex and avoids the quadratic feasible set gives an intersection cut: follow each cone ray to the boundary of the convex set, then cut off the simplex formed by those ray intersections. Larger ray steps give smaller cut coefficients. The choice of the convex set therefore matters, even when the vertex, basis, and violated constraint are fixed. Intersection cuts for integer programming go back to Balas ([Balas, 1971](#)).

Quadratic constraints permit many such sets. A set selected from the violated point need not give the best objective bound on the corner. A set giving that bound need not yield the strongest bound after the LP is reoptimized. A family whose best single cut is weak may improve when all its cuts are imposed, but even that closure can remain weaker than the nonlinear corner. Finally, good cuts at individual vertices do not by themselves imply convergence of successive LP relaxations. These are different questions, and answering one does not settle the others.

This paper develops them in a common framework. Our principal examples are the bilinear inequality $w - xy \leq 0$ and the homogeneous determinant equation $ad - bc = 0$. Both arise in lifted formulations of quadratic optimization. The bilinear inequality also isolates a difficulty that is easy to hide in a larger model: its maximal free sets can extend far beyond the part of the corner that determines the objective bound. Studying this geometry gives exact limits on restricted set

selection, quantitative guarantees when the corner is suitably scaled, and conditions for repeated cuts to converge.

1.1 Prior work and the question studied here

The relation between valid inequalities and free sets is established in the cut-generating-function literature. A convex free neighborhood of the origin defines a gauge and hence an inequality for a corner relaxation. [Conforti et al. \(2015\)](#) develop this correspondence and explain why not every valid inequality is generated by an ordinary, finite-valued cut-generating function. Their Theorem 6.3 gives sufficiency when the projected rays positively span the ambient space. [Cornuéjols et al. \(2015\)](#), Theorem 1.1, prove exact domination of every valid inequality under the more general condition that the translated feasible set lies in the ray cone. [Kılınç-Karzan and Steffy \(2016\)](#), Proposition 5, recover this result, while their Corollary 2 proves sufficiency of relaxed cut-generating functions without that cone condition. A relaxed function can take the value $+\infty$, and its level set need not be a neighborhood of the origin.

These distinctions are relevant to the unrestricted benchmark in this paper. Equality of the best intersection-cut bound and the corner bound is a supremum statement. An ordinary free neighborhood need not attain that supremum. [Kılınç-Karzan and Yang \(2017\)](#) study sufficient and necessary conditions for ordinary cut-generating functions outside the cone-containment setting. Its Examples 4.2 and 4.3 illustrate the difference between transverse and tangential approach to a boundary point. Our general transversality criterion gives a sufficient condition for differentiable sublevel sets, stated at every minimizing contact of the corner. It applies without the cone-containment condition, and its proof uses the geometry of these contacts directly. The unrestricted exactness and dominance results serve as background; we do not claim them as new.

Choosing deeper cuts also has a substantial history. [Eckstein and Nediak \(2005\)](#) formulate a distance-based deepest convexity-cut problem over structured free sets for a finite mixed integer point set. [Poirrier and Yu \(2019\)](#) bound Euclidean cut depth for tableau corners and identify set-dependent depth as a possible criterion for selecting lattice-free sets. [Xavier et al. \(2021\)](#) minimize the largest coefficient in a multi-row lattice-free construction. Our one-cut bound is the reciprocal of the largest normalized coefficient $a_j(C)/\omega_j$. These objectives coincide when the reduced costs are equal; with unequal costs, ours is a weighted version. Coefficient optimization therefore has a direct precedent. [Balas and Margot \(2013\)](#) study a complementary choice: a fixed free set with a tighter polyhedral relaxation than a basis cone. Thus optimizing a free set, or seeking a deeper intersection cut, is not a new general idea. Our selection criterion is the objective bound on a fixed corner, and our restricted families exclude a continuous quadratic feasible set. Euclidean depth enters separately when we study successive rounds.

Family closures have also been studied extensively for lattice-free sets. [Averkov et al. \(2018\)](#), Theorem 5, relate constant-factor approximation by a family closure to approximation of each target cut by a single family member, under their closedness and representation hypotheses. That result concerns lattice-free corner polyhedra at a fixed apex. Our closure statements concern continuous quadratic feasible sets and specific determinant families. Their geometric counterexamples and factor classifications do not follow by transferring the lattice-free theorem. The common point is that combining cuts and approximating a target closure must be distinguished from obtaining one exact cut.

For quadratic constraints, [Muñoz and Serrano \(2022\)](#) construct and analyze quadratic-free sets used to generate intersection cuts. [Muñoz et al. \(2025\)](#) characterize maximal free sets for homogeneous quadratic inequalities, and [Muñoz et al. \(2026\)](#) treat the inhomogeneous case. These characterizations describe the available sets. They do not imply that a set chosen from the

violated point optimizes a prescribed corner objective. Our low-dimensional classification is a consequence of this geometry and Lorentz transformations, rather than a replacement for the general characterization.

Outer-product formulations give another source of quadratic-free sets. [Bienstock et al. \(2020\)](#), Lemma 22 and Theorem 23(iv), construct maximal outer-product-free sets for 2×2 minors. Their rotation family is contained in the determinant orbit studied here. Their discussion also explicitly notes that a choice that is best in a violation sense need not produce the deepest cut. We enlarge that rotation family to a three-parameter orbit, distinguish it from the transformed point rule, and optimize its corner bound by small semidefinite feasibility problems. The existence of the rotation subfamily is prior work.

[Chmiela et al. \(2023\)](#) implement quadratic and minor intersection cuts in SCIP. The experiments in their earlier report ([Chmiela et al., 2020](#)) concern root bounds; the best reported configuration combines quadratic intersection cuts with minor cuts. A separate bilinear implementation studied by [Fischetti and Monaci \(2020\)](#) has heterogeneous computational effects, including improvements on selected instances and losses in its overall comparison. These studies motivate careful separation of local cut strength, root relaxation strength, and branch-and-bound work. Our computational analysis makes that separation explicit and reports implementation and numerical qualifications that affect the conclusions.

1.2 Results and contributions

Let $\bar{s}$ be the projected corner vertex, let p_j be its projected rays, and let $\omega_j > 0$ be their reduced costs. The nonlinear corner bound is

$$z_K(\omega) = \inf\{\omega^T \lambda : \lambda \geq 0, \bar{s} + \sum_j \lambda_j p_j \in S\}.$$

If a free set C has ray steps $\alpha_j(C)$, its one-cut bound is $\min_j \omega_j \alpha_j(C)$, with infinite steps handled as specified in [theorem 2.1](#). This formula turns a geometric set-selection problem into an objective-specific one.

Our main results concern the restrictions that remain after this unrestricted benchmark is understood.

1. **Bilinear orbit bounds and their limits.** For $M(x, y, w) = \begin{pmatrix} w & x \\ y & 1 \end{pmatrix}$, the sets $C_F = \{s : \text{sym}(F^T M(s)) \succeq 0\}$, $\det F > 0$, form a three-parameter family. We study this family $\mathcal{A}$ and its closed upward completions $\mathcal{B}$. Rational corners give strict gaps for both families, although an unrestricted free set attains the corner bound. Tangent edges force the orbit parameter onto a pencil, which explains several of these obstructions. A support-one example, whose minimizing ray vector has exactly one positive component, shows that the obstruction is not confined to tangent edges.
2. **Quantitative guarantees.** Scale each ray by the distance corresponding to the true corner bound. A bilinear measure D of the resulting ray size relative to vertex violation gives a positive lower bound for $z_{\mathcal{A}}/z_K$ and $z_{\mathcal{B}}/z_K$. Its worst-case order $1/D$ is sharp ([theorems 4.2 and 4.5](#) and [proposition 4.4](#)). For SCIP’s fixed point rule in a specified quadratic representation, the sharp order is $\min\{1, 1/(\kappa D)\}$, where κ measures conditioning of the scaled three-ray matrix ([theorem 4.6](#) and [proposition 4.7](#)). These statements have different invariance properties; the point-rule bound is not a representation-independent solver guarantee.
3. **Contact, attainment, and closures.** At a support-one bilinear contact, interval conditions characterize orbit supremum equality and distinguish it from attainment ([theorem 4.9](#)). Taking every cut from a restricted family can improve a single-cut bound without recovering the true corner bound. Exact certificates establish strict fixed-corner closure gaps. Upward completion

can change an infinite approximation factor into exactness at one corner, yet neither $\mathcal{A}$ nor $\mathcal{B}$ has a uniform positive approximation factor over all corners.

4. **Homogeneous minors.** For $ad - bc = 0$, the same determinant orbit consists of maximal free cones. The transformed point rule has two parameters; the rotation family has one, and their intersection at a given vertex is its polar-factor set. Full-dimensional rational examples prove strict orbit gaps, including a support-one example. Complete rational matrix witnesses certify the headline gaps ([theorem 6.7](#)).
5. **Successive cuts and computation.** A uniform positive cut depth at vertices with a fixed positive violation ensures convergence on a compact relaxation. Uniform pointedness of the basis cones supplies a sufficient geometric condition for specified rules. An explicit infinite sequence of maximal bilinear-free sets has uniformly pointed basis cones and fixed violation, but its LP bounds converge below the nonlinear optimum. Archived experiments then compare the different strength criteria. Better corner-bound selection does not consistently yield better reoptimized LP bounds, fewer rounds, or better full-solve performance.

To the best of our knowledge, the particular quantitative guarantees, contact criteria, rational restricted-family and closure obstructions, and explicit nonconvergent sequence stated here are not contained in the prior work reviewed above. This claim concerns these precise statements, not the general use of quadratic-free sets, determinant transformations, free-set optimization, or cutting-plane convergence arguments. The rank and inertia bounds, corner algorithms, and low-dimensional geometry also provide tools for deriving and checking the examples. We identify their established ingredients where they are used.

1.3 How the results fit together

The comparison is always made at a specified level. The following table records the objects that must be distinguished.

Object	Question answered
Nonlinear corner bound z_K	What can the violated constraint enforce on the chosen corner?
Best single family cut	How much of that bound can one selected set enforce?
Fixed-corner family closure	How much can all sets in that family enforce on the same corner?
LP reoptimization	What remains after imposing a cut together with all other LP rows?
Successive rounds	Do new corners and cuts approach the nonlinear optimum?
Branch-and-bound	How do separation cost, numerical behavior, and the changed search tree affect the solve?

[Section 2](#) establishes the corner benchmark and computational reductions. [Section 3](#) introduces the quadratic geometry and bilinear orbit. [Section 4](#) proves the quantitative and contact results, and [section 5](#) treats fixed-corner closures. [Section 6](#) gives the homogeneous-minor extension. [Section 7](#) separates sufficient convergence conditions from counterexamples. [Section 8](#) presents the archived computational evidence. The appendices contain complete technical proofs and the certificate specification. The electronic companion retains the exact certificate data and the compact experimental records.

2 The corner bound and unrestricted intersection cuts

Fix a closed set $S \subseteq \mathbb{R}^k$, a point $\bar{s} \notin S$, and a matrix $P = [p_1, \dots, p_N] \in \mathbb{R}^{k \times N}$. The nonnegative coordinates λ describe a corner $\bar{s} + P\mathbb{R}_+^N$. They may be the coordinates of a simplicial LP basis before projection onto the constraint variables; the projected columns of P need not be independent. For reduced costs $\omega \geq 0$, define

$$X = \{\lambda \geq 0 : \bar{s} + P\lambda \in S\}, \quad z_K(\omega) = \inf_{\lambda \in X} \omega^T \lambda, \quad (1)$$

with the infimum of the empty set equal to $+\infty$.

A closed convex set C is S -free if $\text{int } C \cap S = \emptyset$. We use sets with $\bar{s} \in \text{int } C$. Their ray steps and cut coefficients are

$$\alpha_j(C) = \sup\{t \geq 0 : \bar{s} + tp_j \in C\}, \quad a_j(C) = 1/\alpha_j(C), \quad 1/(+\infty) = 0.$$

The intersection cut and its corner value are

$$a(C)^T \lambda \geq 1, \quad z_C(\omega) = \min_{j: a_j(C) > 0} \frac{\omega_j}{a_j(C)}. \quad (2)$$

If all coefficients vanish, the cut is infeasible and its value is $+\infty$. The restriction $a_j > 0$ avoids an ambiguous product $0 \cdot \infty$ when a zero-cost ray has infinite step.

To see validity, a point with $a(C)^T \lambda < 1$ is a convex combination of finite ray exits and the apex, with positive weight on the apex, plus recession directions corresponding to infinite steps. It therefore lies in $\text{int } C$. In particular, $z_C(\omega) \leq z_K(\omega)$. For positive costs and $t > 0$, introduce the objective simplex

$$T_t = \bar{s} + P\{\lambda \geq 0 : \omega^T \lambda \leq t\} = \text{conv} \left\{ \bar{s}, \bar{s} + \frac{t}{\omega_j} p_j : 1 \leq j \leq N \right\}. \quad (3)$$

Then $z_C(\omega) \geq t$ exactly when $T_t \subseteq C$. Equivalently, $a_j(C) \leq \omega_j/t$ for every j ; the resulting cut dominates the objective-parallel inequality $\omega^T \lambda \geq t$ on the orthant.

Theorem 2.1 (Unrestricted single-cut value). *If $\omega > 0$, every nonempty X has an attained, finite, positive corner value, and*

$$\sup_{C: \bar{s} \in \text{int } C, C \text{ } S\text{-free}} z_C(\omega) = z_K(\omega).$$

This equality includes the value $+\infty$. The supremum is unchanged if only maximal S -free sets are used.

Proof. The closed set X avoids zero, and positive-cost sublevel sets in the orthant are compact. This proves attainment and positivity when X is nonempty. For any finite $0 < t < z_K(\omega)$, the compact set T_t misses S , so $T_t + \delta B$ also misses S for a sufficiently small $\delta > 0$, where B is the closed unit ball. It is a free neighbourhood of the apex and contains T_t , proving a cut value at least t . Let t increase to the corner value, or increase without bound when X is empty. Every full-dimensional free set extends to a maximal one: for a nested chain, the interior of the closed union is the union of the interiors, as a finite simplex around an interior point can be placed in a single chain member. Zorn's lemma applies. Enlargement increases all ray steps. $\square$

The direct proof uses genuine neighbourhoods of the apex. It also explains why equality need only be a supremum. The sufficiency of cut-generating functions under cone hypotheses is established by Cornuéjols et al. (2015), Conforti et al. (2015), and Kılınç-Karzan and Steffy (2016); relaxed cut-generating functions and their sufficiency are treated by Kılınç-Karzan and Steffy (2016). The following precise cone case has a short polyhedral proof, given in appendix A.1.

Proposition 2.2 (The cone containment case). *Suppose $S - \bar{s} \subseteq \text{cone}(P)$. Every valid inequality $c^T \lambda \geq 1$ for X , with $c \geq 0$, is dominated on $\mathbb{R}_+^N$ by one intersection cut from a free neighbourhood of $\bar{s}$. Consequently the best single-cut value is attained for every $\omega \geq 0$. The hypothesis holds in particular when $\text{cone}(P) = \mathbb{R}^k$.*

Without cone containment, a sufficient condition concerns the direction in which the optimum simplex meets the boundary.

Theorem 2.3 (Attainment under transversality). *Let $S = \{s : q(s) \leq 0\}$ with $q \in C^1$, $q(\bar{s}) > 0$, and $\omega > 0$. Suppose $z = z_K(\omega) < \infty$ and every minimizing contact $t = \bar{s} + P\lambda$ satisfies*

$$\nabla q(t)^T(\bar{s} - t) > 0. \quad (4)$$

For sufficiently small $\delta > 0$, $C_\delta = \text{conv}(T_z \cup B(\bar{s}, \delta))$ is S -free and has $z_{C_\delta}(\omega) = z$. Every maximal free extension also attains z .

The condition says that movement from a contact toward the apex enters the forbidden side at a positive first-order rate. Uniformity over the compact contact set allows the apex to be thickened. A full proof, including noninjective P , is in [appendix A.1](#). Multiplying a Fritz John system $\sigma_0 \omega + \sigma P^T \nabla q(t) - \nu = 0$ by a minimizing λ gives $\sigma \nabla q(t)^T(\bar{s} - t) = \sigma_0 z$. Thus a positive objective multiplier implies [equation \(4\)](#). The examples of [Kılınç-Karzan and Yang \(2017\)](#) already show the distinction between transverse and tangential approach; [theorem 2.3](#) states a sufficient smooth condition for the equality case.

Example 2.4 (Zero costs and non-attainment). For $S = \{(x, y) : (x + 1)y + 1 \leq 0\}$, apex zero, rays e_1, e_2 , and $\omega = (0, 1)$, the corner is infeasible and $z_K = +\infty$, but every intersection cut has value zero. Replacing the quadratic by $(x + 1)y + 1 - y^2$ gives the same single-cut value zero and the finite attained corner value $(1 + \sqrt{5})/2$.

With positive costs, take $q(x, y) = x - x^2 + (1 - y)^2$, the same apex and rays, and $\omega = (1, 1)$. The corner value is one, attained only at $(0, 1)$, but no free neighbourhood attains this single-cut value.

These quadratic examples illustrate known sufficiency failures ([Conforti et al., 2015](#); [Kılınç-Karzan and Yang, 2017](#)). Exact proofs are in [appendix A.1](#).

Theorem 2.5 (The unrestricted corner closure). *For every closed S and every P ,*

$$\bigcap_C \{\lambda \geq 0 : a(C)^T \lambda \geq 1\} = \mathcal{D} := \text{cl}(\text{conv } X + \mathbb{R}_+^N),$$

where C ranges over all free neighbourhoods of $\bar{s}$.

Proof. Every cut halfspace is closed, convex, and upward closed, and contains X , so it contains $\mathcal{D}$. If a nonnegative $\hat{\lambda}$ lies outside a nonempty $\mathcal{D}$, strong separation and its recession directions give $u \geq 0$ with $u^T \hat{\lambda} < \inf_X u^T \lambda$. For sufficiently small $\epsilon > 0$, the strict inequality persists with $u + \epsilon \mathbf{1}$, since its infimum over X cannot decrease. [Theorem 2.1](#) supplies a cut whose value exceeds $(u + \epsilon \mathbf{1})^T \hat{\lambda}$, so this point violates that cut. If X is empty, use a positive normal and a target larger than its value at $\hat{\lambda}$. Points outside the orthant are already excluded. $\square$

2.1 How many rays must be considered?

Lemma 2.6 (Independent ray representations). *Let $T \subseteq \mathbb{R}^k$ be closed and $X = \{\lambda \geq 0 : \bar{s} + P\lambda \in T\}$, with $r_0 = \text{rank } P$. Writing $X_J = X \cap \{\lambda_{J^c} = 0\}$,*

$$\text{conv } X = \text{conv} \left(\bigcup_{\substack{|J| \leq r_0 \\ P_J \text{ injective}}} X_J \right) + \{\delta \geq 0 : P\delta = 0\}.$$

For every $\omega \geq 0$, the infimum is the minimum of the support-restricted infima over these J . If attained, it is attained on one such support.

Proof. For any $\lambda \in X$, the pointed fibre polyhedron $\{\mu \geq 0 : P\mu = P\lambda\}$ consists of convex combinations of its vertices plus its recession cone $\{\delta \geq 0 : P\delta = 0\}$. A vertex has independent supported columns, and all fibre points belong to X . This proves the identity. Minimizing a nonnegative cost on a fibre gives a vertex optimum, which proves the value and attainment claims. $\square$

For $q(s) = s^T Qs + b^T s + c$, let $n_+(Q), n_-(Q), n_0(Q)$ denote its inertia. The nonnegative part of the form yields a sharper bound.

Theorem 2.7 (The inertia support bound). *Assume $q(\bar{s}) > 0$, $\omega > 0$, and a feasible corner. Define*

$$\rho(q) = n_+(Q) + n_0(Q) + 1 - \mathbf{1}_{\{b \notin \text{range } Q\}}.$$

Every minimizing ray vector with independent supported columns has at most $\rho(q)$ positive coordinates. Some minimizer therefore uses at most $\min(\text{rank } P, \rho(q))$ rays.

Proof. On an independent support J the positive coordinates μ minimize $\omega_J^T \mu$ subject to $g(\mu) = \mu^T G\mu + 2m^T \mu + g_0 \leq 0$, where $G = P_J^T Q P_J$. The constraint is active. If $\nabla g \neq 0$, KKT gives $\omega_J = -\sigma \nabla g$ with $\sigma > 0$, and second-order necessity makes Q nonnegative on $V = P_J \{d : \omega_J^T d = 0\}$, of dimension $|J| - 1$. This subspace lies in $\nabla q(t)^\perp$. Its intersection with $\ker Q$ is contained in $\ker Q \cap b^\perp$, of dimension $n_0 - \mathbf{1}_{\{b \notin \text{range } Q\}}$. Modulo $\ker Q$, a nonnegative subspace has dimension at most n_+ . These dimensions give the stated bound.

If $\nabla g = 0$, then $g(\mu + d) = d^T G d$. A negative direction belongs to an open symmetric cone, which contains a direction with nonzero cost derivative; one of its signs decreases the objective while staying feasible. Hence $G \succeq 0$. The same dimension argument now applies to the whole $|J|$ -dimensional supported subspace and gives the stronger bound $|J| \leq \rho(q) - 1$. $\square$

For $q = w - xy$, $n_+ = n_0 = 1$ and $b \notin \text{range } Q$, so $\rho = 2$. For strictly concave q in a feasible corner, $\rho = 1$, so a first ray contact attains the corner bound. More generally, if the open complement of a closed S is convex, its closure is the unique maximal full-dimensional free set, and its cut attains the corner value by [theorem 2.1](#). This generic reverse-convex observation does not require the inertia bound. In two variables, $\min(k, \rho) \leq 2$, although ρ itself can equal three.

2.2 Algorithms and complexity

Theorem 2.8 (Fixed-parameter computation). *For rational quadratic data with $q(\bar{s}) > 0$, rational positive costs, and $r = \min(\text{rank } P, \rho(q))$, deciding $z_K(\omega) \leq t$ for rational t takes polynomial time in the input size when r is fixed. A finite positive value can be approximated to relative accuracy ϵ in time polynomial in the input size and $\log(1/\epsilon)$.*

If $\text{rank } P = k$, the search can instead use at most $O(N^{\lfloor k/2 \rfloor})$ independent k -ray cells of a refinement of the fibre-cost normal fan, together with their supports of size at most r . For $k = 3$, at most $2N - 4$ cells and $3N - 6$ two-ray edges suffice.

[Appendix A.2](#) gives the fixed-variable algebraic decision argument, the initial approximation bracket, and the degeneracy-safe normal-fan construction. Generic stationary-point formulas apply on nonsingular faces. At most two rays admit a formula covering all singular and tangential cases.

Proposition 2.9 (An exact two-ray formula). *Fix rays i, j and put $\nu(\theta) = (\theta/\omega_i, (1 - \theta)/\omega_j)$ for $0 \leq \theta \leq 1$. Write*

$$q(\bar{s} + \tau P_{ij} \nu(\theta)) = a(\theta) \tau^2 + b(\theta) \tau + g_0, \quad \Delta(\theta) = b(\theta)^2 - 4g_0 a(\theta),$$

where $g_0 = q(\bar{s}) > 0$. The two-ray value is

$$z_{ij} = \frac{1}{\max_{\substack{0 \leq \theta \leq 1 \\ \Delta(\theta) \geq 0}} u_+(\theta)}, \quad u_+(\theta) = \frac{-b(\theta) + \sqrt{\Delta(\theta)}}{2g_0},$$

provided the maximum is positive; otherwise $z_{ij} = +\infty$. The maximum is obtained among the interval endpoints, real roots of Δ , and roots of $\Delta'^2 - 4b'^2\Delta$ and Δ' , with identically vanishing polynomials treated as in [appendix A.2](#).

Consequently a bilinear corner requires $O(N^2)$ such one- and two-ray formulas. When $\text{rank } P = 3$, a three-dimensional convex-hull computation followed by at most $4N - 6$ formulas suffices. The unrestricted supremum is attained unless some minimizing contact is the tangent first contact of a single ray; the independent-representation argument proving this qualification is in [appendix A.2](#).

Theorem 2.10 (Variable-dimension hardness). *Computing the corner value is NP-hard when the constraint dimension is part of the input, even with $P = I$, $\bar{s} = 0$, and $\omega = \mathbf{1}$. Rational-threshold decision is NP-hard, as is approximation within relative error $1/(5k^2)$ in dimension k . By [theorem 2.1](#), the same statements hold for the unrestricted best single-cut value.*

The proof in [appendix A.2](#) is a Motzkin–Straus reduction. The reduction has unbounded ρ as well as unbounded dimension. A carefully normalized efficacy optimization is also possible with the fixed-parameter oracle; [remark A.2](#) specifies the separation oracle and metric needed for that assertion.

2.3 A fixed-corner closure and a first-round polytope closure differ

Proposition 2.11. *For a polytope $\mathcal{P}$ and closed S ,*

$$\bigcap_{C \text{ } S\text{-free}} \text{conv}(\mathcal{P} \setminus \text{int } C) = \text{conv}(\mathcal{P} \cap S).$$

This operation permits free sets unrelated to an LP basis. In contrast, the rank-one basis intersection-cut closure intersects $\mathcal{P}$ with all intersection cuts from its original feasible bases.

Proposition 2.12 (A rank-one closure gap). *Let $\mathcal{P} = \text{conv}\{(0, 2), (-1/2, 0), (1/2, 0)\}$ and $S = \{s : \|s\| \geq 1\}$. Put*

$$t = \frac{1 + 2\sqrt{13}}{17}, \quad p_{1,2} = \left(\mp \frac{1-t}{2}, 2t \right), \quad v = \left(0, \frac{1 + \sqrt{13}}{6} \right).$$

The rank-one basis intersection-cut closure is $\text{conv}\{(0, 2), p_1, v, p_2\}$, strictly larger than $\text{conv}(\mathcal{P} \cap S) = \text{conv}\{(0, 2), p_1, p_2\}$. A second round at v gives the chord $p_1 p_2$ and the hull.

Appendix A.3 proves both propositions. The example uses all free sets, not a restricted quadratic family. It therefore also shows why the fixed-corner equality in theorem 2.5 cannot imply one-round convergence for a polytope.

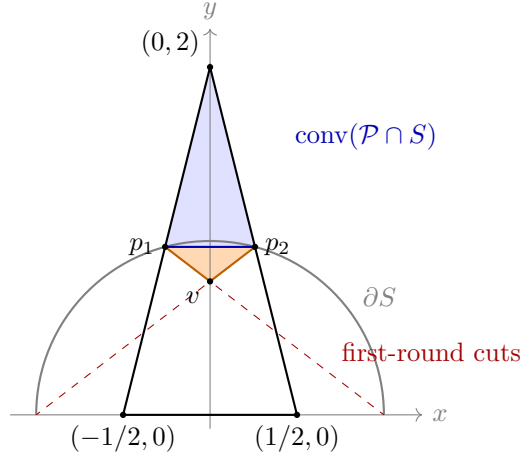


Figure 1: The unrestricted rank-one closure in proposition 2.12. The feasible hull is blue. The first-round cuts also retain the orange triangle below its chord. A cut at the new vertex v removes that triangle.

3 Quadratic orbits and exact obstructions

Quadratic sets can be brought to a difference of squared norms by homogenization and a linear change of coordinates. This does not identify a unique free set: different coordinates can give different cuts. We first describe a signature for which the unrestricted transformation orbit is complete, and then the bilinear signature for which it is not.

3.1 One negative or one positive square

Let $Q_{n,m} = \{(x, y) \in \mathbb{R}^n \times \mathbb{R}^m : \|x\| \leq \|y\|\}$ and $C_\ell = \{(x, y) : \ell^T x \geq \|y\|\}$ for a unit vector $\ell \in \mathbb{R}^n$. An automorphism preserves the homogeneous quadratic form $\|x\|^2 - \|y\|^2$. This geometry is related to the general maximal quadratic-free characterizations of Muñoz et al. (2025, 2026).

Theorem 3.1 (Complete Lorentz orbits). *For $n \geq 2$, every full-dimensional maximal $Q_{n,1}$ -free set is*

$$C_{\gamma_1, \gamma_2} = \{(x, y) : \gamma_1^T x \geq y, \gamma_2^T x \geq -y\}, \quad \|\gamma_i\| = 1, \quad \gamma_1 \neq -\gamma_2. \quad (5)$$

For any fixed unit ℓ , these are exactly the images of C_ℓ under $SO^+(n, 1)$, the determinant-one time-orientation-preserving Lorentz group. For signature $(1, m)$, the only full-dimensional maximal free sets are $\{x \geq \|y\|\}$ and $\{-x \geq \|y\|\}$.

If a quadratic's homogenized nondegenerate part has signature $(n, 1)$ or $(1, m)$, every full-dimensional maximal free affine set is a slice of one of these homogeneous sets, with the radical coordinates included as a product before slicing. Thus the free-direction transformation orbit has single-cut supremum $z_K(\omega)$ for positive costs.

Appendix A.4 gives elementary separation, classification, Lorentz-basis, and slicing proofs. In particular no transitivity theorem is needed. Every indefinite quadratic in two variables has

one of these homogenized signatures. Equality of the supremum does not imply attainment: the positive-cost example in [example 2.4](#) already lies in this class.

For signature $(n, 1)$, testing whether the target simplex fits inside [equation \(5\)](#) is two independent second-order cone feasibility problems in γ_1, γ_2 . The relaxation $\|\gamma_i\| \leq 1$ remains free; the classification proof extends any such full-dimensional pair to a maximal unit pair. Hence the relaxation preserves the best value. Signature $(1, m)$ needs only two candidates.

The implemented point rule is a different restriction. In transformed coordinates it chooses $\ell = x'/\|x'\|$, where (x', y') is the homogenized apex, rather than optimizing the free direction.

Proposition 3.2 (The transformed point rule). *For signature $(n, 1)$, at a fixed apex $u = (x, y)$ outside $Q_{n,1}$, the plain point rule under all transformations produces exactly the pairs in [equation \(5\)](#) for which*

$$u = a(\gamma_1, 1) + b(\gamma_2, -1), \quad a, b > 0. \quad (6)$$

For $n = 2$ this is a one-parameter family. Writing $\gamma_1 = (\cos \theta, \sin \theta)$, its second direction is determined by

$$a = \frac{\|x\|^2 - y^2}{2(\gamma_1^T x - y)}, \quad b = a - y, \quad \gamma_2 = \frac{x - a\gamma_1}{b},$$

on the domain where $a, b > 0$. The unrestricted pairs have two parameters.

The proof is in [appendix A.4](#). In general the respective dimensions are $n - 1$ and $2(n - 1)$. When both null tangency rays meet an affine slice with positive scaling, [equation \(6\)](#) says that the apex lies on the chord between their two finite tangency points. There are also maximal affine sets whose contacts are only asymptotic: for $S = \{(x, y) : xy \geq 1\}$, the quadrant $\{x \geq 0, y \leq 0\}$ is maximal free and misses S entirely. Its two homogeneous tangency rays lie at infinity, so no affine apex satisfies [equation \(6\)](#). Neither the plain point rule nor an enlargement of one of its sets produces this quadrant at any apex. [Example A.3](#) supplies the exact proof. This exclusion concerns generation of sets; it alone asserts no strict gap for the point-rule supremum at a particular corner.

Even a fixed eigen-coordinate choice can lose almost the entire bound.

Proposition 3.3 (A fixed rule with vanishing ratio). *Let $S = \{(x, y) : y^2 \geq x^2 + 1\}$, $\bar{s} = (x_0, 0)$, rays $(-1, 0), (0, 1)$, and costs $(\epsilon, 1)$, where $0 < \epsilon < 1$ and $x_0 \geq \epsilon/\sqrt{1 - \epsilon^2}$. The eigen-coordinate point rule gives*

$$\left\{ (x, y) : |y| \leq \frac{x_0 x + 1}{\sqrt{1 + x_0^2}} \right\}, \quad z_{\text{rule}} = \min \left\{ \epsilon(x_0 + 1/x_0), \sqrt{1 + x_0^2} \right\}.$$

The strip $|y| \leq 1$ belongs to the same free-direction family and has value one, whereas $z_K = \epsilon x_0 + \sqrt{1 - \epsilon^2}$. With $\epsilon = x_0^{-2}$ and $x_0 \rightarrow \infty$, $z_{\text{rule}}/z_K \rightarrow 0$ and the strip's ratio tends to one.

[Appendix A.4](#) proves the formulas. Even the closure of the entire constant-direction family can have a vanishing ratio on pointed corners; [proposition A.4](#) proves that separate assertion. Neither statement concerns the unrestricted transformation orbit in [theorem 3.1](#).

3.2 The determinant model for a bilinear constraint

An indefinite quadratic of rank two is a product of two independent linear forms plus an affine term. If its linear term belongs to the range of its quadratic matrix, only two affine coordinates matter

and [theorem 3.1](#) applies. Otherwise a third independent affine coordinate puts it in the form $w - xy$ or its negative. We use

$$q(s) = w - xy, \quad S = \{q \leq 0\}, \quad M(s) = \begin{pmatrix} w & x \\ y & 1 \end{pmatrix}, \quad M_0(p) = \begin{pmatrix} p_w & p_x \\ p_y & 0 \end{pmatrix}, \quad E = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}. \quad (7)$$

The homogeneous form $\det M$ has signature $(2, 2)$, and E increases the w coordinate. The opposite bilinear side follows by $(x, y, w) \mapsto (-x, y, -w)$. Throughout the bilinear corner statements, $q(\bar{s}) > 0$ and $\omega > 0$. For this S , the cone-containment hypothesis in [proposition 2.2](#) holds exactly when $\text{cone}(P) = \mathbb{R}^3$. Indeed every (x, y, w) is the midpoint of $(x + a, y + a, w)$ and $(x - a, y - a, w)$, both in S for sufficiently large a , so $\text{conv } S = \mathbb{R}^3$. Any convex cone containing $S - \bar{s}$ must therefore be the whole space. The opposite bilinear side follows by the stated symmetry.

Proposition 3.4 (The bilinear orbit). *The determinant-preserving linear maps on 2×2 matrices are $M \mapsto AMB^T$ and $M \mapsto AM^TB^T$, with $\det A \det B = 1$. The homogeneous constant-direction free cone is, in suitable coordinates, $\{M : \text{sym}(M) \succeq 0\}$. Its full orbit is*

$$\hat{C}_F = \{M : \text{sym}(F^T M) \succeq 0\}, \quad \det F > 0,$$

modulo positive scaling of F . Its affine slice is C_F . These cones are full dimensional and determinant-free. At a nonzero rank-one matrix $M = ab^T$, membership is exactly $F^T a \in \mathbb{R}_+ b$. On the slice, their feasible contacts form the increasing Möbius graph

$$y = \frac{F_{11}x + F_{21}}{F_{12}x + F_{22}}, \quad F_{12}x + F_{22} > 0, \quad w = xy.$$

The matrix identities and the automorphism classification are proved in [appendix A.5](#). These are three-parameter sets. The one-parameter family (14a) of [Bienstock et al. \(2020\)](#) is exactly the rotation subfamily $F^T \in SO(2)$; [proposition A.5](#) gives the identities and shows why nonrotation orbit members lie outside both of their displayed families.

Proposition 3.5 (Upward completion). *For $\det F > 0$, define*

$$B_F = \text{cl}(C_F + \mathbb{R}_+ e_w), \quad Z_F = \text{sym}(F^T E).$$

The corresponding homogeneous completion retains exactly the inequalities $\langle Fvv^T, M \rangle \geq 0$ for which $v^T Z_F v \geq 0$. On the affine slice,

$$s \in B_F \iff \exists \tau \in [0, q(s)] : \text{sym}(F^T [M(s) - \tau E]) \succeq 0. \quad (8)$$

Writing $F = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$, the exceptional class $b = 0, a < 0$ has empty completed slice. Every other B_F is full dimensional, maximal S -free, and contains C_F .

The proof in [appendix A.5](#) uses an exact two-term PSD decomposition and treats closure on the slice explicitly. These completions agree with the applicable maximal completion construction of [Muñoz et al. \(2026\)](#). The constant-direction member in the eigen-coordinates of [Chmiela et al. \(2020\)](#) is their bilinear Case 4 set: its defining function is the support function of the retained spherical cap.

Let $\mathcal{A}$ consist of the orbit slices C_F that contain $\bar{s}$ in their interior, and let $\mathcal{B}$ consist of the nonempty completions B_F with that property. Write $z_{\mathcal{A}}, z_{\mathcal{B}}$ for the corresponding single-cut suprema. Then $z_{\mathcal{A}} \leq z_{\mathcal{B}} \leq z_K$. A completion can contain the apex even when its underlying orbit slice does not.

Proposition 3.6 (Convex feasibility for the best orbit value). *For $t > 0$, consider the nested feasibility system*

$$\text{sym}(F^T M(\bar{s})) \succ 0, \quad \text{sym}\left(F^T M\left(\bar{s} + \frac{t}{\omega_j} p_j\right)\right) \succeq 0 \quad (1 \leq j \leq N).$$

Feasibility at target t is equivalent to the existence of an orbit cut of value at least t . The strict apex condition implies $\det F > 0$ and, by positive scaling, may be replaced by $\text{sym}(F^T M(\bar{s})) \succeq I$. Thus bisection uses $N + 1$ linear matrix inequalities (LMIs) of size two in four scalar variables.

Proof. The endpoint LMIs express $T_t \subseteq C_F$, and the apex LMI expresses interior membership. A positive definite symmetric part has positive determinant, so $\det(F^T M(\bar{s})) > 0$; since $q(\bar{s}) > 0$, this implies $\det F > 0$. Scaling normalizes the smallest apex eigenvalue. Convexity and nestedness follow directly from the affine LMIs and the apex-to-endpoint segments. The supremum need not itself be feasible. $\square$

3.3 A tangent edge forces a pencil

Let t be a feasible boundary point in the relative interior of a nontrivial segment e , with direction d satisfying $\nabla q(t)^T d = 0$ and $\det M_0(d) > 0$. Such a segment occurs when a two-ray minimizer lies in the relative interior of its objective face. Orient d so that $d_x > 0$, and set $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Lemma 3.7 (Tangent-edge rigidity). *If an invertible F has $e \subseteq B_F$, then*

$$F^T = \theta(G_0 + uG_1), \quad G_0 = J \text{adj}(M_0(d)), \quad G_1 = J \text{adj}(M(t)), \quad \theta > 0, \quad u \in \mathbb{R}. \quad (9)$$

The pencil has constant determinant $\det M_0(d) > 0$ and $\text{sym}(G_1 M(t)) = 0$. If the nontrivial segment lies on a ruling, so that $\det M_0(d) = 0$, no invertible F has $e \subseteq B_F$.

Appendix A.6 derives the pencil from the two-sided PSD conditions. Without the chosen orientation the sign condition would be $\theta d_x > 0$. A tangent edge therefore reduces the orbit-containment test to one real parameter, but the remaining simplex vertices can impose incompatible conditions on it.

Theorem 3.8 (A strict gap for both transformation families). *For $S = \{w \leq xy\}$, take $\omega = (1, 1, 1)$ and rays from*

$$\bar{s} = (-9/2, 0, 3/2)$$

to

$$v_1 = (-1, -6, 18), \quad v_2 = (-5, 6, -18), \quad v_3 = (0, 5/2, 5/2).$$

Then $z_K = 1$, attained only at $t = (-3, 0, 0) = (v_1 + v_2)/2$. Neither an orbit slice nor an upward completion contains T_1 , and

$$z_A \leq z_B < z_K.$$

The unrestricted optimum is attained by a maximal free set outside $\mathcal{B}$. Strict inequality and unrestricted attainment persist for all corner data in a sufficiently small open neighbourhood of this example.

The proof in appendix A.6 supplies a nonnegative barycentric identity, scalar certificates excluding every pencil parameter for both families, and a compactness argument excluding degenerate limits. Its transversality value is $3/2$. A second rational instance has a larger certified orbit gap; its exact construction and qualification to $\mathcal{A}$ are in example A.7.

The tangent-edge mechanism does not characterize all failures.

Proposition 3.9 (Support one and transverse edges do not suffice). *For the same S and costs, take $\bar{s} = (-2, 3, 2)$ and rays to*

$$v_1 = (0, 0, 0), \quad v_2 = (6, -2, 1/4), \quad v_3 = (1, -5/2, 1/2).$$

Then $z_K = 1$, attained only at v_1 . All three simplex edges at v_1 are transverse, with positive derivative values $2, 1/4, 1/2$. Nevertheless $z_A < 1$: no nonzero F satisfies the closed PSD conditions at all four simplex vertices. The inactive-ray KKT multipliers are $1/8$ and $1/4$.

[Appendix A.6](#) gives a nonnegative barycentric identity and a small integer dual certificate. The full simplex geometry matters even when the minimizing ray and the inactive-ray multipliers are regular.

Proposition 3.10 (A maximal polyhedral set outside the orbit completion). *For any $a, b \in \mathbb{R}$, the set*

$$W = \{(x, y, w) : x \geq a, \ y \leq b, \ w \geq bx + ay - ab\}$$

is maximal $\{w \leq xy\}$ -free and belongs to neither transformation family. More generally, neither family contains two distinct feasible boundary contacts on a common bilinear ruling.

[Appendix A.6](#) proves maximality by explicit interior mixtures and exclusion by the rank-one contact identity. This illustrates a missing maximal set directly, while [theorem 3.8](#) and [proposition 3.9](#) show that missing sets can affect an objective bound.

4 Depth guarantees and their limits

An orbit set must accommodate all directions of the corner with one convex quadratic geometry. The relevant scale is the size of the corner at its optimal objective level relative to the violation at its vertex. This scale gives a uniform approximation guarantee. It also identifies a regime in which even completed orbit sets can lose almost the entire corner bound.

Throughout this section, $S = \{(x, y, w) : q(x, y, w) = w - xy \leq 0\}$, $q_0 = q(\bar{s}) > 0$, there are finitely many rays p_j , and $\omega_j > 0$. We assume that the feasible corner is nonempty, so that $0 < z_K < \infty$. Define

$$\tilde{p}_j = \frac{z_K}{\omega_j} p_j, \quad T_r = \text{conv}\{\bar{s}, \bar{s} + r\tilde{p}_j : 1 \leq j \leq N\}, \quad T^* = T_1. \quad (10)$$

Every point of T_r has a representation of objective value at most rz_K . Hence $q > 0$ on T_r when $r < 1$, and $q \geq 0$ on T^* . An admissible convex set has cut ratio at least r precisely when its steps along all scaled rays are at least r , equivalently when it contains T_r . We use z_A and z_B for the suprema over the orbit sets and their closed upward completions, respectively, with $\bar{s}$ in their interiors.

4.1 The scale of the optimal simplex

Put

$$\tilde{X} = \max_j |\tilde{p}_{jx}|, \quad \tilde{Y} = \max_j |\tilde{p}_{jy}|, \quad D = \sqrt{\frac{\tilde{X}\tilde{Y}}{q_0}}. \quad (11)$$

Thus D^2 compares the largest possible bilinear curvature scale of the scaled rays with the violation at the vertex. Its reciprocal $\delta = 1/D^2$ is a relative depth, with $\delta = \infty$ when $D = 0$.

Lemma 4.1 (Symmetries of depth). *For $ab > 0$, the affine maps*

$$(x, y, w) \mapsto (ax + u, by + v, abw + avx + buy + uv)$$

and the interchange of x and y preserve S , transport both families onto themselves, and preserve z_K, z_A, z_B when the rays are transported and the costs are kept fixed. They preserve D . The relative discriminant

$$\Delta_{\text{rel}} = \frac{B^2 - 4Aq_0}{B^2 + |4Aq_0|} \quad (12)$$

of a restriction $q(\bar{s} + sp) = As^2 + Bs + q_0$ is also preserved whenever its denominator is nonzero. Euclidean angles, the ray condition number, and the fixed rule studied below need not be preserved.

Proof. The map multiplies q by $ab > 0$. In matrix coordinates it is $M' = LMR^T$, with $L = \begin{pmatrix} a & u \\ 0 & 1 \end{pmatrix}$ and $R = \begin{pmatrix} b & v \\ 0 & 1 \end{pmatrix}$. Writing $G = F^T$, the transformed orbit matrix is $G' = RGL^{-1}$, since $\text{sym}(G'M') = R\text{sym}(GM)R^T$. Its determinant is positive. The map sends e_w to abe_w , so it also transports upward completions. Transposition uses $G' = G^{-1}$ and a congruence by G . Ray coordinates and their objective values are unchanged. Finally $\tilde{X}, \tilde{Y}, q_0$ are multiplied by $|a|, |b|, ab$, respectively; the coefficients in the ray restriction share the multiplier ab . $\square$

These are symmetries of the bilinear model, rather than arbitrary affine changes of coordinates. A constant ray restriction has no defined relative discriminant. A zero discriminant means a repeated root; a grazing contact on the forward ray additionally requires $B < 0$. These qualifications avoid using a ray angle as a substitute for vertex depth.

For $t > 0$, the parabolic cylinder centered at the vertex is

$$C_t = \left\{ s : q(s) \geq \frac{(t(x - \bar{x}) - (y - \bar{y})/t)^2}{4} \right\}. \quad (13)$$

It is the epigraph of a convex quadratic and the orbit set with $G = F^T = \begin{pmatrix} t & -t\bar{x} + \bar{y}/t \\ 0 & 1/t \end{pmatrix}$. Indeed its determinant inequality is [equation \(13\)](#), and the second diagonal entry of its PSD matrix is the positive constant $1/t$. It equals its own upward completion, and its interior contains $\bar{s}$. Along a scaled ray, define

$$\ell_j = \frac{\nabla q(\bar{s})^T \tilde{p}_j}{q_0}, \quad b_j(t) = \frac{t\tilde{p}_{jx} + \tilde{p}_{jy}/t}{\sqrt{q_0}}.$$

The cylinder slack, divided by q_0 , is $1 + \ell_j s - b_j(t)^2 s^2 / 4$, and its step is

$$\alpha_j(t) = \frac{2}{\sqrt{\ell_j^2 + b_j(t)^2} - \ell_j}. \quad (14)$$

A zero denominator means an infinite step. Thus a single scalar parameter t already gives a cut with a useful uniform bound.

Theorem 4.2 (Depth guarantee). *With the hypotheses above, without a rank assumption on the projected rays,*

$$\begin{aligned} \frac{z_B}{z_K} &\geq \frac{z_A}{z_K} \geq \rho_{\text{par}} := \sup_{t>0} \min_j \alpha_j(t) \geq f(D), \\ f(D) &= \begin{cases} \frac{2}{\sqrt{(1+D^2)^2 + 4D^2} + 1 + D^2}, & 0 \leq D \leq 1, \\ ((1+\sqrt{2})D)^{-1}, & D \geq 1. \end{cases} \end{aligned} \quad (15)$$

The two branches agree at $D = 1$. For $D \leq 1$, the simpler estimate $f(D) \geq (1 + 2D^2)^{-1}$ also holds. In particular $f(D) \geq \min\{1/3, 0.41/D\}$ for $D > 0$, and $z_A = z_B = z_K$ as suprema when $D = 0$.

Proof. Suppose first that $D > 0$, and choose $t = \sqrt{\tilde{Y}/\tilde{X}}$. Then $|b_j(t)| \leq 2D$. Write

$$g_j(s) = \frac{q(\bar{s} + s\tilde{p}_j)}{q_0} = 1 + \ell_j s + d_j s^2, \quad d_j = -\frac{\tilde{p}_{jx}\tilde{p}_{jy}}{q_0}, \quad |d_j| \leq D^2.$$

Since $g_j(1) \geq 0$, we have $\ell_j \geq -1 - D^2$. If $D \geq 1$, evaluating at $1/D \leq 1$ instead gives $\ell_j \geq -D - d_j/D \geq -2D$. For $\ell_j \geq 0$, the denominator in [equation \(14\)](#) is at most $|b_j| \leq 2D$. For $\ell_j < 0$ and $D \leq 1$, it is at most

$$\sqrt{(1 + D^2)^2 + 4D^2} + 1 + D^2 \leq 2 + 4D^2;$$

the intermediate square-root bound follows by squaring, with difference $8D^4$. For $D \geq 1$, the denominator is at most $2D(1 + \sqrt{2})$. These bounds prove the result for positive depth.

If $D = 0$, all d_j vanish. Feasibility and the scaled corner optimum give $\min_j \ell_j = -1$: minimizing $\sum_j \lambda_j$ subject to $1 + \sum_j \ell_j \lambda_j \leq 0$ has value one. Let $t \rightarrow 0$ if $\tilde{Y} = 0$, or $t \rightarrow \infty$ if $\tilde{X} = 0$. Then every $b_j(t)$ tends to zero. Negative ℓ_j give limiting steps $-1/\ell_j$, while nonnegative ℓ_j give infinite limiting steps. The finite-ray minimum tends to one. Both family values are bounded above by z_K , proving their supremum equality. This limiting argument does not assert attainment. $\square$

In fixed coordinates, a Euclidean depth condition can imply this bound: $\text{dist}(\bar{s}, S) \leq q_0$ by vertical projection, and $\tilde{X}\tilde{Y} \leq \text{diam}(T^*)^2$. Consequently $\delta \geq \text{dist}(\bar{s}, S)/\text{diam}(T^*)^2$ when the product is positive. No condition on the ray matrix is needed in the theorem.

Corollary 4.3 (A discriminant margin). *Suppose $D \geq 1$. Assume that every ray missing S on which q initially decreases has relative discriminant at most $-\mu$, where $0 < \mu \leq 1$. Put $\gamma = \sqrt{(1 - \mu)/(1 + \mu)}$ and $\gamma_D = \max\{\gamma, 1/D\}$. Then*

$$\frac{z_A}{z_K} \geq \rho_{\text{par}} \geq \frac{1}{D(\gamma_D + \sqrt{1 + \gamma_D^2})}.$$

As $D \rightarrow \infty$, its leading constant ranges from $\sqrt{2} - 1$ as $\mu \downarrow 0$ to one when $\mu = 1$.

Proof. A decreasing restriction missing S has $d_j > 0$ and negative discriminant. The margin gives $\ell_j^2 \leq 4\gamma^2 d_j$, hence $-\ell_j \leq 2\gamma D$. A restriction meeting S has its first positive root at least one. If $d_j > 0$, both roots are at least one and $-\ell_j \leq 2$; if $d_j \leq 0$, the positive first root gives $-\ell_j \leq 1$. Thus every negative ℓ_j satisfies $-\ell_j \leq 2D\gamma_D$. Use $|b_j| \leq 2D$ in [equation \(14\)](#). Nonnegative ℓ_j give at least $1/D$, which is at least the stated bound. $\square$

4.2 Sharp order, including completed sets

The next family keeps its ray matrix fixed and makes the vertex approach the surface. Its two horizontal rays miss S with the largest possible negative relative discriminant. These strong ray margins do not prevent loss.

Proposition 4.4 (Fixed rays and vanishing depth). *For $0 < \varepsilon < 1$, let*

$$\bar{s} = (0, 0, \varepsilon), \quad p_1 = (1, -1, 0), \quad p_2 = (1, -2, 0), \quad p_3 = (1, 1, 1), \quad \omega = (1, 1, 1).$$

Then $z_K = z_0 = (1 + \sqrt{1 + 4\varepsilon})/2$, uniquely attained on ray three, with a transversal crossing. The relative discriminants are $-1, -1, +1$, the ray matrix is fixed and nonsingular, and $D = z_0\sqrt{2/\varepsilon}$. Nevertheless $z_B/z_K \rightarrow 0$ as $\varepsilon \downarrow 0$. The lower bound in [theorem 4.2](#) applies.

Proof. Expansion gives

$$q(\bar{s} + P\lambda) = \varepsilon + \lambda_3 - \lambda_3^2 + \lambda_2\lambda_3 + (\lambda_1 + 2\lambda_2)(\lambda_1 + \lambda_2).$$

Feasibility forces $\lambda_3 \geq z_0$, and objective value z_0 forces the other coordinates to vanish. The ray restrictions give the discriminants and depth directly. The completed-set limit argument proving vanishing is given in [appendix B.2](#). It is analytic and supplies no rate by itself. $\square$

Theorem 4.5 (Exact computer-assisted sharpness bounds). *On the family in [proposition 4.4](#), for every $0 < \varepsilon < 1$,*

$$\frac{z_A}{z_K} \leq \frac{z_B}{z_K} \leq \frac{137\sqrt{\varepsilon}}{z_0} = \frac{137\sqrt{2}}{D}.$$

For

$$0 < \varepsilon \leq \varepsilon_1 = \frac{53661932375105209}{2934348356061848092900},$$

an explicit completed set, defined by a rational matrix after normalization, gives

$$z_B/z_K \geq (1367/10)\sqrt{\varepsilon}/z_0.$$

For $0 < \eta < 1$, $k \geq 1$, and $L > 0$, consider instead

$$\bar{s} = (0, 0, 1), \quad p_1 = (1, -1, -2(1-\eta)), \quad p_2 = (1, -k, -2\sqrt{k}(1-\eta)), \quad p_3 = (1, 1, L), \quad \omega = (1, 1, 1). \quad (16)$$

Here $L < z_K < L + 1/L$ and $D = \sqrt{k} z_K$. The following exact bounds hold; the upper bounds are computer-assisted, and the lower bounds are obtained from printed rational witnesses:

η	k	Dz_B/z_K upper, all $L > 0$	Dz_B/z_K lower, $L \geq 17/5$
1/1000	4	197/100	2461/1250
1/100	4	5/2	6117/2500
1/1000	9/4	63/40	15621/10000
1/1000	49/25	77/50	18851/12500

Every upper bound also bounds Dz_A/z_K .

Proof. For [equation \(16\)](#), put $m = \lambda_1 + \sqrt{k}\lambda_2$. Cauchy–Schwarz gives $(\lambda_1 + \lambda_2)(\lambda_1 + k\lambda_2) \geq m^2$, and expansion yields

$$q \geq (m - (1 - \eta))^2 + 2\eta - \eta^2 + L\lambda_3 - \lambda_3^2.$$

Thus feasibility forces $\lambda_3 > L$, while ray three alone reaches S before $L + 1/L$. The two missing ray restrictions have minimum $2\eta - \eta^2$ and relative discriminant $((1 - \eta)^2 - 1)/((1 - \eta)^2 + 1)$. The third discriminant is $+1$. The upper bounds and rational lower witnesses are established in [appendices B.3](#) and [B.4](#). $\square$

For each family separately, define its asymptotic worst-case constant by

$$C_A^* = \lim_{D_0 \rightarrow \infty} \inf_{D \geq D_0} D \frac{z_A}{z_K}, \quad C_B^* = \lim_{D_0 \rightarrow \infty} \inf_{D \geq D_0} D \frac{z_B}{z_K},$$

where the infima range over the corners in this section. The limits exist by monotonicity. The analytic lower bound and the exact computer-assisted upper bound imply

$$\sqrt{2} - 1 \leq C_A^* \leq C_B^* \leq \frac{77}{50} = 1.54.$$

Thus the order $1/D$ is sharp for both families, and the analytic constant is within a factor $1.54(1 + \sqrt{2}) < 3.72$ of either worst-case constant. The constants need not be equal. Bounded depth gives a uniform guarantee; large depth permits bad corners, while some large-depth corners remain exact.

4.3 A fixed point rule and its coordinate dependence

Consider the Case-4 rule applied to the constraint exactly as $w - xy \leq 0$: unit coefficient on w , no additional linear terms in x, y , and source constant $\kappa_{\text{SCIP}} = 0$. All coordinates below are those of this representation. The rule chooses the unit vector proportional to $((\bar{x} - \bar{y})/2, (\bar{w} + 1)/2)$ in the positive Sylvester coordinates. Set

$$N_0^2 = (\bar{x} - \bar{y})^2 + (\bar{w} + 1)^2, \quad V(d) = (\bar{x} - \bar{y})d_w - (\bar{w} + 1)(d_x - d_y).$$

The uncompleted orbit set is

$$C_{\text{fix}} = \left\{ s : 4N_0^2 q(s) \geq V(s - \bar{s})^2, (\bar{w} + 1)(w + 1) + (\bar{x} - \bar{y})(x - y) \geq 0 \right\}. \quad (17)$$

The actual completed set is $B_{\text{fix}} = C_{\text{fix}} + \mathbb{R}_+ e_w$. Write z_{fix} for its cut value. The first positive root of $4N_0^2 q(\bar{s} + sp) = s^2 V(p)^2$ gives the uncompleted step. A completed step must instead allow vertical lowering; it can be larger or infinite. The rotation derivation and an explicit example distinguishing the two steps are in [appendix B.6](#).

Theorem 4.6 (Fixed-rule guarantee). *For any finite ray family, both the uncompleted and completed rule values satisfy*

$$\frac{z_{\text{fix}}}{z_K} \geq \min \left\{ \frac{1}{2}, \frac{1}{2D}, \frac{\sqrt{q_0}}{\max_j \|(\tilde{p}_{jx} - \tilde{p}_{jy}, \tilde{p}_{jw})\|} \right\}.$$

Reciprocals of zero are interpreted as infinite. If the $3 \times N$ scaled ray matrix has full row rank and spectral condition number κ , then

$$\frac{z_{\text{fix}}}{z_K} \geq \min \left\{ \frac{1}{2}, \frac{1}{\sqrt{2N} \kappa D} \right\}.$$

In particular the constant is $1/\sqrt{6}$ for three independent rays.

Proof. Put $s_0 = \min\{1/2, 1/(2D)\}$. The polynomial g_j used above is at least $1/4$ on $[0, s_0]$. Indeed, for $d_j \leq 0$, concavity gives $g_j(s) \geq 1 - s$; for $d_j > 0$ with a forward root, both roots are at least one and $g_j(s) \geq (1 - s)^2$. If there is no forward root and $\ell_j < 0$, $\ell_j > -2\sqrt{d_j}$, so $g_j(s) \geq (1 - \sqrt{d_j}s)^2 \geq (1 - Ds)^2$ on this interval. If $\ell_j \geq 0$, the assertion is immediate. The determinant inequality in [equation \(17\)](#) therefore holds through the additional length $N_0\sqrt{q_0}/|V(\tilde{p}_j)|$. Since $|V(p)| \leq N_0\|(p_x - p_y, p_w)\|$, this proves the first bound. The PSD matrix starts positive definite at $\bar{s}$ and stays so before its first determinant zero; hence the trace branch cannot change first.

For the second bound, $\|(p_x - p_y, p_w)\| \leq \sqrt{2}\|p\| \leq \sqrt{2}\|\tilde{P}\|_2$. Full row rank gives $\sigma_{\min}(\tilde{P}) \leq \|\text{row}_x(\tilde{P})\| \leq \sqrt{N}\tilde{X}$, and likewise with y . Hence $\sigma_{\min} \leq \sqrt{N}D\sqrt{q_0}$ and $\|\tilde{P}\|_2 \leq \kappa\sqrt{N}D\sqrt{q_0}$. Since full row rank requires $N \geq 3$ and $\kappa \geq 1$, the term $1/(\sqrt{2N}\kappa D)$ also does not exceed $1/(2D)$. Upward completion can only increase the cut value. $\square$

Proposition 4.7 (Sharp loss in κD). *Given any $\kappa \geq 1$ and $D > 0$, put $v = \kappa D$, $u = \sqrt{1 + v^2}$, and take unit costs and*

$$\bar{s} = (u, -u, -1), \quad p_1 = (vD, 0, 0), \quad p_2 = (0, -vD, 0), \quad p_3 = (0, 0, -v^2).$$

This corner has exactly depth D , condition number κ , and $z_K = z_A = z_B = 1$, with the family optimum attained. All three relative discriminants are $+1$. Nevertheless both fixed-rule sets give

$$\frac{z_{\text{fix}}}{z_K} = \frac{2}{1 + \sqrt{1 + \kappa^2 D^2}} < \frac{2}{\kappa D}.$$

Thus for three rays the worst fixed-rule ratio at prescribed κ, D has order $\min\{1, 1/(\kappa D)\}$.

The complete calculation is in [appendix B.6.1](#). A simpler rescaling obstruction, proved in [proposition B.2](#), keeps $D = 1$ and an exact cylinder while the fixed-rule value tends to zero. The constant $+1$ in the point rule introduces a coordinate scale. Changing the constraint representation, for example to $2w - 2xy \leq 0$ or $w - xy - c \leq 0$, changes the rule. These statements therefore do not apply unchanged after rescaling, shifting, or to the two-variable Case-2 rule. They concern the set geometry before numerical safeguards or additional strengthening, and do not imply a solver performance improvement.

4.4 Contact geometry: supremum and attainment

Suppose now that the corner optimum has a unique contact point t^* with $q(t^*) = 0$. Translate the bilinear coordinates by a symmetry so that $t^* = 0$. For a vertex v of T^* , its new coordinates are (u_{vx}, u_{vy}, h_v) , where

$$u_v = v - t^*, \quad h_v = \nabla q(t^*)^T u_v, \quad q(v) = h_v - u_{vx}u_{vy}.$$

The height h_v measures transversality to the tangent plane. Assume that t^* is one of the scaled-ray vertices and $h_v > 0$ for every other vertex. Uniqueness then implies $q(v) > 0$ at those other vertices.

Every orbit set through the contact point has the form

$$C_{\alpha, \beta} = \{(x, y, w) : 4\alpha q \geq (\alpha x - y - \beta w)^2, \alpha w + \beta x + 1 \geq 0\}, \quad \alpha > 0, \beta \in \mathbb{R}. \quad (18)$$

A completion containing the contact has a generating orbit set through it: the allowable lowering at a point with $q = 0$ is zero. For a point with positive height h , membership is equivalent to $\beta \in I_v(\alpha)$, where

$$I_v(\alpha) = \left[\frac{\alpha u_{vx} - u_{vy} - 2\sqrt{\alpha q(v)}}{h_v}, \frac{\alpha u_{vx} - u_{vy} + 2\sqrt{\alpha q(v)}}{h_v} \right]. \quad (19)$$

Thus all noncontact vertices impose intervals on a common scalar parameter.

Lemma 4.8 (Approximation from a closed contact set). *If an orbit set contains T^* , with $\bar{s}$ possibly on its boundary, then $z_A = z_K$ as a supremum. No rank assumption is needed for this implication.*

Proof. Write its matrix as G with $\det G > 0$, and fix $0 < r < 1$. Set $G_\epsilon = G + \epsilon M(\bar{s})^{-1}$. At $\bar{s}$ this adds ϵI to the PSD matrix. At any vertex $v_r = (1 - r)\bar{s} + rv$ of T_r , a null vector of its PSD matrix is a common null vector of the endpoint matrices. Along the segment, write $GM(s) = \text{sym}(GM(s)) + k(s)J$, with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. On such a nonzero common null vector z , $GM(s)z = k(s)Jz$. Since $q(s) > 0$ on the contracted segment, $M(s)$ is invertible and $k(s)$ cannot vanish. It retains its sign. Therefore

$$M(\bar{s})^{-1}M(v_r)z = \frac{k(v_r)}{k(\bar{s})}z,$$

so the perturbation is positive on the PSD kernel. This makes every contracted vertex matrix positive definite for one sufficiently small positive ϵ , while $\det G_\epsilon > 0$ by continuity. Hence an admissible orbit set contains T_r and has cut ratio at least r . Let $r \uparrow 1$. Full details of the kernel and perturbation step are given in [appendix B.7](#). $\square$

Theorem 4.9 (Closed and strict contact intervals). *Under the unique support-one contact and positive-height assumptions above,*

1. $z_A = z_K$ as a supremum if and only if, for some $\alpha > 0$, all intervals $I_v(\alpha)$, $v \neq t^*$, have a common point;

2. a member of $\mathcal{A}$ attains z_K if and only if such a common point lies in $\text{int } I_{\bar{s}}(\alpha)$;
3. if no closed intersection exists for any $\alpha > 0$, then $z_{\mathcal{A}} < z_K$.

Proof. An orbit set through zero has $G = \begin{pmatrix} \alpha & 0 \\ \beta & 1 \end{pmatrix}$ after positive scaling, by rank-one PSD membership at $M(0) = e_2 e_2^T$. Its determinant and trace give [equation \(18\)](#). For $h > 0$, the first diagonal entry is $\alpha h > 0$, so the determinant inequality already implies the trace branch. Solving it for β gives [equation \(19\)](#). Strict membership of the vertex is precisely interior membership in its interval. This proves the attainment criterion, and [lemma 4.8](#) proves the sufficiency for supremum equality.

For necessity, normalize an exactness-approximating sequence of matrices and take a nonzero limit G . Its determinant is nonnegative, and its PSD inequalities contain T^* . If the determinant is positive, the intervals follow as above. If it is zero, PSD membership at zero gives $G = \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}$ with $d \geq 0$ and $ad = 0$. If $d > 0$, then $a = 0$ and PSD membership at every positive-height vertex forces $y_v = -(c/d)h_v$. Set $\beta = c/d$ and choose $\alpha > 0$ sufficiently small. The determinant slack is $4\alpha q(v) - \alpha^2 x_v^2 > 0$ at every noncontact vertex. If $d = 0$, PSD membership at $\bar{s}$ forces $a > 0$, and the determinant equation forces $x_v = (c/a)h_v$. Set $\beta = \alpha c/a$ and choose α sufficiently large. The slack is then $4\alpha q(v) - y_v^2 > 0$. Uniqueness gives $q(v) > 0$, and finiteness gives one common choice of α . Either singular case therefore produces an attaining orbit set. This proves necessity and the strict-gap assertion. $\square$

The use of closed intervals in the first part is essential. They characterize an exact supremum, while strict membership at the LP vertex characterizes an attaining set. The theorem concerns $\mathcal{A}$; allowing independent vertical lowering in $\mathcal{B}$ requires further analysis.

Corollary 4.10 (A quantitative cylinder condition). *If, for some $t > 0$,*

$$4h_v \geq (tu_{vx} + u_{vy}/t)^2 \quad (v \neq t^*),$$

with strict inequality at $\bar{s}$, then a parabolic cylinder through t^ attains $z_{\mathcal{A}} = z_{\mathcal{B}} = z_K$. In particular this holds when*

$$H_{\text{cyl}} := \sup_{t>0} \min_{v \neq t^*} \frac{4h_v}{(tu_{vx} + u_{vy}/t)^2} > 1,$$

where a zero denominator means an infinite ratio. A cruder sufficient condition is $h_v \geq XY$ at every other vertex, strictly at $\bar{s}$, where $X = \max_{v \neq t^} |u_{vx}|$ and $Y = \max_{v \neq t^*} |u_{vy}|$.*

Proof. Set $\beta = 0$ and $\alpha = t^2$ in [equation \(18\)](#). Its determinant inequality is exactly $(tu_x + u_y/t)^2 \leq 4h$. The set is an upward-closed cylinder, so it belongs to both families. For $X, Y > 0$, take $t = \sqrt{Y/X}$ and use $(t|u_x| + |u_y|/t)^2 \leq 4XY$. If $XY = 0$, finite positive heights allow a sufficiently small or large finite t instead. $\square$

Proposition 4.11 (Sharp cylinder threshold). *For every $0 < d < 7/128$, take unit costs, $t^* = 0$, and scaled-ray vertices*

$$\bar{s} = (3/2, -1/2, 1/4 - d), \quad v_2 = (1/2, 3/2, 1 - d), \quad v_3 = (3, 1, 4 - d),$$

with rays from $\bar{s}$ to t^, v_2, v_3 . Then $z_K = 1$, the unique minimizer is t^* , all incident edges have positive tangent height, and $H_{\text{cyl}} \geq 1 - 4d$, but $z_{\mathcal{A}} < 1$. Thus no threshold below one can replace the strict sufficient threshold in [corollary 4.10](#).*

Proof. At $d = 0$, the minimum of q on the opposite triangular face is $7/128$, and its height is at least $1/4$; the explicit face calculation is in [appendix B.8](#). For a point θv toward zero,

$$q_d(\theta v)/\theta = (1 - \theta)w_0(v) + \theta q_0(v) - d \geq 7/128 - d > 0 \quad (0 < \theta \leq 1).$$

Hence zero is the unique feasible point of T^* and $z_K = 1$. At $t = 1$, the three height ratios are $1 - 4d, 1 - d, 1 - d/4$. The explicit interval separation in [appendix B.8](#) holds for all $0 < d \leq 1/8$ and rules out every closed intersection. Apply [theorem 4.9](#). $\square$

Proposition 4.12 (Angles alone do not ensure exactness). *For every $\theta < \pi/2$, there is a support-one corner with $z_A < z_K$ whose edges at its unique contact point all make angles at least θ with the tangent plane.*

Proof. Use any instance of [proposition 4.11](#), centered at its contact. The symmetry $(x, y, w) \mapsto (kx, ky, k^2w)$ preserves the family values, depth, and H_{cyl} . An edge (u_x, u_y, h) with $h > 0$ has sine of its angle with the tangent plane equal to

$$\frac{k^2h}{\sqrt{k^2(u_x^2 + u_y^2) + k^4h^2}},$$

which tends to one. There are finitely many edges, so one k suffices. $\square$

For a two-sided tangent edge through t^* with direction d satisfying $d_w = 0$ and $d_x d_y < 0$, PSD membership forces $\alpha = -d_y/d_x$ in [equation \(18\)](#). This is the tangent pencil in [lemma 3.7](#). Positive-height vertices then impose the same intervals with α fixed. Zero-height edge points must also satisfy $\alpha x + y = 0$ and $\beta x + 1 \geq 0$; they cannot be checked by dividing by their height. This distinguishes tangent rigidity from the strictly positive-height support-one criterion above.

5 Closures at a fixed corner

Combining cuts can improve a bound even when every individual cut is suboptimal. The relevant question is therefore whether the intersection of all cuts from a family recovers the corner-hull dominant. Throughout this section the vertex and the projected rays remain fixed. Cuts obtained after reoptimization belong to a different closure.

For a nonempty family $\mathcal{F}$ of closed convex S -free sets containing $\bar{s}$ in their interior, put

$$V = \{a(C) : C \in \mathcal{F}\} \subseteq \mathbb{R}_+^N, \quad \mathcal{C}_{\mathcal{F}} = \{\lambda \geq 0 : a^T \lambda \geq 1 \quad (a \in V)\}.$$

Write $z_{1,\mathcal{F}}(\omega) = \sup_{C \in \mathcal{F}} z_C(\omega)$ and $z_{\text{cl},\mathcal{F}}(\omega) = \inf_{\lambda \in \mathcal{C}_{\mathcal{F}}} \omega^T \lambda$. Validity gives

$$\mathcal{D} \subseteq \mathcal{C}_{\mathcal{F}}, \quad z_{1,\mathcal{F}} \leq z_{\text{cl},\mathcal{F}} \leq z_K. \quad (20)$$

For the one-cut expression we use positive costs, so no convention for $0 \cdot \infty$ is needed.

Lemma 5.1 (Coefficient description). *Let $\widehat{V} = \text{cl}(\text{conv } V + \mathbb{R}_+^N)$ and $U = \text{cl}(V + \mathbb{R}_+^N)$. For $z > 0$ and $\omega \geq 0$,*

$$z_{\text{cl},\mathcal{F}}(\omega) \geq z \iff \omega/z \in \widehat{V}.$$

For $\omega > 0$, $z_{1,\mathcal{F}}(\omega) \geq z$ if and only if $\omega/z \in U$. Consequently, the closure bound equals the best single-cut bound for every positive cost vector if and only if U is convex.

Proof. Every vector of $\widehat{V}$ has scalar product at least one with every point of $\mathcal{C}_{\mathcal{F}}$. If $\omega/z \notin \widehat{V}$, strict separation gives $\mu^T \omega/z < \beta \leq \mu^T a$ for all $a \in \widehat{V}$. Upward closure forces $\mu \geq 0$, and therefore $\beta > 0$. The point μ/β belongs to $\mathcal{C}_{\mathcal{F}}$ and has cost less than z .

For a single cut, $z_C(\omega) \geq z$ means $a(C) \leq \omega/z$. A supremum of at least z therefore gives, for each $\epsilon \in (0, 1)$, a vector $a \in V$ with $a \leq \omega/((1 - \epsilon)z)$. Conversely, if $\omega/z = \lim(a_n + d_n)$ with $d_n \geq 0$, positivity of ω implies $a_n \leq \omega/((1 - \epsilon)z)$ for all sufficiently large n . These observations prove the characterization by U . If U is convex, then $U = \widehat{V}$. If the two bounds agree for all positive costs, every strictly positive point of $\widehat{V}$ belongs to U ; adding $\epsilon \mathbf{1}$ and taking limits proves $\widehat{V} = U$. $\square$

The next theorem identifies the cuts that can contribute to an exact bound. It permits limits of sets and does not require attainment of the best single-cut value.

Theorem 5.2 (Exactness from tight limit cuts). *Let S be closed, $\omega > 0$, and $z = z_K(\omega) \in (0, \infty)$. Then $z_{\text{cl}, \mathcal{F}}(\omega) = z$ if and only if there exist $m \leq N$, vectors $a^{(1)}, \dots, a^{(m)} \in \text{cl } V$, and weights $\theta_i > 0$ with $\sum_i \theta_i = 1$ such that*

$$\sum_i \theta_i a^{(i)} \leq \omega/z. \quad (21)$$

Whenever this holds, every selected vector is tight at every corner minimizer λ^ , and the slack in equation (21) is orthogonal to every such minimizer.*

Proof. The reverse implication follows from lemma 5.1 and validity. For the forward implication, lemma 5.1 supplies

$$b_n = \sum_{i=1}^{N+1} \theta_{n,i} a_{n,i} + d_n \longrightarrow \omega/z, \quad a_{n,i} \in V, \quad d_n \geq 0, \quad \theta_{n,i} \geq 0, \quad \sum_i \theta_{n,i} = 1,$$

where finite-dimensional Carathéodory gives $N + 1$ slots. Take a subsequence on which every weight converges, say to θ_i . If $\theta_i > 0$, nonnegativity gives $0 \leq a_{n,i} \leq b_n/\theta_{n,i}$ componentwise, so that slot is bounded. Pass to a further subsequence on which each such vector converges to $a^{(i)} \in \text{cl } V$. Slots with zero limiting weight can have unbounded vectors. Dropping their nonnegative contributions preserves the required inequality. The retained limiting weights still sum to one.

Positive costs and closedness ensure the existence of a minimizer in X . Limits of valid cut vectors remain valid, so

$$1 = \omega^T \lambda^*/z \geq \sum_i \theta_i (a^{(i)})^T \lambda^* \geq \sum_i \theta_i = 1.$$

Both inequalities are equalities. Every selected vector is tight, and the slack is orthogonal to λ^* . This argument applies to every minimizer. The selected vectors lie in the affine hyperplane $a^T \lambda^* = 1$, of dimension at most $N - 1$, because $\lambda^* \neq 0$. Carathéodory within that hyperplane reduces their number to at most N without changing their convex combination. $\square$

Proposition 5.3 (A smooth active face). *Suppose $S = \{s : \tilde{g}(s) \leq 0\}$, where $\tilde{g}$ is C^1 near the projected corner. Let $\omega > 0$, $z = z_K(\omega) \in (0, \infty)$, and let λ^* be a minimizer with support J . Put $g(\lambda) = \tilde{g}(\bar{s} + P\lambda)$ and assume $\nabla_J g(\lambda^*) \neq 0$. Every valid linear inequality $a^T \lambda \geq 1$ tight at λ^* satisfies $a_J = \omega_J/z$.*

Let $T = \{a \in \text{cl } V : a_J = \omega_J/z\}$. Then

$$z_{\text{cl}, \mathcal{F}}(\omega) = z \iff \omega_{J^c}/z \in \text{conv}\{a_{J^c} : a \in T\} + \mathbb{R}_+^{J^c}. \quad (22)$$

At most $|J^c| + 1$ tight limit vectors are needed on the right.

Proof. The minimizer satisfies $g(\lambda^*) = 0$: if the value were negative, a small move toward zero would preserve feasibility and lower the cost. On the relative interior of the face supported on J , the implicit function theorem makes $g = 0$ a smooth hypersurface near λ^* . For every tangent vector d there is a curve in that hypersurface with derivative d . The valid tight inequality has a local minimum on that curve, so $a_J^T d = 0$. Thus a_J is a multiple of $\nabla_J g(\lambda^*)$. The same applies to the valid tight inequality $\omega^T \lambda/z \geq 1$. Evaluation at λ^* fixes both multiples: their scalar products with λ_J^* equal one. Hence $a_J = \omega_J/z$.

Necessity of equation (22) follows from theorem 5.2. For sufficiency, any convex combination on the right, extended by its fixed coordinates on J , is dominated by ω/z ; theorem 5.2 applies. Carathéodory in $\mathbb{R}^{J^c}$ gives the stated number of vectors. $\square$

Corollary 5.4 (At most one unused ray). *Under the hypotheses of [proposition 5.3](#), if $|J^c| \leq 1$, then*

$$z_{\text{cl},\mathcal{F}}(\omega) = z_K(\omega) \iff z_{1,\mathcal{F}}(\omega) = z_K(\omega).$$

Proof. All selected tight limit cuts agree with ω/z on J . If no coordinate remains, any one of them equals ω/z . If one coordinate remains, a convex average below its target has at least one constituent below that target. Thus some $a \in \text{cl } V$ satisfies $a \leq \omega/z$, and [lemma 5.1](#) gives best-single-cut exactness. The converse follows from [equation \(20\)](#). $\square$

For $q = w - xy$ and three projected rays spanning $\mathbb{R}^3$, a support-two minimizer satisfies the regularity condition in [proposition 5.3](#). Indeed, if its two independent directions were tangent to $q = 0$, they would span the tangent plane. Projection of this plane onto (x, y) is an isomorphism, and the quadratic term $-d_x d_y$ is indefinite there. At a stationary feasible contact, a negative quadratic direction and its opposite are both locally feasible. Minimality would force the positive objective to vanish on an open set of directions, a contradiction. Thus [corollary 5.4](#) transfers every strict single-cut obstruction in a support-two direction to the closure. It does not assert equality of the two suboptimal bounds.

Proposition 5.5 (A dimension bound on the gain). *For $\omega > 0$ with $z_{\text{cl},\mathcal{F}}(\omega) \in (0, \infty)$,*

$$z_{1,\mathcal{F}}(\omega) \geq z_{\text{cl},\mathcal{F}}(\omega)/N.$$

The factor N is attained by nonnegative coefficient families.

Proof. Put $z = z_{\text{cl},\mathcal{F}}(\omega)$. By [lemma 5.1](#), ω/z belongs to the boundary of $\hat{V}$: membership of ω/z' for $z' > z$ would contradict the definition of z . A supporting normal $\mu \neq 0$ can be chosen nonnegative, since $\hat{V}$ is upward closed. The bounded-aggregate argument in the proof of [theorem 5.2](#) gives $b = \sum_i \theta_i a^{(i)} \leq \omega/z$, with $a^{(i)} \in \text{cl } V$. Support and nonnegativity imply

$$\mu^T \omega/z \leq \mu^T b \leq \mu^T \omega/z.$$

Each selected vector therefore lies in the supporting hyperplane. Its dimension is at most $N - 1$, so at most N vectors suffice. Some weight is at least $1/N$, and then $a^{(i)} \leq N\omega/z$. Apply [lemma 5.1](#). For $V = \{e_1, \dots, e_N\}$ and $\omega = \mathbf{1}$, the one-cut bound is one and the closure bound is N . $\square$

5.1 Bilinear families and exact counterexamples

We use the orbit family $\mathcal{A}$ and the completion family $\mathcal{B}$ of [propositions 3.4](#) and [3.5](#). A completion may contain the vertex in its interior even when its uncompleted orbit set does not. The transformed point rule gives a smaller family, denoted $\mathcal{BP}$. Writing $\Xi = F^T M(\bar{s})$, its orbit parameters are exactly $\Xi = \Sigma \succ 0$ symmetric; $\mathcal{BP}$ contains their upward completions. The transformations here are all determinant-preserving homogeneous automorphisms, including transposition. They are not restricted to permutations. The parameter derivation appears in [lemma C.1](#). For comparison, let $\mathcal{P}$ denote the corresponding uncompleted point-rule family. Containment of sets and inclusion of families give

$$\mathcal{C}_{\mathcal{B}} \subseteq \mathcal{C}_{\mathcal{A}} \subseteq \mathcal{C}_{\mathcal{P}}, \quad \mathcal{C}_{\mathcal{B}} \subseteq \mathcal{C}_{\mathcal{BP}} \subseteq \mathcal{C}_{\mathcal{P}}. \quad (23)$$

Theorem 5.6 (Strict closure loss). *Let*

$$\bar{s} = (-9/2, 0, 3/2), \quad v_1 = (-1, -6, 18), \quad v_2 = (-5, 6, -18), \quad v_3 = (0, 5/2, 5/2), \quad p_j = v_j - \bar{s}.$$

Then $z_K(\mathbf{1}) = 1$, with unique minimizer $(1/2, 1/2, 0)$, and the closures of $\mathcal{A}, \mathcal{B}, \mathcal{BP}$, and $\mathcal{P}$ all strictly contain $\mathcal{D}$. In particular,

$$\begin{aligned} (67/250, 41/100, 151/500) &\in \mathcal{C}_{\mathcal{B}}, & (3/11, 41/99, 10/33) &\in \mathcal{C}_{\mathcal{B}}, \\ (51/350, 0, 17/150) &\in \mathcal{C}_{\mathcal{BP}}, & (1/5, 0, 1/6) &\in \mathcal{C}_{\mathcal{BP}}. \end{aligned}$$

In the direction **1** the certified brackets are

$$0.97538 \leq z_{\mathcal{A}} \leq z_{\text{cl}, \mathcal{A}} \leq z_{\text{cl}, \mathcal{B}} \leq 49/50, \quad 2539/10000 \leq z_{\mathcal{BP}} \leq z_{\text{cl}, \mathcal{BP}} \leq 136/525.$$

The same lower bound 0.97538 holds for $z_{\mathcal{B}}$.

Proof. The corner geometry and the strict inequality $z_{\mathcal{B}}(\mathbf{1}) < 1$ are proved in [theorem 3.8](#). The support-two minimizer is regular: $P_{\{1,2\}}^T \nabla q = (-3/2, -3/2)$ at its image. By [corollary 5.4](#), exactness of any of the four closures would imply best-single-cut exactness, contradicting [theorem 3.8](#) and [equation \(23\)](#). The displayed points and numerical brackets have exact finite certificates described in [appendix C.2](#). Every point of $\mathcal{D}$ has coordinate sum at least one, whereas the four displayed points have sums $49/50, 98/99, 136/525$, and $11/30$. $\square$

The larger orbit example in [theorem 3.8](#) has $\bar{s} = (-1/2, 1/2, 1)$, endpoints $(1/2, -4, -1/2)$, $(-10, 3, 24)$, and $(9/2, 1/2, 15/2)$, and a support-two minimizer with unit costs. Its strict $\mathcal{A}$ single-cut loss therefore also proves strict $\mathcal{A}$ closure loss. Its numerical $\mathcal{B}$ loss is not an exact certificate for that family.

Proposition 5.7 (Strict improvement with support one). *At the corner*

$$\bar{s} = (-2, 3, 2), \quad v_1 = (0, 0, 0), \quad v_2 = (6, -2, 1/4), \quad v_3 = (1, -5/2, 1/2), \quad p_j = v_j - \bar{s},$$

unit costs give a unique corner minimizer e_1 and

$$0.9838 \leq z_{\mathcal{A}} \leq 0.9839 < 0.9953611 \leq z_{\text{cl}, \mathcal{A}} \leq 999/1000 < 1 = z_K.$$

Proof. The corner value and the single-cut bracket are proved in [proposition 3.9](#). Three rational orbit sets and a rational LP dual prove the closure lower bound; an exact finite certificate proves $(24/25, 1/100, 29/1000) \in \mathcal{C}_{\mathcal{A}}$. All data and certificate implications appear in [appendices C.2](#) and [C.3](#). $\square$

Thus the closure can be strictly better than every single cut and still fail to be exact. The lower bound also holds for $\mathcal{B}$ by [equation \(23\)](#). The upper bound $999/1000$ has only been certified for $\mathcal{A}$; the analogous $\mathcal{B}$ upper bound remains a numerical observation.

5.2 Finite and unbounded approximation factors

For a fixed corner with $X \neq \emptyset$, define

$$\rho_{\mathcal{F}} = \sup_{\omega > 0} \frac{z_K(\omega)}{z_{\text{cl}, \mathcal{F}}(\omega)}.$$

Equivalently, it is the least $\rho \geq 1$ such that $\mathcal{C}_{\mathcal{F}} \subseteq (1/\rho)\mathcal{D}$, with value infinity if there is no such ρ . Strict positive-cost separation proves this equivalence, as well as $\mathcal{C}_{\mathcal{F}} = \mathcal{D}$ if and only if all positive-cost bounds are exact. The gain factor in [proposition 5.5](#) compares closure and single-cut bounds; $\rho_{\mathcal{F}}$ compares the closure with the corner bound.

Theorem 5.8 (Unbounded fixed-corner factors). *Let $\epsilon > 0$ in every cost vector below. At the corner of [theorem 5.6](#), every $\mathcal{BP}$ cut has $a_1 \geq 7/2$, and this bound is attained. Hence $(2/7, 0, 0) \in \mathcal{C}_{\mathcal{BP}}$ and, for $\omega_\epsilon = (\epsilon, 1, 1)$,*

$$z_{\text{cl}, \mathcal{BP}}(\omega_\epsilon) \leq 2\epsilon/7, \quad z_K(\omega_\epsilon) \geq 367/759.$$

In particular $\rho_{\mathcal{BP}} = \rho_{\mathcal{P}} = \infty$.

Now take the vertex $(1, -1, 1)$ and projected rays $e_x, -e_y, e_w, -e_w$. They are the projections of the independent rays $e_x, -e_y, e_w + e_z, -e_w + e_z$ in four variables. At this corner,

$$\mathcal{D} = \{\lambda \geq 0 : \lambda_4 \geq 2\}, \quad z_K(\omega) = 2\omega_4.$$

For $\omega_\epsilon = (\epsilon, \epsilon, \epsilon, 1)$,

$$\begin{aligned} z_{\text{cl}, \mathcal{A}}(\omega_\epsilon) &\leq 60\epsilon, & \rho_{\mathcal{A}} &= \infty, \\ z_{\text{cl}, \mathcal{BP}}(\omega_\epsilon) &\leq 4(\sqrt{2} + 1)\epsilon, & \rho_{\mathcal{BP}} &= \infty, \\ \mathcal{C}_{\mathcal{B}} &= \mathcal{D}, & \rho_{\mathcal{B}} &= 1. \end{aligned}$$

Moreover, every $\mathcal{BP}$ parameter $\Sigma \succ 0$ satisfies

$$a_1 + a_2 \geq (\sqrt{2} - 1)/2,$$

with equality exactly when Σ is a positive multiple of I . Consequently, $(t, t, 0, 0) \in \mathcal{C}_{\mathcal{BP}}$ exactly when $t \geq 2(\sqrt{2} + 1)$.

Proof. Exact witnesses and full proofs are in [appendices C.4](#) and [C.5](#). At the four-ray corner, expansion gives $q = 2 + \lambda_1 + \lambda_2 + \lambda_1\lambda_2 + \lambda_3 - \lambda_4$. The point $2e_4 \in X$ and the inequality $\lambda_4 \geq 2$ give the stated dominant. An integer matrix certificate puts $(20, 20, 20, 0)$ in $\mathcal{C}_{\mathcal{A}}$. The $\mathcal{BP}$ coefficient inequality and its attainment put $(2(\sqrt{2} + 1), 2(\sqrt{2} + 1), 0, 0)$ in $\mathcal{C}_{\mathcal{BP}}$. The completion with $F^T = J^T$, $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, is

$$\{x \geq 0, y \leq 0, w \geq 1 - 2\sqrt{-xy}\},$$

and gives the exact cut $\lambda_4/2 \geq 1$. All assertions of unboundedness follow by letting ϵ decrease to zero in the positive-cost directions displayed above. $\square$

At the four-ray corner [theorem 4.2](#) also gives $(2 - \sqrt{2})\epsilon \leq z_{\mathcal{A}}(\omega_\epsilon)$ for $0 < \epsilon \leq \sqrt{2}$. This auxiliary range is necessary: the bound $z_{\mathcal{A}} \leq z_K = 2$ precludes that linear lower estimate for all $\epsilon > 0$. The corner is removed immediately by bound propagation if a solver knows $x \geq 1$, $y \leq -1$, and $w \leq 1$; it establishes a geometric distinction between the families.

Proposition 5.9 (A finite-factor class). *If every projected ray first meets S at a finite step t_j , then any member $C \in \mathcal{F}$ gives*

$$\rho_{\mathcal{F}} \leq \max_j \frac{t_j}{\alpha_j(C)} < \infty.$$

Proof. The first-hit points belong to X , so $z_K(\omega) \leq \min_j \omega_j t_j$. Since C is S -free, $0 < \alpha_j(C) \leq t_j < \infty$; and

$$z_{\text{cl}, \mathcal{F}}(\omega) \geq \min_j \omega_j \alpha_j(C) \geq \min_j \frac{\alpha_j(C)}{t_j} \min_j \omega_j t_j.$$

$\square$

Corollary 5.10 (No uniform factor across corners). *There is no positive uniform lower bound for $z_{\text{cl}, \mathcal{A}}/z_K$ or $z_{\text{cl}, \mathcal{B}}/z_K$, even with three rays, positive unit costs, and a fixed vertex $(0, 0, 1)$. The same conclusion holds for $\mathcal{BP}$.*

Proof. The three-ray construction in [proposition 4.4](#) has $z_{\mathcal{B}}/z_K \rightarrow 0$. By [proposition 5.5](#), $z_{\text{cl},\mathcal{B}}/z_K \leq 3z_{\mathcal{B}}/z_K \rightarrow 0$. The rescaling $(x, y, w) \mapsto (x/\sqrt{\epsilon}, y/\sqrt{\epsilon}, w/\epsilon)$ maps its vertex to $(0, 0, 1)$ and preserves the corner values and family ratios. The inclusion relations give the conclusions for $\mathcal{A}$ and $\mathcal{BP}$. If the quantitative single-cut estimate $z_{\mathcal{B}}/z_K \leq 137\sqrt{\epsilon}/z_0$, with $z_0 = (1 + \sqrt{1 + 4\epsilon})/2$ and $0 < \epsilon < 1$, is used, the closure estimate is $z_{\text{cl},\mathcal{B}}/z_K \leq 411\sqrt{\epsilon}/z_0$. The quantitative estimate is [theorem 4.5](#); the qualitative conclusion needs only [proposition 4.4](#). $\square$

At the three-ray corner of [theorem 5.6](#), a further exact certificate gives $\rho_{\mathcal{A}} \geq \rho_{\mathcal{B}} > 1.1166$; see [appendix C.6](#). Whether either fixed-corner factor is finite there remains open. Numerical estimates near 1.128 do not establish finiteness.

5.3 The unrestricted closure and later corners

By [theorem 2.5](#), all S -free sets together recover $\mathcal{D}$ at one corner. Thus the preceding fixed-corner gaps result from restricting the set family. This equality of sets gives exact unrestricted closure bounds for nonnegative costs as well as positive costs; it does not extend positive-cost single-cut sufficiency or attainment to zero costs.

The rank-one closure of a polyhedron uses its original bases. It need not equal the hull of its nonlinear feasible points, even when the unrestricted family is used at every basis. The disk example in [proposition 2.12](#) proves this and closes its gap at a newly created basis in the second round. The distinction is geometric: each original basis contributes the dominant of its own corner, whereas the polyhedron has additional constraints outside that corner.

The bilinear records exhibit the same distinction. Reoptimization with one orbit cut per round reached the original corner bound after three cuts at [theorem 5.6](#) and after two at [proposition 5.7](#). These are numerical trajectories, not finite convergence theorems. The complete retained survey and its one-sided bounds appear in [appendix C.7](#).

[Averkov et al. \(2018\)](#) study constant-factor approximation by lattice-free closures at a fixed apex. Under their closure and target-representation hypotheses, approximation of a target cut by the entire family is equivalent, up to constants, to approximation by an individual member. Our smooth-face criterion concerns exactness in one objective direction and uses the support of a corner minimizer. Its hypotheses do not give a direct application of their theorem to the quadratic families. The coefficient description and the dimension bound above are elementary finite-dimensional aggregation statements; the bilinear obstructions and certified support-one improvement identify what happens for the restricted quadratic families.

6 The orbit obstruction for homogeneous minors

A non-principal 2×2 minor with four distinct row and column indices has four independent entries. Write

$$M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad S_{\det} = \{M : \det M = 0\}.$$

Every rank-one matrix is uv^T , so this is exactly the projection of the outer-product equations onto those entries. After a column interchange, we may assume $\det \bar{M} > 0$. Throughout this section $\omega > 0$, the projected rays are matrices P_j , and the finite corner bound is

$$z_K = \min \left\{ \omega^T \lambda : \lambda \geq 0, \det \left(\bar{M} + \sum_j \lambda_j P_j \right) = 0 \right\}.$$

For $z \geq 0$, put $T_z = \text{conv}\{\bar{M}, \bar{M} + (z/\omega_j)P_j : j = 1, \dots, N\}$. The determinant has signature $(2, 2)$: the coordinates

$$x = \left(\frac{a+d}{2}, \frac{c-b}{2}\right), \quad y = \left(\frac{d-a}{2}, \frac{b+c}{2}\right) \quad \text{give} \quad \det M = \|x\|^2 - \|y\|^2.$$

Its polar form is $B(M, N) = \frac{1}{2} \text{tr}(\text{adj}(M)N)$.

Lemma 6.1. *Let C be closed and convex with $\bar{M} \in \text{int } C$. Then C is $S_{\det}$ -free if and only if $C \subseteq \{\det \geq 0\}$, and if and only if it is $\{\det \leq 0\}$ -free. Every maximal such set is a cone. The corner bounds for $\{\det = 0\}$ and $\{\det \leq 0\}$ agree.*

Proof. The determinant is positive on the connected set $\text{int } C$ if it never vanishes there, and $C = \text{cl}(\text{int } C)$. Conversely, every singular matrix has matrices of negative determinant arbitrarily close to it: use a direction on which its nonzero determinant gradient is negative, or $\text{diag}(1, -1)$ at zero. Thus $\text{int}\{\det \geq 0\} = \{\det > 0\}$. Homogeneity gives $\text{cl}(\text{cone } C) \subseteq \{\det \geq 0\}$; this closed convex cone is free and contains C , proving the maximality assertion. Finally a negative determinant reached at $\lambda \geq 0$ must first vanish on the segment from zero to λ , at no greater objective cost. $\square$

The polar-factor choice used by the minor intersection-cut formula (Bienstock et al., 2020; Chmiela et al., 2020) is a member of a larger finite-dimensional family. Define

$$C_F = \{M : \text{sym}(F^T M) \succeq 0\}, \quad \det F > 0.$$

Proposition 6.2 (Polar-factor choice). *Let $\bar{M} = UH$ be the polar decomposition, where $U \in SO(2)$ and $H \succ 0$ is symmetric. The set*

$$\{M : \|y(M)\| \leq x(\bar{M})^T x(M) / \|x(\bar{M})\|\}$$

equals C_U . It is the rotation-family member chosen by $(\lambda_1, \lambda_2) \propto (\bar{a} + \bar{d}, \bar{b} - \bar{c})$ in the outer-product-free inequality $\lambda_1(a+d) + \lambda_2(b-c) \geq \|(b+c, a-d)\|$. For $\det \bar{M} < 0$, a column interchange yields C_W , with W the orthogonal polar factor of $\bar{M}$ and $\det W = -1$.

Proof. For the rotation R_ϕ , the condition $\text{sym}(R_\phi^T M) \succeq 0$ is $(\cos \phi, \sin \phi)^T x(M) \geq \|y(M)\|$. Choose $(\cos \phi, \sin \phi) = x(\bar{M}) / \|x(\bar{M})\|$. Then $R_\phi^T \bar{M}$ is symmetric, has positive trace and positive determinant, and is therefore positive definite. Uniqueness of the polar decomposition gives $R_\phi = U$. The stated outer-product-free inequality has precisely these coordinates. If Π interchanges columns, the polar decomposition of $\bar{M}\Pi$ is $(W\Pi)(\Pi^T H \Pi)$, and congruence by Π gives the last assertion. $\square$

Proposition 6.3 (Orbit and point rule). *Let G_+ consist of $M \mapsto AMB^T$ and $M \mapsto AM^T B^T$, with $A, B \in GL_2(\mathbb{R})$ and $\det A \det B > 0$.*

1. *Each C_F is a maximal $S_{\det}$ -free cone, and $C_F = C_{F'}$ exactly when $F' = cF$ for $c > 0$. The G_+ -orbit of C_U is the three-parameter family of all C_F .*
2. *Under $M \mapsto AMB^T$, its parameter changes by $F'^T = BF^T A^{-1}$. Transposition sends C_F to $C_{F^{-1}}$.*
3. *Applying the polar-factor choice in every G_+ coordinate system gives precisely*

$$\mathcal{P}(\bar{M}) = \{C_F : F^T \bar{M} \text{ is symmetric positive definite}\}.$$

This two-parameter family is equivariant under G_+ .

4. *The one-parameter rotation family $\{C_R : R \in SO(2)\}$ intersects $\mathcal{P}(\bar{M})$ in the single set C_U .*

The proof in [appendix D.1](#) includes a direct maximality argument. Thus these homogeneous orbit sets need no maximal-completion step. The full family of maximal determinant-free sets can be larger than this orbit. The general homogeneous classification uses nonexpansive sphere maps $\Gamma : S^1 \rightarrow S^1$, subject to its convex-hull condition ([Muñoz et al., 2025](#)); the orbit is a finite-dimensional subfamily.

The best member of each of these families has a small semidefinite feasibility formulation. The parameter spaces for F^T are, respectively, $\mathbb{R}^{2 \times 2}$, $\{G : G\bar{M} \text{ is symmetric}\}$, $\text{span}\{I, J\}$, and $\mathbb{R}U^T$, where $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$.

Proposition 6.4 (Family optimization and exact certificates). *For any of these linear spaces L , a parameter $G \in L$ gives a family member with $z_{C_F}(\omega) \geq z$ if and only if*

$$\text{sym}(G\bar{M}) \succ 0, \quad \text{sym}(GV) \succeq 0 \quad (V \text{ a vertex of } T_z), \quad G = F^T.$$

Feasibility is monotone in z and uses at most $N + 1$ symmetric 2×2 blocks and four scalar variables. An upper certificate consists of positive definite symmetric matrices Y_V , one for each vertex including $\bar{M}$, such that

$$\sum_V \text{tr}(GVY_V) = 0 \quad \text{for every } G \in L. \quad (24)$$

Its existence implies $\sup_{F^T \in L} z_{C_F}(\omega) \leq z$.

Proof. The first conditions are exactly $\bar{M} \in \text{int } C_F$ and $T_z \subseteq C_F$; positive definiteness at the apex also forces $\det F > 0$. Smaller simplices are contained in T_z . If a feasible G and the certificate both existed, the left side of [equation \(24\)](#) would be a sum of nonnegative matrix inner products, with the apex term strictly positive. This contradicts its being zero. $\square$

Proposition 6.5 (Support and tangent edges). *There is a corner minimizer whose supported projected rays are linearly independent. Every such minimizer has support at most three; support three requires $\bar{M}$ to lie in the span of its three rays. A support-two minimizer lies in the relative interior of an edge of T_{z_K} , with edge direction D_{edge} satisfying $\text{tr}(\text{adj}(M_*)D_{\text{edge}}) = 0$ and $\det D_{\text{edge}} \geq 0$.*

The proof in [appendix D.1](#) uses the signature $(2, 2)$ and keeps the independence hypothesis explicit. Duplicating one supported ray can produce other minimizers with larger support.

A tangent edge with $\det D_{\text{edge}} > 0$ constrains the entire orbit parameter to one projective degree of freedom.

Lemma 6.6 (Tangent-edge pencil). *Let $M_* = ab^T \neq 0$ and let D_{edge} satisfy $\text{tr}(\text{adj}(M_*)D_{\text{edge}}) = 0$, $\det D_{\text{edge}} > 0$. Set $G_0 = J \text{adj}(D_{\text{edge}})$ and $G_1 = J \text{adj}(M_*)$. Then $G_1 a = 0$, $G_0 a = c_0 b$ for some $c_0 \neq 0$, and $\det(G_0 + tG_1) = \det D_{\text{edge}}$ for all t . If an orbit set contains $M_* + sD_{\text{edge}}$ for all $|s| \leq \varepsilon$, then*

$$F^T = \theta(G_0 + tG_1), \quad \theta c_0 > 0.$$

Consequently each additional vertex imposes an interval condition on t . Disjoint intervals obstruct attainment, but do not by themselves prove a strict supremum gap. The following rational certificates do prove one.

Theorem 6.7 (Full-dimensional minor counterexamples). *The rational four-ray corners A , B and $S1$ in [table 8](#) have $\omega = (1, 1, 1, 1)$, independent rays, and $z_K = 1$ with a unique minimizer. A and B have tangent-edge minimizers; $S1$ has a support-one minimizer with strictly positive inactive-ray multipliers and transverse incident edges. Their orbit suprema satisfy*

$$z_{\text{orbit}}(A) \leq \frac{9}{20}, \quad z_{\text{orbit}}(B) \leq \frac{461}{500}, \quad z_{\text{orbit}}(S1) \leq \frac{779}{1000}.$$

For A , an unrestricted maximal free set does attain the corner bound one. The strict gap for A persists on an open neighborhood of its apex, rays and positive objective weights.

[Appendix D.2](#) gives exact face calculations and the complete rational dual matrices. In particular, support one and transversality alone do not guarantee orbit exactness.

Corollary 6.8 (Bilinear slices and exact comparisons). *Embed a bilinear corner by $M = \begin{pmatrix} w & x \\ y & h \end{pmatrix}$, with apex $(\bar{w}, \bar{x}, \bar{y}, 1)$ and rays $(p_{jw}, p_{jx}, p_{jy}, 0)$. Its corner bound and minor orbit bound equal, respectively, the bilinear corner bound and family- $\mathcal{A}$ bound. For the three rational bilinear examples of [theorem 3.8](#) and [proposition 3.9](#), including the second tangent-edge instance, exact orbit brackets are $[0.9753853, 0.9753864]$, $[0.8347663, 0.8347772]$, and $[0.9838465, 0.9838468]$. Thus A and $S1$ have larger gaps than these three selected bilinear examples, whereas B has a smaller gap than the second bilinear example.*

Proof. On $h = 1$, $\det M = w - xy$, so [lemma 6.1](#) identifies the corner bounds. The slice of C_F has exactly the bilinear family- $\mathcal{A}$ ray steps. The rational parameters and dual matrices in [appendix D.3](#) prove both ends of the displayed brackets and all the strict comparisons. $\square$

There is therefore no ordering of the gaps across all minor and bilinear corners. Since every bilinear corner embeds as a minor corner, the infimum minor ratio is no greater than the infimum bilinear ratio; this does not compare full-dimensional corners under specified conditioning bounds.

Proposition 6.9 (Contact restriction). *If a closed convex free set C with $\bar{M} \in \text{int } C$ satisfies $z_C = z_K < \infty$, then $T_{z_K} \subseteq C$ and every corner minimizer belongs to $C \cap S_{\det}$. For an orbit set,*

$$C_F \cap S_{\det} = \{F^{-T}bb^T : b \in \mathbb{R}^2\}.$$

This contact cone has dimension two, whereas $S_{\det}$ has dimension three. If the unique minimizer is on ray j , the polar-factor set can be exact only if $x(P_j)$ is parallel to $x(\bar{M})$, a single linear condition on that ray.

Proof. Exactness implies $\alpha_j \geq z_K/\omega_j$ for every ray, hence simplex containment. For a rank-one ab^T , positive semidefiniteness of $\text{sym}((F^T a)b^T)$ is equivalent to $F^T a = cb$ with $c \geq 0$. Absorb c into b to obtain the contact formula. At a support-one contact, $U^T M_*$ and $U^T \bar{M}$ are symmetric; their difference forces the skew coordinate of $U^T P_j$ to vanish, which is the stated condition. $\square$

Proposition 6.10 (Variable scaling). *For $0 < \delta \leq 1/4$, take*

$$\bar{M} = \text{diag}(1, \delta), \quad (P_1, P_2, P_3, P_4) = (E_{12}, -E_{11}, -\delta E_{22}, -\delta E_{21}), \quad \omega = (1, 1, 1, 1).$$

Then $z_K = 1$, the polar-factor set gives $2\sqrt{\delta}$, and both the point-rule family and the orbit attain one. Every rotation-family set gives at most $((16+2\sqrt{73})/3)\sqrt{\delta}$. This is the image under $M \mapsto \text{diag}(1, \delta)M$ of a corner where the polar-factor choice is exact.

The calculations in [appendix D.4](#) show that both the polar-factor choice and the full rotation family can lose arbitrarily much under one variable scaling, while the equivariant point-rule and orbit families remain exact.

Proposition 6.11 (Positive-determinant principal minors). *On the symmetric matrix space, let $\det \bar{M} > 0$. If $\text{tr } \bar{M} > 0$, the positive semidefinite cone is the unique maximal determinant-free set containing $\bar{M}$ in its interior; if $\text{tr } \bar{M} < 0$, it is the negative semidefinite cone. The polar-factor set restricted to this space is that cone and its intersection cut attains z_K .*

Proof. The two components of $\{\det > 0\}$ on symmetric matrices are the positive and negative definite cones. A connected free interior containing $\bar{M}$ stays in its component, so its closure lies in the corresponding semidefinite cone. That cone is convex and free, proving uniqueness. Every T_z with $z < z_K$ stays in the same component, so the cone's ray steps give $z_C \geq z_K$; validity gives the reverse inequality. The polar factors are respectively I and $-I$. $\square$

These conclusions use the full rank-one equation. If one entry is diagonal, the outer-product projection instead has closure $\{\det M = 0, a \geq 0\}$, which can permit larger free sets. Negative-determinant principal minors and such bound-aware enlargements therefore require a separate analysis; the precise projection and the principal-minor distinction are recorded in [appendix D.4](#).

7 Successive cuts and reoptimization

A corner bound measures improvement in the current cone. After a cut is added, reoptimization can select a different vertex and a different cone. Thus local bound attainment does not determine the progress of a sequence of LP relaxations. We first give a geometric condition that does guarantee convergence. We then show why a ratio of local objective bounds cannot replace that condition, and construct a sequence of maximal sets whose LP values remain strictly below the nonlinear optimum.

7.1 The loop and a sufficient depth condition

Let $P_0 = \{v \in \mathbb{R}^n : Av \leq b\}$ be a polytope contained in $[-B, B]^n$, and let E be a finite set of bilinear terms. For $e = (i, j, k) \in E$, put

$$S_e = \{v : v_k = v_i v_j\}, \quad F = P_0 \cap \bigcap_{e \in E} S_e \neq \emptyset, \quad z^* = \min_{v \in F} c^T v.$$

The value z^* also equals the minimum over $\text{conv } F$. In round r , solve the LP over P_r and choose an optimal vertex v^r with an optimal nonsingular basis of tight rows. Its basis cone is

$$K_r = v^r + \text{cone}\{r_1^r, \dots, r_n^r\}, \quad P_r \subseteq K_r.$$

If $v^r \in F$, stop. Otherwise, for every violated term choose a closed convex set containing v^r in its interior and free of the side of the bilinear inequality that v^r violates. Add its intersection-cut halfspace $H_{r,e}$ and retain every earlier cut. The sets may be defined in the three coordinates of the term and lifted to $\mathbb{R}^n$. Every added cut is valid for $P_r \cap S_e$, so

$$F \subseteq P_{r+1} \subseteq P_r \subseteq P_0.$$

In particular, $c^T v^r$ is nondecreasing and bounded above by z^* . This is an exact loop: it does not discard cuts or stop at a positive feasibility tolerance.

For a cut with steps $\alpha_j \in (0, \infty]$, define

$$d_{r,e} = \text{dist}(v^r, K_r \cap H_{r,e}), \quad \delta_{r,e} = \min_{j: \alpha_j < \infty} \alpha_j \|r_j^r\|, \quad \gamma_r = \text{dist} \left(0, \text{conv} \left\{ \frac{r_j^r}{\|r_j^r\|} : 1 \leq j \leq n \right\} \right). \quad (25)$$

The quantity $d_{r,e}$ is the distance to the retained part of the cone, which can exceed the distance to the cut hyperplane. The basis cone is pointed, so $\gamma_r > 0$. A positive lower bound independent of r is a stronger condition. At least one step is finite: if every step were infinite, convexity and the interior vertex would imply $K_r \subseteq \text{int } C$, contradicting the nonempty feasible set in K_r .

Theorem 7.1 (Convergence under uniform cut depth). *Suppose that for every $t > 0$ there is $\eta(t) > 0$ such that, whenever $\text{dist}(v^r, S_e) \geq t$, the cut added for term e has $d_{r,e} \geq \eta(t)$. Then the loop either stops at an optimal point of F , or its LP values converge to z^* and every accumulation point of its vertices belongs to F and is optimal.*

Proof. If the loop stops, its point is feasible and its LP value is at most z^* , so it is optimal. Otherwise compactness of P_0 supplies an accumulation point $\hat{v}$ and a subsequence $v^{r_\ell} \rightarrow \hat{v}$. If $\hat{v} \notin S_e$, closedness of S_e gives $t = \text{dist}(\hat{v}, S_e)/2 > 0$. For all sufficiently large ℓ , $\text{dist}(v^{r_\ell}, S_e) \geq t$, and round r_ℓ adds a cut of depth at least $\eta(t)$. For $m > \ell$, retention of that cut gives

$$v^{r_m} \in P_{r_m} \subseteq P_{r_\ell+1} \subseteq K_{r_\ell} \cap H_{r_\ell,e}.$$

Consequently $\|v^{r_m} - v^{r_\ell}\| \geq \eta(t)$, contradicting the convergence of the subsequence. Thus every accumulation point is in F . The monotone LP values have a limit $z \leq z^*$, and continuity gives $c^T \hat{v} = z$. Feasibility implies $z \geq z^*$. $\square$

The proof uses compactness, closed feasibility sets, valid retained cuts, and a deep cut for each term that remains violated. It does not require positive reduced costs, maximal sets, or bilinear structure. Those features enter the following sufficient conditions for depth.

Lemma 7.2 (Step length and cone geometry). *For every cut in [equation \(25\)](#), $d_{r,e} \geq \gamma_r \delta_{r,e}$.*

Proof. Write $u_j = r_j^r / \|r_j^r\|$ and $\mu_j = \lambda_j \|r_j^r\|$ for a point $v^r + \sum_j \lambda_j r_j^r \in K_r \cap H_{r,e}$. The cut implies

$$1 \leq \sum_{j: \alpha_j < \infty} \frac{\mu_j}{\alpha_j \|r_j^r\|} \leq \frac{\sum_j \mu_j}{\delta_{r,e}}.$$

Since $\sum_j \mu_j > 0$, the definition of γ_r also gives

$$\|\sum_j \mu_j u_j\| \geq \gamma_r \sum_j \mu_j \geq \gamma_r \delta_{r,e}.$$

Taking the infimum proves the claim. $\square$

Corollary 7.3 (Uniform pointedness and steps). *If $\gamma_r \geq \gamma_0 > 0$ in every round and, for every $t > 0$, $\delta_{r,e} \geq \eta_0(t) > 0$ whenever $\text{dist}(v^r, S_e) \geq t$, then the conclusion of [theorem 7.1](#) holds.*

Proof. Apply [lemma 7.2](#) with $\eta(t) = \gamma_0 \eta_0(t)$. $\square$

Corollary 7.4 (One selected term per round). *Instead add a cut for just one violated term e_r per round. The conclusion of [theorem 7.1](#) still holds under either of these conditions:*

- (a) e_r maximizes $\text{dist}(v^r, S_e)$, and the depth hypothesis holds for the cut actually added.
- (b) Each term has pairwise distinct indices, e_r maximizes $q_e(v^r) = |v_k^r - v_i^r v_j^r|$, $\gamma_r \geq \gamma_0 > 0$, and $\delta_{r,e_r} \geq \beta q_{e_r}(v^r)$ for a fixed $\beta > 0$.

Proof. Use the subsequence and the term e in the proof of [theorem 7.1](#). In (a), the selected distance is at least t , so the cut added has depth at least $\eta(t)$. In (b), changing only the coordinate v_k^r to $v_i^r v_j^r$ produces a point of S_e , whence $q_e(v^r) \geq \text{dist}(v^r, S_e)$. The selected violation is therefore at least t , and [lemma 7.2](#) gives depth at least $\gamma_0 \beta t$. Every later vertex satisfies the selected cut, so the same contradiction applies. $\square$

An arbitrary choice of a violated term does not ensure the cut needed in this proof. The corollary makes no claim for a selection rule that can indefinitely postpone a term.

7.2 Rules that provide a step bound

For this subsection the indices of each term are pairwise distinct. Write $s = (x, y, w)$ for its three coordinates, let $\bar{s}$ be the projected LP vertex, and put $q = |\bar{w} - \bar{x}\bar{y}| > 0$. A ball contained in the chosen set gives a step bound: if $C \supseteq B(\bar{s}, \rho)$ and p_j is the projected ray, then

$$\alpha_j \|r_j^r\| \geq \rho,$$

because $\|p_j\| \leq \|r_j^r\|$; a zero projected ray has infinite step. A lower bound on steps still needs cone geometry to yield a lower bound on cut depth.

Lemma 7.5 (Uniform steps for bilinear rules). *Suppose $P_0 \subseteq [-B, B]^n$, and set*

$$L_B = \left(B^2 + \frac{(B+1)^2}{4} \right)^{1/2}, \quad T_B = \sqrt{3B^2 + 1}.$$

The following bounds hold for either violated side of a bilinear term:

- (a) *The default Case-4 bilinear set of Chmiela et al. (2020), defined explicitly in equation (34), gives $\delta_{r,e} \geq q/(2\sqrt{2}L_B)$.*
- (b) *The orbit set with $F^T = M_\sigma(\bar{s})^{-1}$ gives $\delta_{r,e} \geq q/T_B$, where*

$$M_+(s) = \begin{pmatrix} w & x \\ y & 1 \end{pmatrix}, \quad M_-(s) = \begin{pmatrix} x & w \\ 1 & y \end{pmatrix},$$

and σ is chosen so $\det M_\sigma(\bar{s}) = q > 0$.

- (c) *Fix $0 < \theta \leq 1$. Suppose an idealized rule returns an orbit set whose value $\min_j u_j \alpha_j$ is at least θ times the supremum of that value over all orbit sets containing $\bar{s}$ in their interior. With $u_j = \|r_j^r\|$, it gives $\delta_{r,e} \geq \theta q/T_B$. With $u_j = \|r_j^r\|(\hat{\omega}_j + \tau)$, $0 \leq \hat{\omega}_j \leq 1$, and fixed $\tau > 0$, it gives*

$$\delta_{r,e} \geq \frac{\theta \tau q}{(1 + \tau)T_B}.$$

The constants θ and τ are independent of the round and term.

The proof in [appendix E.1](#) exhibits a ball in each reference set and compares the idealized objectives with the ball supplied by (b). The two objectives in (c) correspond to the geometric-step rule and the Euclidean reduced-cost rule with a positive perturbation, respectively. Since $q \geq \text{dist}(v^r, S_e)$, these bounds establish the step hypothesis of [corollary 7.3](#), and the residual bound in [corollary 7.4\(b\)](#). Thus these rules converge under uniform pointedness. The assertion for the optimized rules assumes a uniform multiplicative accuracy θ . A fixed absolute bisection error does not imply that accuracy when the supremum can be small; it supplies no such guarantee for a numerical implementation by itself.

7.3 Exact local values can accompany small depth

Proposition 7.6 (Bound-optimal sets can be shallow). *Let $S = \{(x, y, w) : w \leq xy\}$, $\bar{s} = (0, 0, 1)$, and take the ordered rays e_y, e_x, e_w with positive reduced costs $\omega = (1, \varepsilon, 1)$, $0 < \varepsilon \leq 1$. Then $z_K(\omega) = 2\sqrt{\varepsilon}$. Every closed convex S -free set containing $\bar{s}$ in its interior satisfies $\alpha_y \alpha_x \leq 4$. Consequently, a set satisfying $z_C(\omega) \geq \theta z_K(\omega)$, $0 < \theta \leq 1$, has*

$$\delta \leq \frac{2\sqrt{\varepsilon}}{\theta}, \quad d \leq \frac{2\sqrt{\varepsilon}}{\theta}.$$

Moreover, maximal sets attain $z_C(\omega) = z_K(\omega)$ while $d \rightarrow 0$, even though $\text{dist}(\bar{s}, S) = 1$ and the cone is fixed and uniformly pointed.

Proof. A point in the cone is $(\lambda_x, \lambda_y, 1 + \lambda_w)$. Its membership in S requires $\lambda_x \lambda_y \geq 1 + \lambda_w$. Minimization sets $\lambda_w = 0$, and AM-GM gives

$$z_K(\omega) = \min_{\lambda_x \lambda_y \geq 1} (\lambda_y + \varepsilon \lambda_x) = 2\sqrt{\varepsilon}.$$

For finite positive $t_y \leq \alpha_y$ and $t_x \leq \alpha_x$, the midpoint of $(0, t_y, 1)$ and $(t_x, 0, 1)$ belongs to C . If $t_y t_x > 4$, moving slightly from that midpoint toward the interior point $\bar{s}$ produces a point in $\text{int } C \cap S$, a contradiction. Thus $t_y t_x \leq 4$ for all such choices. Since both steps are positive, neither can be infinite, and $\alpha_y \alpha_x \leq 4$. The assumed bound implies $\varepsilon \alpha_x \geq 2\theta\sqrt{\varepsilon}$, and hence $\alpha_y \leq 2\sqrt{\varepsilon}/\theta$. The ray intercept is in the retained cone, so $d \leq \alpha_y$ as well.

For attainment, take

$$C_\varepsilon = \left\{ w \geq \frac{(\sqrt{\varepsilon}x + y/\sqrt{\varepsilon})^2}{4} \right\}.$$

These are orbit sets with $F^T = \text{diag}(\sqrt{\varepsilon}, 1/\sqrt{\varepsilon})$, and they are maximal by [theorem 7.8](#). Their steps are $(2\sqrt{\varepsilon}, 2/\sqrt{\varepsilon}, \infty)$ in the stated order, so the local ratio is exactly one. The resulting halfspace is $\sqrt{\varepsilon}x + y/\sqrt{\varepsilon} \geq 2$, whose closest point to $\bar{s}$ lies in the cone. Therefore

$$d = \frac{2\sqrt{\varepsilon}}{\sqrt{1 + \varepsilon^2}} \rightarrow 0.$$

If $w \leq 0$, a point of S is at distance at least one from $\bar{s}$. If $w > 0$, then $x^2 + y^2 \geq 2xy \geq 2w$, so its squared distance is at least $2w + (w - 1)^2 = 1 + w^2$. The point $(0, 0, 0) \in S$ attains distance one. $\square$

The cone in this example has $\gamma = 1/\sqrt{3}$. It is also the basis cone at the unique LP optimum of the fixed box relaxation $P_0 = [0, 2]^2 \times [1, 2]$ for the objectives $\varepsilon x + y + w$; the nonlinear feasible set is nonempty. Thus fixed box bounds and uniform pointedness do not convert the raw local ratio z_C/z_K into an objective-independent depth guarantee. The objective changes with ε , and its smallest positive reduced cost tends to zero. This proposition does not prove that a bound-optimal rule lacks a depth modulus along one fixed problem, or that its loop fails to converge.

7.4 Bound attainment and ties

Proposition 7.7 (Ties forced by a supported corner minimizer). *Let $\omega \in \mathbb{R}_{++}^N$, let S be closed, and suppose the corner minimum is attained at $\lambda^* \geq 0$, with $\bar{s} + P\lambda^* \in S$ and $\omega^T \lambda^* = z_K(\omega) \in (0, \infty)$. Put $J = \text{supp } \lambda^*$. If a valid intersection cut attains $z_C(\omega) = z_K(\omega)$, then*

$$\omega_j \alpha_j = z_K(\omega) \quad (j \in J), \quad \lambda^* \in \text{relint conv}\{\alpha_j e_j : j \in J\}.$$

This simplex belongs to the optimal face of the corner LP after the cut. If $|J| \geq 2$, each intercept basis in that simplex has an additional zero reduced cost. Thus the corner LP has a dual tie and its optimal face contains a feasible corner point.

Proof. Validity and bound attainment give

$$1 \leq \sum_{j \in J} \frac{\lambda_j^*}{\alpha_j} \leq \sum_{j \in J} \frac{\omega_j \lambda_j^*}{z_K(\omega)} = 1.$$

Every summand with $j \in J$ has positive coefficient, so equality forces $1/\alpha_j = \omega_j/z_K(\omega)$. The positive numbers λ_j^*/α_j sum to one and express λ^* as a relative interior convex combination of the intercepts. Every intercept has value $z_K(\omega)$, the minimum of the corner LP. At an intercept basis indexed by $i \in J$, the reduced cost of another variable $j \in J$ is $\omega_j - (z_K(\omega)/\alpha_j) = 0$. $\square$

Only ties on the support of λ^* are forced. The proposition does not imply ties when that support has size one, and it concerns the corner LP, whose intercepts need not belong to the full current relaxation. It does not establish a cause of slower reoptimization.

7.5 A suboptimal limit with maximal sets

Theorem 7.8 (Stalling on uniformly pointed cones). *Fix $0 < \varepsilon < 1$, $X \geq 2/\sqrt{\varepsilon}$, $Y \geq \sqrt{\varepsilon}$, and $W > 1$. For*

$$\min\{\varepsilon x + y + w : 0 \leq x \leq X, 0 \leq y \leq Y, 1 \leq w \leq W, w = xy\},$$

the optimum is $z^ = 1 + 2\sqrt{\varepsilon}$. For $a > 0$, every set*

$$C_a = \{(x, y, w) : w \geq (ax + y/a)^2/4\}$$

is maximal for $S = \{w \leq xy\}$ and is the image of the default Case-4 set at $(0, 0, 1)$ under $(x, y, w) \mapsto (x/a, ay, w)$. Given any strictly increasing sequence

$$0 = \xi_0 < \xi_1 < \dots \longrightarrow \xi_\infty < 2/\sqrt{\varepsilon},$$

start from the box relaxation and use $C_{2/\xi_{r+1}}$ in round r . The unique LP optimum in round r is

$$v^r = (\xi_r, 0, 1), \quad c^T v^r = 1 + \varepsilon \xi_r.$$

The cones have a uniform positive pointedness bound, but the cut depths tend to zero. The LP values converge to $1 + \varepsilon \xi_\infty < z^$, although every vertex has violation one. The same conclusion holds when the nonlinear constraint is $w \leq xy$.*

The construction keeps its steps along one ray arbitrarily small. Its cuts have the explicit form

$$\frac{x}{\xi_{r+1}} + \frac{\xi_{r+1}y}{4} \geq 1.$$

The next vertex moves only from $(\xi_r, 0, 1)$ to $(\xi_{r+1}, 0, 1)$. [Appendix E.2](#) proves maximality, all basis and cut identities, and a uniform separating-vector bound for the cones. For example, $\varepsilon = 1/4$ and $\xi_r = 2(1 - 2^{-r})$ give limit $3/2$ and optimum 2 . Any value strictly between the initial LP value and z^* can be obtained by choosing ξ_∞ .

At the initial corner, in contrast, the bound-optimal set $C_{\sqrt{\varepsilon}}$ gives the cut $\varepsilon x + y \geq 2\sqrt{\varepsilon}$. Together with $w \geq 1$ this already gives the true optimum, which the feasible point $(1/\sqrt{\varepsilon}, \sqrt{\varepsilon}, 1)$ attains. Thus the example also admits a bound-optimal choice that closes the gap in one round.

This is an adversarial choice among maximal sets. The sets are images of the initial default set; the construction does not follow the default rule at subsequent vertices. Nor does it require the initial relaxation to contain McCormick inequalities. It proves failure for an unrestricted maximal-set selection rule in the stated loop, while convergence of the natural default and bound-optimal rules without the sufficient geometric hypotheses remains open.

8 Computational evidence and its limits

The computations distinguish four questions: whether a formula reproduces the implemented ray intersections, how strong a cut is on one corner, what happens after LP reoptimization, and whether a rule improves a solver run. A positive answer to an earlier question does not imply a positive answer to a later one. We report retained experiments and corrected records; no new optimization experiment was conducted for this paper. The companion contains the compact records and their provenance. [Appendix F](#) gives cohort definitions and numerical conventions.

Table 1: Implementation and rejection diagnostics. Each row has its own denominator. Rescaling passes concern the first restriction piece only.

Observation	Count	Fraction
Classified step discrepancies	1,211/610,193	0.20%
Numerical zero-rate face meets S	3,347/4,578	73.1%
Generated cuts, all MINLPLib attempts	29,022/76,914	37.7%
Dynamism aborts, all attempts	38,105/76,914	49.5%
Free-basis-status aborts, all attempts	9,784/76,914	12.7%
First-piece rescaling passes, sample	1,728/1,799	96.1%
First-piece rescaling passes, full diagnostic	31,926/34,834	91.7%

8.1 Implementation fidelity and numerical corner values

The source audit and instrumented runs concern SCIP 10.0.3. Across 610,193 ray comparisons from 7,945 ray-bearing attempts, the corrected set model differs from SCIP’s step on 1,211 rays (0.1985%). The retained classifications are 819 model-cutoff cases, 31 SCIP infinity classifications, 307 conservative bisection cases, and 54 rounding or root-formula cases. The set case agrees in every attempt. The relative scaling of the Case 4 eigen-coordinate blocks matters when $\kappa_{\text{SCIP}} \neq 0$: an incorrect scaling need not describe an S -free set. The bilinear experiments used $\kappa_{\text{SCIP}} = 0$ and are unaffected by this formula pitfall. They required a separate corner-oracle correction described below.

On the sampled MINLPLib corners, numerical degeneracy often leaves the corner-bound criterion without a positive quantity to optimize. Of 4,578 ray-bearing attempts, 3,347 (73.1%) have a numerically zero-rate face meeting S . The classification treats $\omega_j \leq 10^{-9} \max_i \omega_i$ as zero. Another 164 records have all costs in the zero class. Reachability distinguishes zero corner value from infinite corner value in the latter class.

The remaining 1,067 records, from 26 instances, have a positive value under the reduced-space numerical test. Both z_C and z_K in the reported ratios use costs floored at the same threshold. The median SCIP-set ratio z_C/z_K is 0.923 per corner and 0.818 with equal instance weighting; 371 records (34.8%) have ratio below 0.5. These are numerical ratios. Support-at-most-two calculations, generic three-ray KKT enumeration, and higher-dimensional solver estimates have different guarantees; their counts and limitations appear in [Appendix F.3](#).

In 183 of the 371 low-ratio records, the determining numerically zero-rate ray exits the SCIP set but has no detected one-ray intersection with S . Four additional records have approximately 10^{-14} drift in the dumped LP values. Their exact dumped-data ray restrictions reach S only after steps of order 10^{18} . Whether this drift occurs at the true LP corner is undetermined. Those four records do not establish a geometric gap.

Consistent ray rescaling changes the intersection parameter inversely and preserves the mathematical cut after conversion to the original coordinates. The inspected implementation does not normalize these rays. Rescaling makes many recorded first-piece dynamism tests pass ([Table 1](#)). This is a diagnostic of a noninvariant rejection test. It does not count valid cuts recovered by a complete implementation: later pieces and numerical safeguards remain untested. The observed LP-seen rates are lower bounds, since a cut may enter and leave between snapshots. The attempt counts describe the instrumented binary, including its overhead on time-limited runs.

Table 2: Corrected one-cut experiment: mean fraction of the single-term gap. The orbit search attains the numerical corner value to bisection resolution on these 120 corners.

Quantity	4 variables, 4 products ($n = 47$)	6 variables, 8 products ($n = 73$)
SCIP-set corner increment	0.724	0.616
Orbit/corner increment	0.783	0.675
SCIP-cut LP re-solve	0.895	0.870
Orbit-cut LP re-solve	0.931	0.910
Objective-parallel corner cut, LP re-solve	0.783	0.675

Table 3: Mean root-gap fraction closed by the external loop. Intervals are 95% paired- t intervals for orbit minus SCIP model after round 10.

Size (n)	SCIP r1	Orbit r1	SCIP/orbit r10	Difference r10
4×4 (60)	0.757	0.761	0.974 / 0.966	-0.008 [-0.016, +0.001]
6×8 (60)	0.754	0.755	0.968 / 0.957	-0.011 [-0.019, -0.002]
8×12 (50)	0.804	0.799	0.974 / 0.969	-0.005 [-0.010, +0.000]
10×20 (50)	0.705	0.699	0.937 / 0.912	-0.025 [-0.045, -0.005]

8.2 A corner optimum and an LP re-solve are different quantities

The one-cut experiment starts from a McCormick LP, selects its most violated bilinear term, and compares cuts after re-solving the full LP. Its denominator is the gap between the initial LP and a global solve enforcing that one term. The retained sample has 47 small and 73 larger corners. Ten old corner values were infeasible underestimates caused by antiparallel projected rays. Rational lower and upper checks correct those ten values; [Appendix F.2](#) identifies them.

The objective-parallel cut gains exactly the corner increment. Tilted cuts can gain more through other LP rows. The orbit cut improves on the SCIP-model cut by more than 0.01 of the single-term gap on 62 corners, and loses by more than 0.01 on 22. Thus even numerical corner attainment does not give LP dominance. The numerical minimum orbit/corner ratio is about 0.999999989; this is not a collection of 120 exact certificates. SCIP’s ratio is below 0.9 on 27 corners and below 0.5 on three.

The minor experiment leads to the same distinction for signature $(2, 2)$. It contains 86 and 71 LP corners from random 3×3 and 4×4 models. On all 157, the orbit search attains the numerical corner value, whereas SCIP’s set does so on 12. Yet the maximum-margin choice decreases mean full-LP gap closure relative to SCIP by 0.0758 and 0.0651. The orbit-optimal set nearest SCIP has mean differences +0.0225 and +0.0057; the comparisons do not establish a mean advantage. These are single-minor LP re-solves, without branch-and-bound integration. The exact minor examples in [Theorem 6.7](#) establish geometric obstructions independently of these frequencies.

8.3 Later rounds and explored policies

The larger experiment is an external root cutting loop. At each round it re-solves the LP, forms its corner, and cuts violated bilinear terms. There are 220 paired constructed instances: 60 each at sizes 4×4 and 6×8 , and 50 each at 8×12 and 10×20 . Here the denominator is the gap to a global solve enforcing all bilinear equations. The implemented orbit search floors reduced costs and uses floating-point bisection. We call it the orbit rule without claiming exact optimization for raw zero reduced costs.

Table 4: Paired mean changes in root-gap fraction inside SCIP 10.0.3. The comparisons within each seed use the same eligible cohort.

Comparison	Seed 0 ($n = 251$)	Seed 1 ($n = 250$)	Seed 2 ($n = 248$)
Point rule minus disabled cuts	+0.114	+0.104	+0.106
Corner criterion minus point rule	−0.008	−0.007	−0.010
Efficacy criterion minus point rule	+0.001	+0.003	−0.004

Let Δ_r denote the pooled orbit-minus-SCIP difference after round r . The estimates and 95% paired intervals are

$$\begin{aligned}\Delta_1 &= -0.001, & [-0.016, +0.014], \\ \Delta_{10} \text{ and } \Delta_{20} &= -0.012, & [-0.017, -0.006].\end{aligned}$$

The antiparallel-ray correction changes the first-round orbit result by +0.011 [+0.004, +0.019] against the bug-affected implementation, but the round-10 difference is −0.001 [−0.003, +0.002]. It therefore does not explain the later-round reversal.

Switching after one orbit root round to SCIP-model rounds already leaves a pooled round-10 deficit −0.006 [−0.011, −0.002]. Cross-evaluations of retained states after early schedules report different amounts of subsequent progress under the tested rules. Bound attainment and ties alone do not explain the loss: nonattaining, low-tie policies also lose. Shorter steps on many rays are a plausible common feature. The intervention contrasting a majority-ray selection rule with random orbit selection is inconclusive and does not hold orbit exposure constant. These observations do not identify a causal mechanism.

The efficacy policy chooses, per term, the largest Euclidean efficacy among the SCIP cut, the orbit cut, two cost-perturbed orbit choices, and a Euclidean-depth orbit choice. Selected after examining the records, it has pooled advantage 0.004 in AUC, defined as the mean gap fraction at rounds 1–10, with Holm-adjusted $p = 0.025$. The Holm family has 14 complete-cohort rule comparisons against SCIP and is applied separately to AUC and round 10; the latter adjusted p is 0.33. Comparisons of switching from orbit to SCIP-model rounds against continuing orbit also depend on the tested family and statistical test ([Appendix F.4](#)). Both findings need fresh data. The convergence hypotheses of the mathematical results are not verified by these finite trajectories.

8.4 Selection inside SCIP

The inside-SCIP experiment searches alternative admissible directions using 24 half-circle samples with local refinement in two dimensions, and projected subgradient ascent in higher dimensions. This is a heuristic search. Root-gap closure is measured against recorded reference values on complete, comparable cases; its denominator differs from the one-term experiment above.

The corner criterion provides no consistent improvement in this sample. In a full summary scan of 305 instances and 109,389 corners, 81.6% of the floored criterion minima occur at numerically zero-rate rays. The positive surrogate then measures a flooring convention rather than a positive true corner gain. Degeneracy alone does not explain every loss. The raw individual corners from this full scan were not retained, so the companion preserves its summaries rather than a full reconstruction.

Full solves do not repair the inference. Row capture in the patched build changes some tree paths; the subsequent 120 stock reruns were made in a different batch. Their CPU timing comparison is uncontrolled and is excluded from the performance conclusions. The 300 debug-solution runs are a separate validity audit with incomplete objective mappings and unresolved diagnostics, not ordinary

full-solve benchmarks or a clean validity certificate ([Appendix F.6](#)). The evidence establishes neither a stock-SCIP timing benefit nor a generally better selection rule.

9 Consequences for cut selection

The unrestricted corner bound is a useful reference value. It says what one quadratic constraint can enforce on a chosen basis cone, and it separates a loss caused by the relaxation from a loss caused by the free set family. For low-dimensional constraints, the support and inertia results make this reference value accessible through small subproblems. For larger projected dimension, the complexity result explains why one should not expect a general efficient exact selection procedure.

The determinant orbit provides a small optimization problem and a geometric explanation for its failures. Its parameters can be searched by semidefinite feasibility, its contacts can force a one-dimensional pencil, and exact certificates can prove that every parameter misses a target bound. This makes the family useful even when it is not exact: it supplies a controlled baseline against which a point rule, a completion, or a larger family can be assessed. The minor examples show that this role extends beyond a single bilinear lifting variable.

The scaled-depth guarantee is conditional in an explicit and verifiable way. If D is bounded across a collection of bilinear corners, the orbit retains a positive fraction of each corner bound. If D grows, the sharp examples show that a fixed fraction cannot be promised. The additional conditioning term for a fixed point rule explains a further loss that can occur without changing the bilinear constraint. Translation and rescaling can alter that rule, so its guarantee should be applied to the actual quadratic representation used in the implementation.

Completion and closure address different weaknesses. Upward completion can remove artificial finite exits and can make a family exact at a corner where the uncompleted orbit has an infinite approximation factor. A closure can combine cuts with complementary coefficient patterns. Neither operation supplies general exactness: the certified closure examples retain a gap, and the sharp family shows why there is no uniform guarantee independent of scaled vertex depth. A better single-cut value and a better closure value should therefore be measured separately.

For repeated cuts, local objective improvement is insufficient. The convergence theorem requires a positive depth whenever violation stays away from zero. The explicit stalling sequence retains fixed violation and well-behaved basis cones while its depths vanish. It demonstrates a failure of arbitrary maximal-set selection. It does not establish that SCIP’s default rule stalls, nor that the implemented approximate orbit search satisfies the theorem’s uniform-depth hypotheses. These are distinct assertions requiring distinct evidence.

The numerical comparisons reinforce this separation. Stronger corner bounds can lead to weaker full-LP bounds after reoptimization, and small changes in successive corners can change later progress. Reduced-cost floors used in a practical search also replace the exact objective criterion by a surrogate. The retained solver studies provide no consistent improvement from the tested corner-bound and efficacy searches. The full-solve records contain implementation and timing qualifications that prevent a general recommendation about enabling intersection cuts. They remain useful evidence about the tested rules and about the measurements needed in future implementations.

Several extensions would require additional results rather than a stronger reading of the present ones. The support-one criterion for the uncompleted orbit does not characterize exactness of every upward completion. The depth guarantee for $\mathcal{B}$ does not automatically give the same guarantee for its transformed point-rule subfamily. The three-parameter determinant orbit does not describe every maximal free set of signature $(2, 2)$. Finally, the idealized convergence rule does not certify an approximate implementation without a uniform approximation and cone-geometry bound. The

results proved here delimit these questions and provide exact examples and quantitative tests for addressing them.

A Proofs and certificates for the foundations

A.1 Cone containment, transversality, and boundary failures

Proof of [proposition 2.2](#). If S is empty, the whole space gives an infeasible intersection cut, which dominates any valid inequality. Otherwise work in $V = \text{span } P$. For a valid $c \geq 0$, the dual polyhedron $D_c = \{y \in V : P^T y \leq c\}$ is nonempty and pointed: its lineality is $\ker(P^T) \cap V = \{0\}$. It has finitely many vertices $y^1, \dots, y^m$. The fibre LP has an attained finite minimum for every $v \in \text{cone}(P)$, and duality and the existence of a vertex optimum give

$$\phi(v) = \min_{\lambda \geq 0: P\lambda = v} c^T \lambda = \max_{1 \leq i \leq m} (y^i)^T v.$$

Define the full-dimensional polyhedron

$$C = \{\bar{s} + v + u : v \in V, u \in V^\perp, (y^i)^T v \leq 1 \text{ for every } i\}.$$

It contains a neighbourhood of the apex. A point of S has $u = 0$ and $v \in \text{cone}(P)$, and validity gives $\phi(v) \geq 1$; hence it is not in $\text{int } C$. Along ray p_j , every $0 \leq t \leq 1/c_j$ lies in C , and the entire ray lies there if $c_j = 0$. Thus $a_j(C) \leq c_j$, proving domination.

For $0 < z_K(\omega) < \infty$, apply this result to $c = \omega/z_K(\omega)$. If $z_K = 0$, every free neighbourhood has value zero by validity. If $z_K = +\infty$, cone containment forces $S = \emptyset$, so the whole-space set attains that value. This proves the stated attainment claim even when ω has zero entries.

When $\text{cone}(P) = \mathbb{R}^k$ and $0 < z = z_K(\omega) < \infty$, the cost image $V_z = \bar{s} + P\{\lambda \geq 0 : \omega^T \lambda \leq z\}$ itself is a closed polyhedron, a free neighbourhood, and an attaining set. To see interior membership, represent each $\pm e_i$ by nonnegative ray vectors; their maximum finite cost bounds the cost of any sufficiently small vector. If $s \in \text{int } V_z$, extend the segment from $\bar{s}$ through s slightly within V_z and rescale its ray representation to obtain a representation of s with cost strictly below z . Thus s cannot be feasible. Its original ray steps are at least z/ω_j , or infinite for zero costs. $\square$

Proof of [theorem 2.3](#). Put $T = T_z$ and $M = T \cap S$. Every representation of a point of M with objective at most z has objective exactly z . Moreover $q = 0$ on M : a point with $q < 0$ could be scaled slightly toward the apex to produce a feasible ray vector of smaller cost. In particular, $q \geq 0$ on T . The contact set M is compact. Continuity and the strict hypothesis give a neighbourhood U of M , a neighbourhood W of $\bar{s}$, and $\eta > 0$ such that

$$\nabla q(\xi)^T(b - t) \geq \eta$$

whenever $t, \xi \in U$ are sufficiently close to each other and $b \in W$. For example, first take a positive minimum of $\nabla q(t)^T(\bar{s} - t)$ on M , and then use uniform continuity on compact neighbourhoods.

Suppose for arbitrarily small $\delta_n \downarrow 0$ there are $p_n \in C_{\delta_n} \cap S$ outside M . Write $p_n = \theta_n b_n + (1 - \theta_n)t_n$, with b_n in the apex ball, $t_n \in T$, and $0 \leq \theta_n \leq 1$. A subsequence has $p_n \rightarrow p \in M$, $t_n \rightarrow t \in T$, and $\theta_n \rightarrow \theta$. If $\theta > 0$, a ray representation of t with cost at most z gives a representation of $p = \theta \bar{s} + (1 - \theta)t$ with cost at most $(1 - \theta)z < z$. This contradicts $p \in S$ without requiring P to be injective. Thus $\theta = 0$ and $t_n \rightarrow p \in M$. When $\theta_n = 0$, $p_n = t_n$ already belongs to M , a contradiction. Otherwise the segment mean value theorem gives, for large n ,

$$q(p_n) = q(t_n) + \theta_n \nabla q(\xi_n)^T(b_n - t_n) \geq \theta_n \eta > 0,$$

again a contradiction. Therefore $C_\delta \cap S = M$ for all sufficiently small δ . If a point of M were interior to C_δ , it would be an interior local minimum of q , so its gradient would vanish, contrary to the strict hypothesis. Thus C_δ is free. It contains T_z , and validity now gives $z_{C_\delta} = z$. $\square$

Proof of [example 2.4](#). For $q = (x + 1)y + 1$, the positive orthant has $q \geq 1$, so the corner is infeasible. Any free neighbourhood contains $B(0, \delta)$. If its first ray step were infinite, e_1 would be a recession direction, and $(2/\delta, -\delta/2)$ would be interior to the set. But its quadratic value is $-\delta/2 < 0$. The first step is therefore finite, so the zero-cost first ray makes every cut value zero. For $q = (x + 1)y + 1 - y^2$, the same point has value $-\delta/2 - \delta^2/4 < 0$. Orthant feasibility requires $y^2 - y - 1 \geq 0$, so $y \geq (1 + \sqrt{5})/2$, attained at $x = 0$.

For $q = x - x^2 + (1 - y)^2$, any orthant point of cost at most one has $0 \leq x \leq 1$ and $q \geq 0$. Equality forces $(x, y) = (0, 1)$; this point is feasible, so the corner value is one with a unique minimizer. If a cut had value at least one, its second step would be at least one and $(0, 1)$ would lie in its defining free set. For sufficiently small $s > 0$, $(-s, 0)$ lies in the apex ball. The point

$$(1 - s)(0, 1) + s(-s, 0) = (-s^2, 1 - s)$$

is then interior to the convex set, while its quadratic value is $-s^4 < 0$. This proves non-attainment. $\square$

A.2 Algorithmic details

Proof of [theorem 2.8](#). By [lemma 2.6](#) and [theorem 2.7](#), a feasible optimum uses at most r rays. Thus $z_K \leq t$ exactly when, for one of the $O(N^r)$ supports J , the fixed-variable existential formula

$$\exists \mu \in \mathbb{R}^J : \quad \mu \geq 0, \quad \omega_J^T \mu \leq t, \quad q(\bar{s} + P_J \mu) \leq 0$$

holds. Its number of variables and polynomial degrees are bounded for fixed r . Fixed-variable real-algebraic decision therefore has polynomial bit complexity ([Renegar, 1992](#), Theorem 1.1). The same formulas without the objective bound detect infeasibility.

For completeness, a finite positive value has a polynomial-bit initial bracket. On each fixed-size support, real quantifier elimination with the objective threshold left free yields univariate polynomials of bounded degree and coefficient bit size polynomial in the input size. A finite endpoint is a root of one of those polynomials. Standard root-size bounds place a positive endpoint between 2^{-B} and 2^B for a polynomially bounded B . There are only polynomially many supports. Bisection on a common such bracket, using the decision test, yields relative error ϵ with polynomially many calls in the input size and $\log(1/\epsilon)$.

For the normal-fan refinement, assume $\text{rank } P = k$ and consider $D_\omega = \{y : P^T y \leq \omega\}$. It is pointed and contains zero in its interior. Fibre duality gives

$$\phi(v) = \min_{\lambda \geq 0 : P\lambda = v} \omega^T \lambda = \max_{y \in D_\omega} y^T v \quad (v \in \text{cone}(P)).$$

For a vertex y , $\phi(v) = y^T v$ on its normal cone, generated by the active columns p_j . Perturb the right-hand side by a sufficiently small generic lexicographic perturbation, keeping every entry positive, to obtain a simple polyhedron. Each active independent k -set J gives a vertex depending continuously on that perturbation. There are finitely many such sets. If its limit were infeasible for the original polyhedron, it would remain infeasible under small perturbations. Thus every perturbed vertex limits to an original vertex, and its normal cone $\text{cone}(P_J)$ lies in that original normal cone. The perturbed vertex cones cover $\text{cone}(P)$ because every bounded dual objective there has a vertex

optimum. On such a cell, $\phi(P_J\mu) = y^T P_J\mu = \omega_J^T \mu$, with the original costs. The cell problems therefore preserve the original value; their optima use at most r generators by [theorem 2.7](#).

A pointed unbounded polyhedron can be truncated by one additional facet while keeping all its vertices. The upper bound theorem for the resulting polytope gives $O(N^{\lfloor k/2 \rfloor})$ cells ([McMullen, 1970](#)). For $k = 3$, a bounded simple polyhedron with at most N facets has at most $2N - 4$ vertices. In the unbounded case the truncation has at most $N + 1$ facets and $2N - 2$ vertices, at least three on the new facet, so the same bound for original vertices is valid. The normal-fan edge graph embeds on the sphere, has at most N ray-direction vertices, and is simple after zero and duplicate directions are removed. Its planar bound is $3N - 6$ edges. $\square$

Remark A.1 (Generic stationary points). On a face F of a support subproblem, write $g(\mu) = \mu^T G_F \mu + 2m_F^T \mu + g_0$. For a regular stationary point with nonsingular G_F ,

$$\mu = -G_F^{-1}(m_F + \tau \omega_F), \quad \tau^2 = \frac{m_F^T G_F^{-1} m_F - g_0}{\omega_F^T G_F^{-1} \omega_F}, \quad \tau > 0.$$

The denominator must be nonzero, and candidates must satisfy the coordinate and active-constraint conditions. Enumerating faces gives the generic closed-form solution. Singular faces, abnormal stationary points, and zero denominators require the exact algebraic decision method or, with at most two rays, the next formula.

Proof of [proposition 2.9](#). Every nonzero nonnegative two-ray vector is $\tau\nu(\theta)$, with cost $\tau > 0$. Substituting $u = 1/\tau$ gives $g_0 u^2 + b(\theta)u + a(\theta) \leq 0$. Since $g_0 > 0$, the greatest feasible positive reciprocal radius is its larger root $u_+(\theta)$. The compact discriminant-feasible set gives the stated maximum; a positive larger root itself is feasible.

Away from $\Delta = 0$, $u_+ = (-b' + \Delta'/(2\sqrt{\Delta}))/g_0$, so interior stationary points satisfy $\Delta' = 2b'\sqrt{\Delta}$ and hence $\Delta'^2 - 4b'^2\Delta = 0$. This polynomial has degree at most two, because b is affine and Δ is quadratic. Interval boundaries occur at zero or one or at a root of Δ . Squaring may add candidates, but evaluating the actual larger root discards no valid maximum. Including roots of Δ' also handles the $b' = 0$ case directly. If the stationary polynomial vanishes identically and $b' \neq 0$, Δ is a square of an affine function, so $\sqrt{\Delta}$ and u_+ are piecewise affine; maxima occur at boundaries or kinks $\Delta = 0$. If $b' = 0$, identically vanishing stationarity means Δ is constant, and the root is constant. If $\Delta \equiv 0$, the root is affine. All exceptional cases are therefore covered by finitely many endpoint and quadratic-root evaluations. If no positive larger root exists, the two-ray subproblem is infeasible. $\square$

Bilinear attainment qualification. The gradient of $q = w - xy$ never vanishes. Suppose a minimizing contact t fails [equation \(4\)](#). Choose an optimal representation of that same point with independent supported columns using its fibre LP. If the supported gradient were nonzero, the regular KKT system on its positive support would give a positive objective multiplier and thus positive transversality. Therefore the supported gradient vanishes. The abnormal case of the inertia proof bounds its support by $\rho - 1 = 1$. The contact is consequently a single ray's first feasible point, and its directional gradient is zero. If no such minimizing contact exists, all minimizing contacts satisfy the transversality hypothesis, so [theorem 2.3](#) applies. $\square$

Proof of [theorem 2.10](#). For a graph G on k vertices, let A_G have zero diagonal and entries one on its edges, and let $\omega(G)$ denote its clique number. Set $q(s) = \mu - s^T A_G s$, $P = I$, $\bar{s} = 0$, and $\omega = \mathbf{1}$. Writing $s = \tau\nu$, $\nu \geq 0$, $\mathbf{1}^T \nu = 1$, the Motzkin–Straus identity ([Motzkin and Straus, 1965](#)) gives

$$\max_{\nu} \nu^T A_G \nu = 1 - \frac{1}{\omega(G)}, \quad z_K = \sqrt{\frac{\mu \omega(G)}{\omega(G) - 1}}.$$

An edgeless graph has infinite value and can be recognized directly. For a rational integer threshold $\omega_0 \geq 2$, use $\mu = \omega_0(\omega_0 - 1)$. Then $z_K \leq \omega_0$ exactly when $\omega(G) \geq \omega_0$, proving rational-threshold NP-hardness. With $\mu = 1$ put $f(h) = \sqrt{h/(h-1)}$. For consecutive clique sizes,

$$\frac{f(h)}{f(h+1)} = \left(1 - \frac{1}{h^2}\right)^{-1/2} \geq 1 + \frac{1}{2k^2}.$$

For $\delta = 1/(5k^2)$ and $k \geq 2$, $(1 + \delta)/(1 - \delta) < 1 + 1/(2k^2)$. The relative-error intervals around all possible finite values are disjoint, so a δ -approximation determines the clique number. The matrix $-A_G$ is indefinite whenever G has an edge. $\square$

Remark A.2 (A specified efficacy optimization). In ray-coordinate Euclidean norm, a normalized valid cut $a^T \lambda \geq 1$ has efficacy $1/\|a\|$. Thus maximize efficacy by minimizing $\|a\|$ over $\mathcal{V} = \{a \geq 0 : a^T \lambda \geq 1 \text{ on } X\}$. For fixed r , the support-value identity extends from positive to nonnegative costs by perturbation: for any $a \geq 0$, $z_K(a + \epsilon \mathbf{1}) \downarrow z_K(a)$, and the finite union of supports of size at most r has the same infimum. The fixed-variable algebraic procedure can therefore test $\exists \lambda \in X : a^T \lambda < 1$ and return a violating point. Such a point gives the separating inequality $\lambda^T c \geq 1$ for all $c \in \mathcal{V}$. Algebraic sample points can be approximated to the accuracy of a weak separation oracle.

If X is nonempty, write $z_1 = z_K(\mathbf{1}) > 0$. The vector $(2/z_1)\mathbf{1}$ lies in $\mathcal{V}$ with a ball of radius $1/z_1$: for every $\lambda \in X$, $\|\lambda\| \leq \mathbf{1}^T \lambda$ and $(2/z_1)\mathbf{1}^T \lambda - 1 \geq \|\lambda\|/z_1$. A feasible vector of norm $2\sqrt{N}/z_1$ bounds the optimum. Set $a_0 = (2/z_1)\mathbf{1}$ and $R = 4\sqrt{N}/z_1$, and introduce the bounded epigraph

$$\mathcal{E} = \{(a, t) : a \in \mathcal{V}, \|a\| \leq t \leq 2R\}.$$

Its linear objective t has the same minimum as the norm problem. It contains the ball of radius $1/(2z_1)$ about (a_0, R) : the a component stays inside the established feasible ball, the norm constraint has initial slack $R/2$, and a joint displacement of that radius changes $\|a\| - t$ by at most $\sqrt{2}/(2z_1)$. The upper constraint has slack R . The whole epigraph lies in the ball of radius $2\sqrt{2}R$ about zero. The weak separation oracle for $\mathcal{V}$, together with elementary separation for the norm and linear constraints, therefore supplies a weak separation oracle for this bounded body. Weak separation-to-linear-optimization equivalence (Grötschel et al., 1988, Corollary 4.2.7) minimizes t to prescribed accuracy in polynomial time for fixed r . For rational bit encoding, choose a rational b with $0 < b \leq z_1 \leq 2b$, replace a_0 by $(2/b)\mathbf{1}$ and R by $4\lceil\sqrt{N}\rceil/b$, and use radius $1/(2b)$ for the inner ball and $3R$ for the outer ball. The same estimates apply; the preceding algebraic size bound controls their bit sizes. This concerns a stated ray-coordinate metric. An original-variable metric must be supplied through its linear map; it is not implicit in this assertion. If X is empty, the normalized-cut efficacy problem degenerates and is excluded. The efficacy supremum of genuine intersection cuts has this same value: perturb a valid nonnegative vector to be positive, and apply [theorem 2.1](#) at target $1 - \epsilon$ to obtain cut coefficients bounded by the perturbed vector divided by $1 - \epsilon$. Let both perturbations vanish. Attainment by a genuine intersection cut does not follow.

A.3 The two polytope closures

Proof of propositions 2.11 and 2.12. Every set in the intersection defining the nonlocal closure contains $\mathcal{P} \cap S$. If this intersection is nonempty, its convex hull is compact. A point p of $\mathcal{P}$ outside that hull can be strictly separated by an inequality $a^T s \geq b$. Choose $a^T p < b' < b$. The compact set $\mathcal{P} \cap \{a^T s \leq b'\}$ misses closed S . Its sufficiently small ball enlargement is free and contains that compact set in its interior. All remaining points of $\mathcal{P}$ have $a^T s > b'$, so their convex hull excludes p . If $\mathcal{P} \cap S$ is empty, a small enlargement of all of $\mathcal{P}$ is free and removes every point.

For the triangle example, the closed unit disk is the unique maximal full-dimensional free set: every free interior lies in the open disk, so every full-dimensional closed free set lies in its closure. The top vertex belongs to S and yields no cut. At $(-1/2, 0)$ the basis rays point toward $(0, 2)$ and $(1/2, 0)$. The first exit is $p_1 = (-1/2, 0) + t((0, 2) - (-1/2, 0))$, where $17t^2 - 2t - 3 = 0$ gives the displayed positive root. The second exit is $(1, 0)$, beyond the triangle. The disk cut is the line through these two exits and dominates every other cut at that basis. The other lower vertex gives its reflection. Their intersection is

$$(0, 2t/(3/2 - t/2)) = \left(0, \frac{1 + \sqrt{13}}{6}\right).$$

The resulting polygon has vertices $(0, 2), p_1, v, p_2$. The feasible part of the triangle lies above the chord $p_1 p_2$: below height $2t$, its horizontal sections lie strictly inside the disk. All three vertices of the triangle above that chord are feasible, so the hull is precisely $\text{conv}\{(0, 2), p_1, p_2\}$. Since $(1 + \sqrt{13})/6 < 2t$, the first closure is strictly larger. At its new vertex v , the two active cuts give rays through p_1 and p_2 , so the next disk cut is their chord and removes the remaining gap. $\square$

A.4 Lorentz classification and the point rule

Proof of [theorem 3.1](#). First let $m = 1$. The forbidden homogeneous set is the union of the two closed convex Lorentz cones $Q_+ = \{\|x\| \leq y\}$ and $Q_- = \{\|x\| \leq -y\}$. Separate the interior of a full-dimensional free convex set from each cone. Because the cones contain zero and are homogeneous, the resulting halfspaces can have zero right-hand side and take the form

$$v^T x \geq y, \quad w^T x \geq -y, \quad \|v\|, \|w\| \leq 1.$$

Indeed a nonzero functional nonpositive on Q_+ has the form $(v, -1)$ after positive normalization, with $\|v\| \leq 1$; the other cone gives the second inequality. Their intersection contains the given free set and is free. Full dimension requires $v + w \neq 0$.

With the Lorentz form $B(\xi, \eta) = \xi^T \text{diag}(I_n, -1)\eta$, put $A = (v, -1)$ and $B_0 = (w, 1)$. The quadratic values of A and B_0 are nonpositive, whereas the value at $A + B_0$ is $\|v + w\|^2 > 0$. Consequently $A + tB_0$ and $B_0 + sA$ are null for some $0 \leq t, s < 1$. Their halfspaces still contain the original intersection, since the normals are nonnegative combinations. Their last entries are respectively negative and positive; normalizing those entries gives the unit-direction pair in [equation \(5\)](#). It is full dimensional, hence $\gamma_1 \neq -\gamma_2$.

Every such pair is maximal. Let $k_1 = (\gamma_1, 1)$, $k_2 = (\gamma_2, -1)$, and $h = 1 + \gamma_1^T \gamma_2 > 0$. At k_1 the first inequality is tight and the second has slack h . If an added point p violates the first inequality, then $(k_1 - \theta p)/(1 - \theta)$ lies in the interior of the pair for sufficiently small $\theta > 0$. Hence k_1 becomes an interior point of the enlargement and is forbidden. The second inequality has the same argument. Thus a maximal original set equals the pair.

To construct its orbit from a fixed unit ℓ , set $d = \sqrt{2/h}$, $e' = d(k_1 + k_2)/2$, and $\tau' = d(k_1 - k_2)/2$. These are Lorentz-orthonormal, with e' spacelike and τ' future timelike. Map $(\ell, 0)$ to e' and $(0, 1)$ to τ' , then extend by an isometry on their positive definite orthogonal complements. Their dimension is $n - 1 \geq 1$, so choose the extension with determinant one. The resulting map is in $SO^+(n, 1)$ and sends $(\ell, \pm 1)$ to $dk_{1,2}$. Lorentz orthogonality identifies the corresponding supporting hyperplanes and their orientations, so it sends C_ℓ to the target pair.

For $n = 1$, the complement of $Q_{1,m}$ has two connected components, $x > \|y\|$ and $-x > \|y\|$. The connected interior of any full-dimensional convex free set lies in one component; its closure lies in the corresponding closed cone. These cones are free and maximal, proving the classification.

For an affine slice $H = \{u : \ell(u) = -1\}$ of a homogeneous form, let K be a full-dimensional free convex set relative to H . The interior of its positive cone is $\{rk : r > 0, k \in \text{relint}_H K\}$: the map $(r, k) \mapsto rk$ has inverse $u \mapsto (-\ell(u), u/[-\ell(u)])$ on $\{\ell < 0\}$. Homogeneity makes its closed convex cone free. The classification just proved supplies a maximal homogeneous free set containing it. Because the relative interior of K lies in the homogeneous interior, slicing that set back is free and contains K ; maximality of K gives equality.

Finally, radical coordinates require a lineality argument. If $S' + V = S'$ for a linear subspace V , any full-dimensional maximal free set C has $C + V = C$, since $\text{int cl}(C + V) = \text{int } C + V$ misses S' . Write the diagonalized slice as $a^T x + d^T y + h^T z = -1$, with z radical. If $h = 0$, this gives the product with all radical coordinates. If $h \neq 0$, the affine bijection to the nondegenerate coordinates and $h^\perp$ is

$$(x, y, z_\perp) \mapsto \left(x, y, z_\perp - \frac{1 + a^T x + d^T y}{\|h\|^2} h \right).$$

The feasible set becomes $Q_{n,m} \times h^\perp$. Lineality and maximality identify every free set as a product with a maximal homogeneous free set. Equivalently, include the entire radical product before imposing the original slice. $\square$

Proof of proposition 3.2. In transformed coordinates, the apex is

$$(\|x'\|_\ell, y') = \frac{\|x'\| + y'}{2}(\ell, 1) + \frac{\|x'\| - y'}{2}(\ell, -1).$$

Both coefficients are positive. Transformation gives equation (6). Conversely, the explicit Lorentz map in the preceding proof sends $(\ell, \pm 1)$ to $dk_{1,2}$, so its inverse sends an apex satisfying equation (6) to $((a+b)\ell/d, (a-b)/d)$. The normalized spatial direction is exactly ℓ , proving the converse. For $n = 2$, the identity $\|x - a\gamma_1\| = b = a - y$ yields the displayed expression for a , and then b, γ_2 . For any unit γ_1 with $\gamma_1^T x - y > 0$, the formula gives $a > 0$ and $2(a - y)(\gamma_1^T x - y) = \|x - y\gamma_1\|^2 > 0$, so $b > 0$. This establishes the claimed parameterization and its domain. Multiplying the two tangency rays to make their slice coordinates one turns a positive apex combination into a convex combination precisely when both scaling signs are positive. $\square$

Example A.3 (A maximal set with only asymptotic contacts). For $S = \{xy \geq 1\}$, let $K = \{x \geq 0, y \leq 0\}$. If $p \notin K$ has $p_x < 0$, choose a small $\epsilon > 0$ and large $T > 0$ so that the midpoint of p and $(\epsilon, -T) \in \text{int } K$ has two negative coordinates and product greater than one. This midpoint lies in the interior of $\text{conv}(K \cup \{p\})$. If $p_y > 0$, use $(T, -\epsilon)$. Thus K is maximal free, although $K \cap S = \emptyset$.

The homogenized form is $t^2 - xy$, with signature $(2, 1)$. The two null tangency rays of K are $(1, 0, 0)$ and $(0, -1, 0)$. Their span has $t = 0$, so no affine apex with $t = 1$ satisfies the point-rule span condition. More generally any plain point-rule slice contains a finite feasible contact: its apex is a positive combination of its two null tangency rays and has slice coordinate one, so at least one ray has positive slice coordinate and can be scaled to a finite contact. Every enlargement of that set still contains this contact. Since K has none, the exclusion also holds for the affine enlargement procedure of Muñoz and Serrano (2022), even if the apex is allowed to vary.

Proof of proposition 3.3. The two point-rule steps are $x_0 + 1/x_0$ and $\sqrt{1 + x_0^2}$, by substitution along the two rays. The strip's steps are $+\infty$ and one, giving its value one. It is maximal free: it is invariant under horizontal translation, and adding a point with $|y| > 1$ makes a horizontal interior band meet S . Corner minimization reduces to

$$\min_{\lambda_1 \geq 0} \epsilon \lambda_1 + \sqrt{(x_0 - \lambda_1)^2 + 1}.$$

The stationary point has $x_0 - \lambda_1 = \epsilon/\sqrt{1 - \epsilon^2}$ and is feasible under the stated hypothesis. Convexity gives the displayed value. The asymptotic ratios follow on substituting $\epsilon = x_0^{-2}$. $\square$

Proposition A.4 (The constant-direction closure on pointed corners). *For $S = \{y^2 \geq x^2 + 1\}$, there are pointed two-ray corners with positive costs for which the closure of all constant-direction sets has value tending to zero, while the corner value and one Lorentz-transformed split's value tend to one.*

Proof. For $a > 1$, put $r = \sqrt{1 + a^2}$, $h = \sqrt{1 + 2a^2}$, $n = (r, -a)/h$, $\eta = a^{-3}$, and use apex zero, costs $(1, 1)$, and rays $r_1 = (a, r)$, $r_2 = -r_1 + \eta n$. Their determinant is $-\eta h \neq 0$, so the corner is pointed. The boost $(X, Y) = (rx - ay, ry - ax)$ preserves $y^2 - x^2$ and sends the rays to $(0, 1)$ and $(\eta h, -1 - \eta c)$, where $c = 2ar/h$. For ray cost $\tau = \lambda_1 + \lambda_2$, $|Y| \leq (1 + \eta c)\tau$; feasibility requires $|Y| \geq 1$. Thus

$$(1 + \eta c)^{-1} \leq z_K \leq 1.$$

The free oblique split $|Y| \leq 1$ has value $(1 + \eta c)^{-1}$, which tends to one.

All constant-direction sets containing zero are $|y| \leq (ux + 1)/\sqrt{1 + u^2}$, $u \in \mathbb{R}$. Both rays are timelike for this choice of η : the second has squared Lorentz length $1 + 2\eta c - \eta^2/h^2 > 0$. Their cut coefficients are positive and have sum

$$a_1(u) + a_2(u) = (2r + \eta a/h)\sqrt{1 + u^2} - (\eta r/h)u.$$

Its minimum is $L = \sqrt{4r^2 + 4\eta ar/h - \eta^2/h^2}$. Consequently $(1/L, 1/L)$ satisfies every family cut, and the closure value is at most $2/L \rightarrow 0$. This is an analytic pointed perturbation, not an inference from the exactly opposite-ray limit. $\square$

A.5 The determinant orbit and its completion

Proof of proposition 3.4. A two-dimensional linear space all of whose matrices have rank at most one has either a common left factor or a common right factor. Indeed two rank-one matrices with independent left and right factors have a rank-two sum. These two families of planes are the two rulings of the determinant-zero cone. A determinant-preserving bijection preserves or interchanges the rulings: planes within one ruling meet only at zero, whereas planes from opposite rulings meet in a rank-one line. If it preserves them, select images of the four matrix units. Their left and right factors give independent bases a_1, a_2 and b_1, b_2 , with scalar coefficients c_{ij} . The image of $(e_1 + e_2)(e_1 + e_2)^T$ has rank one, so $c_{11}c_{22} = c_{12}c_{21}$. These nonzero coefficients split into row and column factors, giving $M \mapsto AMB^T$. If the rulings are interchanged, compose with transposition. Determinant preservation then imposes $\det A \det B = 1$.

With $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $K = \text{diag}(1, -1)$, and $L = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, the map

$$\Phi(x_1, x_2, y_1, y_2) = x_1 I + x_2 J + y_1 K + y_2 L$$

has determinant $\|x\|^2 - \|y\|^2$, and the symmetric part has eigenvalues $x_1 \pm \|y\|$. Thus the seed cone is $\widehat{C}_I$. Under $M \mapsto AMB^T$, its image is $\text{sym}(BA^{-1}M) \succeq 0$, so $F^T = BA^{-1}$ and $\det F = 1/(\det A)^2 > 0$. Transposition fixes the seed. Conversely choose $A = \gamma I$, $B = \gamma F^T$, with $\gamma^4 \det F = 1$, to obtain every F of positive determinant.

The map $M \mapsto \text{sym}(F^T M)$ is onto the symmetric matrices. Writing $F^T M = H + tJ$, $\det(F^T M) = \det H + t^2$. Hence positive definite symmetric parts have positive determinant, and the whole PSD preimage has nonnegative determinant. This proves full dimension and freeness for $\det F > 0$; a negative determinant reverses the sign on the interior and is not free. For a nonzero

rank-one ab^T , the symmetric part of $(F^T a)b^T$ is PSD exactly when $F^T a$ is a nonnegative multiple of b . Since F is invertible, that multiple is positive for nonzero a, b . With $a = (x, 1)^T$, $b = (y, 1)^T$, this gives the stated graph and positive denominator. Its derivative is $\det F / (F_{12}x + F_{22})^2 > 0$. $\square$

Proposition A.5 (The BCM rotation family). *For $M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $G_t = \begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$,*

$$\text{sym}(G_t M) \succeq 0 \iff \cos t(a + d) - \sin t(b - c) \geq \|(b + c, a - d)\|.$$

Thus BCM's (14a) is exactly the rotation subfamily. If F is not a scalar multiple of a rotation, its orbit cone is no (14a) member. A positive-determinant orbit cone is contained in no (14b) member.

Proof. The PSD trace criterion for a symmetric 2×2 matrix gives the displayed identity: its traceless-part norm is unchanged by the rotation. For $A = F^T$, the lineality of the cone $\text{sym}(AM) \succeq 0$ is $\text{span}(A^{-1}J)$. Equality with a rotation member would make $A^{-1}J$ proportional to $G_t^{-1}J$, forcing A to be a scalar multiple of G_t . The interior of a positive-determinant orbit cone has $ad - bc > 0$. The (14b) inequality instead implies $\|(b + c, a - d)\| \geq \|(a + d, b - c)\|$, that is $ad - bc \leq 0$. These facts exclude containment. For example $F^T = \text{diag}(2, 1/2)$ is outside the rotation family. $\square$

Proof of proposition 3.5. The normal indexed by v is Fvv^T . Its null tangency matrix is

$$F^{-T}(Jv)(Jv)^T.$$

The identity $F^{-T} = JFJ^T / \det F$ gives its lower-right coordinate as $(Fv)_1 v_1 / \det F$. Its sign therefore agrees with $\langle Fvv^T, E \rangle = v^T Z_F v$.

The dual cone of $\widehat{C}_F$ is $F\mathbb{S}_+^2$. The dual of its upward sum consists of FY with $Y \succeq 0$ and $\langle Y, Z_F \rangle \geq 0$. Such a $Y = LL^T$ decomposes into two rank-one kept terms. For $u_\theta = (\cos \theta, \sin \theta)$, put $f(\theta) = (Lu_\theta)^T Z_F (Lu_\theta)$. Then $f(\theta) + f(\theta + \pi/2) = \langle Y, Z_F \rangle \geq 0$, and the difference changes sign as θ increases by $\pi/2$. At an intermediate angle both terms are equal and nonnegative. The two corresponding rank-one terms sum to Y . This works also for singular L . The dual is therefore the cone generated by the kept rank-one normals, proving by bipolarity the homogeneous closed-sum identity.

Closure commutes with this positive slice for a specific reason. If $M_n = A_n + \tau_n E \rightarrow M$ with $A_n \in \widehat{C}_F$, $\tau_n \geq 0$, and $h(M) > 0$, then

$$0 \leq \det A_n = \det M_n - \tau_n h(M_n)$$

bounds τ_n . Passing to a subsequence gives a finite lowering and $M - \tau E \in \widehat{C}_F$. On $h = 1$ it satisfies $0 \leq \tau \leq q(s)$, proving equation (8) and the slice identity. Determinant nonnegativity also makes the completed slice a subset of $\{q \geq 0\}$, hence free when full dimensional.

Now $Z_F = \begin{pmatrix} a & b/2 \\ b/2 & 0 \end{pmatrix}$. If $b = 0, a < 0$, only vectors with first coordinate zero are retained. Since $d < 0$, their inequality is $h \leq 0$, so the slice is empty. Otherwise Z_F has positive directions, dense in its nonnegative directions. Each strictly kept normal has a tangency ray with $h > 0$. Scale its contact to $h = 1$ and perturb its rank-one PSD image to be positive definite; this proves that the original cone and its completion meet the slice in full dimension.

At such a contact $M_* = a_0 b_0^T$, write $F^T M_* = c b_0 b_0^T$, $c > 0$. A dual normal FY active there satisfies $b_0^T Y b_0 = 0$, forcing Y onto the single rank-one ray orthogonal to b_0 . Thus the completed cone's normal cone at the contact is that one ray. The same is true in the slice after restricting the normal, since the cone's interior meets the slice. Any outside point violates a kept inequality and, by density, a strictly kept one. Adding it makes the associated contact interior to the convex enlargement: there is then no nonzero supporting normal at that contact. This contradicts freeness and proves maximality. $\square$

Remark A.6 (The retained-cap support formula). Let $G = \{\beta \in \mathbb{R}^m : \|\beta\| = 1, \beta_{\text{last}} \leq \gamma\}$, where $m \geq 2$ and $-1 \leq \gamma \leq 1$. For $y = (\tilde{y}, y_{\text{last}})$, its support function is

$$\max_{\beta \in G} \beta^T y = \begin{cases} \|y\|, & y_{\text{last}} \leq \gamma \|y\|, \\ \sqrt{1 - \gamma^2} \|\tilde{y}\| + \gamma y_{\text{last}}, & y_{\text{last}} > \gamma \|y\|. \end{cases}$$

Indeed fixing $\beta_{\text{last}} = t$ reduces the maximization to $ty_{\text{last}} + \sqrt{1 - t^2} \|\tilde{y}\|$ for $t \leq \gamma$. The unconstrained optimum is $t = y_{\text{last}}/\|y\|$; otherwise the endpoint γ is optimal. The zero vector and endpoint caps follow by continuity. This is the two-branch function in bilinear Case 4 of [Chmiela et al. \(2020\)](#), with γ their last positive direction component. It explains the identification with the constant-direction retained set.

A.6 The pencil and the exact starting examples

Proof of lemma 3.7. Let $M_* = M(t) = a_0 b_0^T$, with $a_0 = (x_t, 1)^T$ and $b_0 = (y_t, 1)^T$, and let $D = M_0(d)$. At the contact no lowering is possible, so $F^T a_0 = c b_0$, $c > 0$. For small positive and negative s , membership of $t + sd$ gives a lowering $0 \leq \tau(s) \leq s^2 \det D$. Set $Y = \text{sym}(F^T D)$, $Z = \text{sym}(F^T E)$, and $v = J b_0$. The contact inequality is strictly kept, so $v^T Z v > 0$. The null quadratic entry of the lowered PSD matrix is

$$s v^T Y v - \tau(s) v^T Z v \geq 0.$$

The $O(s^2)$ lowering bound for both signs of s forces $v^T Y v = 0$, and then forces $\tau(s) = 0$. A PSD matrix with this zero diagonal entry has zero off-diagonal entry in the basis (b_0, v) . Hence $Y = \mu b_0 b_0^T$ and $\text{sym}(F^T [D - (\mu/c) M_*]) = 0$. Put $u = -\mu/c$. The matrix $D + u M_*$ has determinant $\det D$, because $\det M_* = 0$ and the mixed term is $\nabla q(t)^T d = 0$. It is invertible, so $F^T (D + u M_*) = \theta' J$ and $F^T = \theta J \text{adj}(D + u M_*)$. For $G_0 = J \text{adj} D$, $G_0 a_0 = d_x b_0$; for $G_1 = J \text{adj} M_*$, $G_1 a_0 = 0$. Thus $\theta d_x > 0$, and the chosen $d_x > 0$ gives $\theta > 0$. The determinant and $G_1 M_*$ identities follow immediately.

If $\det D = 0$, the same argument gives $F^T (D + u M_*) = \theta' J$ with a singular left-hand side. Thus $\theta' = 0$, so $D + u M_* = 0$. Its lower-right entry gives $u = 0$, and then $D = 0$, contrary to the segment being nontrivial. $\square$

Proof of theorem 3.8. Let $(\alpha, \beta, \gamma, \delta)$ be nonnegative barycentric coordinates of $(\bar{s}, v_1, v_2, v_3)$. For homogeneous barycentric coordinates, put $Q = w(\alpha + \beta + \gamma + \delta) - xy$. Direct expansion gives the exact identity

$$64Q = 3[16(\beta - \gamma) - 5\alpha + 2\delta]^2 + 21\alpha^2 + 148\delta^2 + 1036\alpha\delta + 192\alpha\gamma + 1280\beta\delta.$$

On the simplex the coordinate sum is one. Equality forces $\alpha = \delta = 0$ and $\beta = \gamma = 1/2$. Thus $T_1 \cap S = \{t\}$, and the invertible ray matrix (determinant 249) gives the unique corner minimizer $(1/2, 1/2, 0)$ and value one.

For $d = v_1 - v_2 = (4, -12, 36)$ the pencil is

$$G(u) = \begin{pmatrix} 12 & 36 \\ -u & 4 - 3u \end{pmatrix}, \quad \det G(u) = 48.$$

At the four vertices the PSD parameter intervals are

$$\begin{array}{c|c} \bar{s} & [12 - 8\sqrt{2}, 12 + 8\sqrt{2}] \\ v_1 & (-\infty, 2] \\ v_2 & [-2, \infty) \\ v_3 & [-13/5 - 2\sqrt{30}/5, -13/5 + 2\sqrt{30}/5]. \end{array}$$

They follow by the two diagonal and determinant tests. The first and last intervals are disjoint, excluding all orbit slices containing T_1 .

For completions set $u_s = (1 - \sqrt{2}/3, 1)^T$, $u_3 = (-3/10 + \sqrt{30}/60, 1)^T$, $r_s = 12 - 8\sqrt{2}$, and $r_3 = -13/5 + 2\sqrt{30}/5 < r_s$. With $A_v(u) = \text{sym}(G(u)M(v))$ and $Z(u) = \text{sym}(G(u)E)$, direct multiplication gives

$$\begin{aligned} u_s^T A_s(u) u_s &= (\sqrt{2}/2)(u - r_s), \\ u_s^T Z(u) u_s &= -(1 - \sqrt{2}/3)u + 44/3 - 8\sqrt{2} > 0 & (u \leq r_s), \\ u_3^T A_{v_3}(u) u_3 &= -(\sqrt{30}/6)(u - r_3), \\ u_3^T Z(u) u_3 &= (3/10 - \sqrt{30}/60)u + 59/50 - 3\sqrt{30}/25 > 0 & (u \geq r_3). \end{aligned}$$

Thus no nonnegative lowering repairs the apex for $u < r_s$, or the third vertex for $u > r_3$. Every real parameter is excluded.

To prove a strict supremum gap, suppose completions have values tending to one. Normalize their parameters by $\|F\|_{\text{Frob}} = 1$ and choose feasible target simplices T_{z_n} with $z_n \rightarrow 1$. The vertex lowerings lie in $[0, q(v_{j,n})]$ and are bounded. Pass to limits $F \neq 0$ and τ_j . Every convex combination of the lowered limiting vertices has nonnegative determinant: it is a limit of combinations in the original homogeneous cones. If $\det F > 0$, the closed PSD conditions put T_1 in its completion, contradicting the pencil certificate.

Otherwise F has rank one. Write $F^T = ab^T$. PSD of $\text{sym}(ab^T M)$ forces $M^T b$ into $\mathbb{R}_+ a$, so all lowered vertices lie in one affine plane Σ . This is a genuine plane: the equation obtained by taking a vector orthogonal to a cannot vanish identically on the entire $h = 1$ slice for nonzero a, b . The lowered midpoint of v_1, v_2 is $M(t) - (\tau_1 + \tau_2)E/2$; its determinant is $-(\tau_1 + \tau_2)/2$. Nonnegativity forces $\tau_1 = \tau_2 = 0$, so $t \in \Sigma$. If Σ is vertical, all original vertices also lie in it, contradicting their affine independence. Otherwise it is $w = \ell(x, y)$ for an affine function ℓ . Convex combinations of the lowered vertices and nonnegative E directions form its epigraph over the projection of T_1 . Their determinants are nonnegative, so $\ell(x, y) \geq xy$ throughout that projection, with equality at $(-3, 0)$. This point is interior: its projection is the strict convex combination with weights $(184, 287, 257, 72)/800$ of the four projected vertices. The derivative at this interior minimum forces ℓ to be the tangent plane to xy there. But $\ell - xy = -(x + 3)y$ changes sign in every neighbourhood of that point. This contradiction excludes rank-one limits and proves $z_B < 1$.

Finally, $\nabla q(t)^T(\bar{s} - t) = 3/2 > 0$, so [theorem 2.3](#) gives unrestricted attainment. For stability, positive costs make nearby bounded objective sublevels uniformly compact. Limits of minimizers give lower semicontinuity of the corner value. A small radial extension of the original minimizer has $q < 0$, giving upper semicontinuity. The value is therefore continuous here, and every nearby minimizer approaches the unique original one. The strict transversality value consequently persists. If no neighbourhood preserved the strict family gap, choose approaching data and family cuts with values approaching the nearby corner values. The same normalized-parameter and bounded-lowering limit would produce one of the two contradictions just proved at the original data. Thus the gap and unrestricted attainment persist on an open neighbourhood. $\square$

Example A.7 (A second rational orbit obstruction). Take costs $\mathbf{1}$, apex $(-1/2, 1/2, 1)$, and

$$v_1 = (1/2, -4, -1/2), \quad v_2 = (-10, 3, 24), \quad v_3 = (9/2, 1/2, 15/2).$$

The unique corner contact is $t = (-1, -3, 3)$, with ray vector $(6/7, 1/7, 0)$ and value one. Here $z_A < 1$. The statement asserts no certified strict gap for the unrestricted completion family.

Proof. Let vertex zero denote the apex. On edge ij use $(1 - \theta)v_i + \theta v_j$, $0 \leq \theta \leq 1$. The exact edge polynomials and minima are

ij	$q(\theta)$	$\min q$
01	$9\theta^2/2 - 17\theta/4 + 5/4$	$71/288$
02	$95\theta^2/4 + 29\theta + 5/4$	$5/4$
03	$4\theta + 5/4$	$5/4$
12	$147\theta^2/2 - 21\theta + 3/2$	0
13	$-18\theta^2 + 87\theta/4 + 3/2$	$3/2$
23	$145\theta^2/4 - 85\theta + 54$	$21/4$

The zero is unique on edge 12 at $\theta = 1/7$. The projected determinants of faces 012, 013, 023, 123 are respectively $-161/4, 45/2, -25/2, -301/4$. Every triangular face therefore has independent x, y directions, so the restricted quadratic Hessian is indefinite and cannot have an interior minimum. The full tetrahedron also has an indefinite Hessian. Thus all minima occur on edges, proving nonnegativity and uniqueness. The ray matrix is invertible, so the corner minimizer is unique.

The tangent pencil gives the intervals

$$K_A(\bar{s}) = [133/12 - 7\sqrt{30}/6, 133/12 + 7\sqrt{30}/6], \quad K_A(v_3) = [-1/14 - 3\sqrt{14}/7, -1/14 + 3\sqrt{14}/7].$$

Their endpoints follow by the diagonal and determinant tests for $G(u) = \begin{pmatrix} 3u+7 & 3u-49/2 \\ -u & 21/2-u \end{pmatrix}$. They are disjoint, since the lower endpoint of the first exceeds 4 whereas the upper endpoint of the second is less than 2. Thus no invertible orbit parameter contains T_1 . If values approached one, normalized parameters would have a nonzero limit. A positive-determinant limit is excluded by the pencil. A rank-one limit forces the four vertices into a plane by the same rank-one PSD identity used above, whereas the ray determinant is $-6083/8 \neq 0$. This proves $z_A < 1$. $\square$

Proof of proposition 3.9. For the stated four vertices, homogeneous barycentric coordinates give the identity

$$512Q = (64\alpha - 79\gamma - 32\delta)^2 + 31\gamma^2 + 512\delta^2 + 1280\alpha\delta + 4032\gamma\delta + 128\beta(8\alpha + \gamma + 2\delta).$$

Equality forces $\alpha = \gamma = \delta = 0$ and $\beta = 1$. Hence $T_1 \cap S = \{v_1\}$. The ray determinant is $67/2$, so the minimizer is uniquely e_1 . At zero $\nabla q = (0, 0, 1)$, giving edge derivatives $2, 1/4, 1/2$. Also $P^T \nabla q = (-2, -7/4, -3/2)$. The KKT multiplier $\sigma = 1/2$ gives the stated inactive multipliers.

For vertices $(\bar{s}, v_1, v_2, v_3)$ respectively, take the positive definite matrices

$$Y_0 = \begin{pmatrix} 92 & -75 \\ -75 & 64 \end{pmatrix}, \quad Y_1 = \begin{pmatrix} 726 & 89 \\ 89 & 12 \end{pmatrix}, \quad Y_2 = \begin{pmatrix} 96 & -64 \\ -64 & 45 \end{pmatrix}, \quad Y_3 = \begin{pmatrix} 20 & 16 \\ 16 & 16 \end{pmatrix}.$$

Their determinants are 263, 791, 224, 64, and direct multiplication gives

$$\sum_{j=0}^3 M(v_j)Y_j = \begin{pmatrix} 334 & -278 \\ 201 & -161 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 89 & 12 \end{pmatrix} + \begin{pmatrix} -360 & 254 \\ -256 & 173 \end{pmatrix} + \begin{pmatrix} 26 & 24 \\ -34 & -24 \end{pmatrix} = 0.$$

If every $\text{sym}(F^T M(v_j))$ were PSD, pairing with the Y_j and summing would give zero. Positive definiteness forces every PSD matrix to be zero. The four affine-independent $M(v_j)$ span all 2×2 matrices, and $\text{sym}(F^T M) = 0$ for all matrices forces $F = 0$. Thus even a nonzero singular F is excluded. If orbit cut values approached one, a Frobenius-normalized parameter sequence would converge to a nonzero matrix satisfying these closed conditions. This contradiction proves the strict supremum gap. $\square$

Proof of proposition 3.10. Use the affine shear $X = x - a$, $Y = y - b$, $W = w - (bx + ay - ab)$. Then $q = W - XY$ and the wedge is $X \geq 0, Y \leq 0, W \geq 0$, whose interior is free. If an outside point has $X < 0$, mix it with an interior wedge point $(\epsilon, -T, 1)$, where ϵ is small and T large. The resulting interior point of the convex enlargement has XY arbitrarily large positive while W stays bounded, hence $q < 0$. For $Y > 0$ use $(T, -\epsilon, 1)$. The remaining outside points have $X \geq 0, Y \leq 0$ and $W < 0$. Mix a weight ϵ of such a point with $(\epsilon^2, -\epsilon^2, \epsilon^2)$. Its W coordinate is negative of order ϵ , whereas XY is of order ϵ^2 , again giving $q < 0$ in the enlargement's interior. This proves maximality.

At any contact $q = 0$, a completion lowering must be zero. Two distinct contacts with the same x have $M_i = ab_i^T$ with the same nonzero a and independent b_i . The rank-one membership identity would force $F^T a \in \mathbb{R}_+ b_1 \cap \mathbb{R}_+ b_2 = \{0\}$, impossible for invertible F . The same applies with y fixed. The wedge contains such contacts, so neither family contains it. $\square$

B Proofs and exact certificates for depth bounds

B.1 Orbit equations and vertical completion

For $G = \begin{pmatrix} g_{11} & g_{12} \\ g_{21} & g_{22} \end{pmatrix}$ with positive determinant, a direct 2×2 identity gives

$$C_G = \left\{ s : \det G q(s) \geq \frac{(g_{11}x - g_{22}y - g_{21}w + g_{12})^2}{4}, \quad g_{11}w + g_{12}y + g_{21}x + g_{22} \geq 0 \right\}. \quad (26)$$

Indeed $\det \text{sym } A = \det A - (A_{12} - A_{21})^2/4$, and a symmetric 2×2 matrix is PSD if and only if its determinant and trace are nonnegative. This describes a trace-selected branch of the scalar quadratic region. It is not a classification by connected components: two branches can meet at an apex.

Lemma B.1 (Exact completion membership). *The sum $B_G = C_G + \mathbb{R}_+ e_w$ is closed. A point s belongs to it if and only if $q(s) \geq 0$ and*

$$\text{sym}(GM(s - \tau e_w)) \succeq 0 \quad \text{for some } 0 \leq \tau \leq q(s). \quad (27)$$

Proof. Equation (26) implies $q \geq 0$ on C_G . If $s = c + \tau e_w$, then $q(c) = q(s) - \tau \geq 0$, giving the stated bound on τ . Conversely a lowered point proves membership. If $s_n = c_n + \tau_n e_w \rightarrow s$, then $0 \leq \tau_n \leq q(s_n)$ is bounded. A subsequence converges to $\tau \geq 0$, and the closed PSD inequality gives $s - \tau e_w \in C_G$. Thus the sum is closed. $\square$

B.2 Analytic collapse of the completed family

We prove the final assertion of proposition 4.4. Suppose it fails. Then, after reducing a positive lower bound if necessary, there are $0 < z < 1$, $\varepsilon_n \downarrow 0$, and admissible completed sets B_{G_n} containing

$$T_n = \text{conv}\{\bar{s}_n, \bar{s}_n + zp_1, \bar{s}_n + zp_2, \bar{s}_n + zp_3\}, \quad \bar{s}_n = (0, 0, \varepsilon_n).$$

Normalize $\|G_n\|_F = 1$ and take a subsequence with $G_n \rightarrow G \neq 0$. The determinant of G is nonnegative. By lemma B.1, each vertex has a lowered point in C_{G_n} with lowering between zero and its q value. Pass to limits of these four bounded lowerings. The limiting lowered points are

$$\ell_0 = 0, \quad \ell_1 = (z, -z, -\tau_1), \quad \ell_2 = (z, -2z, -\tau_2), \quad \ell_3 = (z, z, z - \tau_3),$$

where $0 \leq \tau_1 \leq z^2$, $0 \leq \tau_2 \leq 2z^2$, and $0 \leq \tau_3 \leq z - z^2$.

First suppose $\det G > 0$. The limiting completed set B_G contains the full-dimensional tetrahedron $T_0 = \text{conv}\{0, zp_1, zp_2, zp_3\}$. It is S -free and contains zero. It must lie in $w \geq 0$. Otherwise a point $b \in B_G$ with $b_w < 0$ gives $q(\theta b) < 0$ for sufficiently small positive θ . Taking a convex combination of θb with an interior point of T_0 , with weight sufficiently close to one on θb , produces an interior point of B_G in $\{q < 0\}$, a contradiction. Consequently $C_G \subseteq B_G$ also lies in $w \geq 0$. A point of B_G with $w = 0$ cannot have been lowered by a positive amount, and therefore belongs to C_G .

In particular $0, zp_1, zp_2 \in C_G$. At zero, PSD membership forces the second column of G to be a positive multiple of e_2 . After positive scaling, $G = \begin{pmatrix} \alpha & 0 \\ \beta & 1 \end{pmatrix}$ with $\alpha > 0$. At $w = 0$, the first diagonal entry of $\text{sym}(GM)$ is zero, so PSD membership forces its off-diagonal entry $(\alpha x + y)/2$ to vanish. The two horizontal vertices require both $\alpha = 1$ and $\alpha = 2$, a contradiction.

Now suppose $\det G = 0$. Write $G = ab^T$ with $a, b \neq 0$. The PSD condition for a rank-one product gives $M(\ell_i)^T b \in \mathbb{R}_+ a$. In particular all ℓ_i lie in a nontrivial affine plane. The equation is nontrivial because the condition that $M(s)^T b$ be parallel to a fixed nonzero a cannot hold identically in the three coordinates unless $b = 0$. Their planar projections are not collinear, so this plane has the form $w = Ax + By$. Let

$$U = \text{conv}\{\ell_i : 0 \leq i \leq 3\} + \mathbb{R}_+ e_w.$$

This full-dimensional set is S -free. To see the limiting step explicitly, the corresponding lowered points before taking limits generate subsets $U_n \subseteq B_{G_n}$. A point of $\text{int } U$ has an interior planar projection and positive slack above its lower plane. Expressing its projection with strictly positive convex weights on the limiting vertices, then perturbing those weights and using its positive vertical slack, shows that a neighborhood of the point belongs to U_n for all sufficiently large n . If that point also had $q < 0$, B_{G_n} would fail to be S -free. Thus $Ax + By \geq xy$ on the projection

$$\Delta = \text{conv}\{(0, 0), (z, -z), (z, -2z), (z, z)\}.$$

Interior projections first give this inequality, and continuity gives it on the boundary. Near zero along $(1, -1)$ and $(1, -2)$ it yields $A - B \geq 0$ and $A - 2B \geq 0$. The two lowered horizontal vertices give $A - B = -\tau_1/z \leq 0$ and $A - 2B = -\tau_2/z \leq 0$. Hence $A = B = 0$. Then ℓ_3 lying in the plane requires $\tau_3 = z$, contrary to $\tau_3 \leq z - z^2$. Both cases are impossible. This proves vanishing.

B.3 The finite rational certificate for the upper constants

The upper bounds in [theorem 4.5](#) have one common certificate scheme. In the normalized frame, the vertex is $s_0 = (0, 0, 1)$. For the first family, put

$$P_1 = (\rho, -\rho, 1), \quad P_2 = (\rho, -2\rho, 1), \quad P_3 = (\rho, \rho, 1 + \rho/\sqrt{\varepsilon}).$$

These are the vertices of T_r when $r = \rho\sqrt{\varepsilon}/z_0$. For [equation \(16\)](#), put

$$P_1 = (\rho, -\rho, 1 - 2\rho(1 - \eta)), \quad P_2 = (\rho, -k\rho, 1 - 2\sqrt{k}\rho(1 - \eta)), \quad P_3 = (\rho, \rho, 1 + \rho L),$$

with $r = \rho/z_K$. All test points used below have nonnegative q . Let m_{01}, m_{02}, m_{12} be the three midpoints of s_0, P_1, P_2 . If a completed orbit set contains T_r , it must contain these six test points and its generator must intersect the line $\{(\rho, \rho, h) : h \in \mathbb{R}\}$. These necessary conditions involve neither ε nor L . They apply even when the generating orbit set does not contain s_0 .

For a rational vector v , a point exclusion consists of

$$v^T GM(s)v < 0, \quad v^T GM(s - q(s)e_w)v < 0. \tag{28}$$

The same quadratic form is then negative throughout the allowable lowering interval by affinity, so [lemma B.1](#) excludes s . For the vertical line, its PSD matrix has second diagonal entry $\rho g_{21} + g_{22}$. Its determinant is

$$\det G(h - \rho^2) - \frac{(c - g_{21}h)^2}{4}, \quad c = \rho(g_{11} - g_{22}) + g_{12}.$$

If $g_{21} \neq 0$, completing the square gives its maximum over h as

$$\frac{\det G}{g_{21}^2} \Psi(G), \quad \Psi(G) = \rho g_{21}(g_{11} - g_{22}) + g_{11}g_{22} - \rho^2 g_{21}^2. \quad (29)$$

If $g_{21} = 0$, $\Psi(G) = \det G > 0$. Thus either $\rho g_{21} + g_{22} < 0$ or $\Psi(G) < 0$ excludes the line whenever $\det G > 0$.

Normalize the largest absolute matrix entry to one by positive scaling. This covers every nonzero G by the eight facets $g_{ij} = \pm 1$, with the other three entries in $[-1, 1]$. Each facet is recursively bisected in a free coordinate with longest interval, resolving ties by row-major order. A binary path records the successive lower or upper half intervals. Each leaf has one of the following exact exclusions:

1. $\det G < 0$ at every vertex of the box;
2. a test point and rational vector satisfying both linear inequalities in [equation \(28\)](#) throughout the box;
3. an exact interval upper bound on $\Psi(G)$ that is negative;
4. the upper bound of $\rho g_{21} + g_{22}$ that is negative.

The first test is valid by multilinearity of the determinant in the free entries. The second and fourth are linear box bounds. For the third, write the intervals of g_{11}, g_{21}, g_{22} as I_a, I_c, I_d . Let $\max(II')$ and $\min(II')$ denote the maximum and minimum of the four endpoint products. An upper bound is

$$\rho \max(I_c I_a) - \rho \min(I_c I_d) + \max(I_a I_d) - \rho^2 \min_{c \in I_c} c^2,$$

where the last minimum is zero if $0 \in I_c$ and is the smaller squared endpoint otherwise. All arithmetic is rational.

The complete leaf lists supplied with the evidence companion have the counts in the table. In each facet their paths are prefix-free and satisfy $\sum_{\text{leaves } p} 2^{-|p|} = 1$. A finite prefix-free family gives disjoint box interiors; the exact sum gives coverage up to boundary, and the closed boxes cover the boundary too. All exclusions are strict and hold on those closed boxes. Thus every admissible normalized matrix is excluded from at least one necessary condition. No completed set contains T_r at the listed threshold; monotonicity of T_r then gives the stated supremum upper bound.

Family and ρ	Leaves	Point	Ψ	Det.	Diag.
First family, 137	28699	8664	19944	54	37
$\eta = 1/1000, k = 4, 197/200$	11132	10593	451	74	14
$\eta = 1/100, k = 4, 5/4$	1882	1690	150	27	15
$\eta = 1/1000, k = 9/4, 21/20$	10705	10300	267	76	62
$\eta = 1/1000, k = 49/25, 11/10$	11943	11499	245	93	106

This is a computer-assisted proof with exact certificates. The numerical search used to discover the leaves is not part of the validity argument. The companion includes the rational records and a verifier; an independent verification also checked exact facet volumes, nonoverlap, and the exact maximum of Ψ on each relevant box. The hypotheses and exclusions above are the complete mathematical reduction.

B.4 Printed rational lower witnesses

The lower bounds use the following matrices G , read as exact terminating decimals, and rational heights h_0 . Set $s_0 = (0, 0, 1)$ and use the points P_1, P_2 defined in [appendix B.3](#) at the stated ρ .

Family	ρ	G and h_0
First family	1367/10	$\begin{pmatrix} 1 & -1.678587 \\ 0.007315 & 0.698332 \end{pmatrix}, h_0 = 5417132036/169459$
$\eta = 1/1000, k = 4$	2461/2500	$\begin{pmatrix} 1.00000003 & -2.11841757 \\ 1.01581925 & 0.30393078 \end{pmatrix}, h_0 = 1240951/370546$
$\eta = 1/100, k = 4$	6117/5000	$\begin{pmatrix} 1 & -2.08684938 \\ 0.81738698 & 0.4028837 \end{pmatrix}, h_0 = 4011447/862162$
$\eta = 1/1000, k = 9/4$	5207/5000	$\begin{pmatrix} 1 & -2.43916573 \\ 0.96020732 & 0.5468295 \end{pmatrix}, h_0 = 4025363/954377$
$\eta = 1/1000, k = 49/25$	2693/2500	$\begin{pmatrix} 1 & -2.51367434 \\ 0.9282616 & 0.62838339 \end{pmatrix}, h_0 = 412614/89749$

The complete verification is four 2×2 rational matrix checks per row:

$$\text{sym } G \succ 0, \quad \text{sym}(GM(P_1)) \succeq 0, \quad \text{sym}(GM(P_2)) \succeq 0, \quad \text{sym}(GM(\rho, \rho, h_0)) \succeq 0. \quad (30)$$

For reproducibility without a matrix package, if $G = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $s = (x, y, w)$, the three independent entries to test are

$$A = aw + by, \quad B = (ax + b + cw + dy)/2, \quad E = cx + d.$$

PSD means $A, E \geq 0$ and $AE - B^2 \geq 0$; PD means $A > 0$ and $AE - B^2 > 0$. Substitution of each printed row gives these signs exactly. The finite exact arithmetic records in the companion contain the results. Positive definiteness at s_0 also implies $\det G > 0$.

Upward completion contains (ρ, ρ, h) for every $h \geq h_0$. In the first family, $1 + \rho/\sqrt{\varepsilon} \geq h_0$ precisely when $\varepsilon \leq (\rho/(h_0 - 1))^2$, which is the exact ε_1 in [theorem 4.5](#). In the four tangent families, the individual thresholds $L \geq (h_0 - 1)/\rho$ are

$$\frac{1088006250}{455956853}, \quad \frac{7873212500}{2636922477}, \quad \frac{15354930000}{4969441039}, \quad \frac{807162500}{241694057},$$

all below $17/5$. The completed set then contains every vertex of T_r and hence T_r . Its cut ratio is at least r , yielding every listed lower bound. No optimization or assertion of optimality is needed for these witnesses.

B.5 A dual certificate for an orbit upper bound

A second exact certificate illustrates the dual PSD obstruction and retains an upper bound independent of the completed-set calculation. On the first family in [proposition 4.4](#), it gives

$$z_A/z_K \leq 160\sqrt{\varepsilon}/z_0 \quad (0 < \varepsilon \leq 1/9765625).$$

The completed-family bound in [theorem 4.5](#) is stronger.

Set $\rho = 160$, $u = \sqrt{\varepsilon}/160$, and

$$M_1 = \begin{pmatrix} 1 & 160 \\ -160 & 1 \end{pmatrix}, \quad M_2 = \begin{pmatrix} 1 & 160 \\ -320 & 1 \end{pmatrix}, \quad M_3 = \begin{pmatrix} 1 + 1/u & 160 \\ 160 & 1 \end{pmatrix}.$$

Take

$$Y_1 = \begin{pmatrix} 979/10^6 & -5679/500000 \\ -5679/500000 & 26393/200000 \end{pmatrix}, \quad Y_2 = \begin{pmatrix} 131/250000 & 859/100000 \\ 859/100000 & 35253/250000 \end{pmatrix},$$

and

$$n = 1/10, \quad \vartheta = 140667062516/1953125, \quad \sigma_\infty = -187502/3125,$$

$$\sigma(u) = \sigma_\infty + 160\vartheta u^2, \quad Y_3(u) = \begin{pmatrix} \vartheta u^2 & \sigma(u)u \\ \sigma(u)u & n \end{pmatrix}.$$

Put $Y_0(u) = -(M_1Y_1 + M_2Y_2 + M_3Y_3(u))$. The sum is symmetric identically. The fixed matrices Y_1, Y_2 are PD by their positive first entries and determinants. The remaining signs can be checked without root counting. For a polynomial $p(u) = \sum_k p_k u^k$ and $0 \leq u \leq u_* = 1/500000$, use

$$p(u) \geq p_0 + \sum_{k \geq 1: p_k < 0} p_k u_*^k.$$

Direct expansion of the printed matrices gives the following exact lower bounds on this right-hand expression:

Polynomial	Lower bound
$\vartheta n - \sigma(u)^2$	$2096684649823859428966288734234359/582076609134674072265625000000 > 3600$
$Y_0(u)_{11}$	$24149614711567794247/76293945312500000000 > 3/10$
$\det Y_0(u)$	$176579577419584483919878545141452997/2328306436538696289062500000000000000 > 3/40$

Thus $Y_3(u), Y_0(u) \succ 0$ for $0 < u \leq u_*$. At $u = 0$ the limiting matrix is

$$Y_0(0) = \begin{pmatrix} 441377/10^6 & 20443/62500 \\ 20443/62500 & 558543/10^6 \end{pmatrix}.$$

An admissible G has $\text{sym } G \succ 0$ at the normalized vertex. If it contained all three endpoint matrices, the trace identity

$$0 = \text{tr} \left(G \left(Y_0 + \sum_{i=1}^3 M_i Y_i \right) \right) = \langle \text{sym } G, Y_0 \rangle + \sum_{i=1}^3 \langle \text{sym}(GM_i), Y_i \rangle$$

would have a strictly positive first term and nonnegative remaining terms. This is impossible. The claimed upper bound follows from the printed rational certificate and these elementary polynomial sign bounds.

B.6 Derivation and sharpness of the fixed rule

In Sylvester coordinates,

$$M = x_1 I + x_2 J + y_1 \text{diag}(1, -1) + y_2 \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad (x_1, x_2, y_1, y_2) = \frac{1}{2}(w+1, x-y, w-1, x+y).$$

Let $\sin \theta = (\bar{x} - \bar{y})/N_0$ and $\cos \theta = (\bar{w} + 1)/N_0$. Multiplication by

$$R_\theta = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}$$

rotates the positive coordinates so that $\text{sym}(R_\theta M) \succeq 0$ is the chosen inequality $\lambda^T x \geq \|y\|$. Its determinant is one and its skew affine form is

$$\ell = \cos \theta(x - y) - \sin \theta(w + 1) = -V(s - \bar{s})/N_0.$$

Equation (26) now gives equation (17). Because the vertex has $q_0 > 0$ and $\ell(\bar{s}) = 0$ with positive trace, it is positive definite. Along a ray, the first determinant zero is the uncompleted exit. The completed exit is instead determined by equation (27). For example, at $\bar{s} = (1, 0, 1)$ on $p = e_w$, the uncompleted determinant equation is $20(1 + s) = s^2$, giving $s = 10 + \sqrt{120}$. The completion contains the entire upward ray.

On the first sharp family, $\bar{x} = \bar{y} = 0$ makes the fixed set the upward-closed cylinder $w \geq (x + y)^2/4$. Its ray steps are $\infty, 2\sqrt{\varepsilon}, z_0$. Therefore $z_{\text{fix}} = \min\{2\sqrt{\varepsilon}, z_0\}$, with $2\sqrt{\varepsilon} \leq z_0$ exactly when $\varepsilon \leq 4/9$.

Proposition B.2 (A pure rescaling obstruction). *Let $L \geq 1$, $\bar{s} = (0, 0, 1)$, $p_1 = (L, -1/L, 0)$, $p_2 = (L, 1/L, 0)$, $p_3 = e_w$, and unit costs. Then $z_K = 1$, $D = 1$, the discriminants are $-1, +1, +1$, and*

$$\text{cond}(P) = \begin{cases} \sqrt{2}L, & 1 \leq L \leq \sqrt{2}, \\ L^2, & L \geq \sqrt{2}. \end{cases}$$

The orbit cylinder with $t = 1/L$ is exact, but $z_{\text{fix}} = 2/(L + 1/L)$.

Proof. Expansion gives $q = 1 + \lambda_3 + \lambda_1^2 - \lambda_2^2$, so objective one is feasible only on ray two, with transversal contact. The cylinder $q \geq (x/L - Ly)^2/4$ contains T^* and has the vertex inside. The fixed cylinder $w \geq (x + y)^2/4$ exits on ray two at $2/(L + 1/L)$, no later than on ray one. Its completion is the same cylinder. The rows of P are orthogonal with norms $\sqrt{2}L, \sqrt{2}/L, 1$, giving the stated condition number. The family is the image of $L = 1$ under $(x, y, w) \mapsto (Lx, y/L, w)$. $\square$

B.6.1 Proof of the prescribed condition number and depth example

For the construction in proposition 4.7, expansion gives

$$q(\bar{s} + P\lambda) = v^2(1 - \lambda_3) + uvD(\lambda_1 + \lambda_2) + v^2D^2\lambda_1\lambda_2.$$

Thus $z_K = 1$, uniquely on ray three. The restrictions on the first two rays are $v^2 + uvDs$ and on the third $v^2(1 - s)$, proving the extreme relative discriminants. Also $q_0 = v^2$, $\tilde{X} = \tilde{Y} = vD$, and the singular values are vD, vD, v^2 . Since $v = \kappa D$, their ratio is exactly κ .

Translate the contact point $t^* = (u, -u, -u^2)$ to zero by

$$(x', y', w') = (x - u, y + u, w + ux - uy - u^2).$$

The other vertices become $(0, 0, v^2)$, $(vD, 0, v^2 + uvD)$, and $(0, -vD, v^2 + uvD)$. Use equation (18) with $\alpha = 1$ and $\beta = D/(uD + v)$. At the vertex its determinant slack is $v^2(4 - \beta^2v^2) > 0$, since $\beta v < 1$. At the two other vertices the squared affine form vanishes, and their slack is $4(v^2 + uvD) > 0$. At the contact the slack is zero. The trace branch $w' + \beta x' + 1$ is positive at all four vertices. This orbit set therefore contains T^* with $\bar{s}$ inside, proving attained exactness for both families.

For the fixed rule, $\bar{x} - \bar{y} = 2u$ and $\bar{w} + 1 = 0$, so $C_{\text{fix}} = \{4q \geq (w + 1)^2, x \geq y\}$. The first two rays stay inside forever. On ray three the endpoint solves $4v^2(1 - s) = v^4s^2$, giving $s = 2/(u + 1)$. The vertical section at $(u, -u)$ is $w \in [1 - 2u, 1 + 2u]$. Its upward completion is $w \geq 1 - 2u$, so the downward ray has exactly the same step. This proves the claimed completed and uncompleted values.

The two-variable fixed-rule example in proposition 3.3 does not settle the analogous margin or conditioning question for Case 2. Its horizontal ray restriction is $(x_0 - s)^2 + 1$, with relative discriminant $-4/(8x_0^2 + 4) \rightarrow 0$. Its two scaled ray lengths are z_K/ϵ and z_K , so their condition number is $1/\epsilon$. Both proposed safeguards therefore fail along its vanishing-ratio sequence.

B.7 Full details of the contact approximation

Let $A_0 = \text{sym}(GM(\bar{s}))$ and $A_i = \text{sym}(GM(v_i))$ be PSD. For fixed $0 < r < 1$, put $A_i(r) = (1-r)A_0 + rA_i$. If $z \in \ker A_i(r)$, its quadratic form is a positive combination of two nonnegative quadratic forms, so both vanish. For a PSD matrix A , $z^T A z = 0$ implies $Az = 0$, by diagonalizing A . Hence

$$\ker A_i(r) = \ker A_0 \cap \ker A_i.$$

Such a vector is annihilated by the symmetric part at every point of the segment from $\bar{s}$ to v_i . On the shortened segment to $v_i(r)$, the objective level is below z_K , so $q > 0$ throughout. Write $GM(s) = A(s) + k(s)J$. If the common kernel is nonzero, then $GM(s)z = k(s)Jz$ for a nonzero z . The left side cannot vanish, since $\det G > 0$ and $\det M(s) = q(s) > 0$. Thus $k(s) \neq 0$ and its continuous sign is constant. Comparing the identities at the two shortened endpoints gives

$$M(v_i(r))z = \frac{k(v_i(r))}{k(\bar{s})}M(\bar{s})z.$$

Consequently the symmetric perturbation $H_i = \text{sym}(M(\bar{s})^{-1}M(v_i(r)))$ satisfies

$$z^T H_i z = \frac{k(v_i(r))}{k(\bar{s})} \|z\|^2 > 0 \quad \text{on } \ker A_i(r) \setminus \{0\}.$$

Here is the perturbation argument without a limiting eigenvalue assertion. In an orthogonal decomposition into the positive eigenspace and kernel of $A_i(r)$, its perturbation has blocks

$$\begin{pmatrix} A_+ + \epsilon H_{++} & \epsilon H_{+0} \\ \epsilon H_{0+} & \epsilon H_{00} \end{pmatrix}.$$

The first block remains positive definite for small ϵ . Its Schur complement is

$$\epsilon H_{00} - \epsilon^2 H_{0+} (A_+ + \epsilon H_{++})^{-1} H_{+0},$$

which is positive definite for all sufficiently small positive ϵ because $H_{00} \succ 0$ and the inverse is bounded. If there is no kernel, ordinary continuity suffices. If the whole matrix is zero, the perturbation itself is positive definite by the displayed kernel inequality. There are finitely many vertices, so the minimum of their positive admissible epsilon thresholds is positive. Also $\det(G + \epsilon M(\bar{s})^{-1}) > 0$ for small ϵ . At $\bar{s}$ the symmetric matrix is $A_0 + \epsilon I$. These facts prove [lemma 4.8](#) completely.

For completeness, the singular-limit necessity in [theorem 4.9](#) uses uniqueness rather than dimension. Normalize a sequence with cut ratios tending to one to Frobenius norm one. Closed PSD inequalities pass to a nonzero limit G on every vertex of T^* , and $\det G \geq 0$. If $\det G = 0$, membership at zero gives $G = \begin{pmatrix} a & 0 \\ c & d \end{pmatrix}$ with $d \geq 0$, $ad = 0$. If $d > 0$, the PSD matrix has zero first diagonal entry, so its off-diagonal entry $(ch + dy)/2$ vanishes. Hence $y = -(c/d)h$. Set $\beta = c/d$ and take

$$0 < \alpha < \min_{v \neq 0, x_v \neq 0} \frac{4q(v)}{x_v^2},$$

with an empty minimum interpreted as infinity. Its determinant slack is strictly positive at every noncontact vertex. If $d = 0$, PSD membership at $\bar{s}$, whose height is positive, forces $a \geq 0$. If $a = 0$, the zero first diagonal entry would force $ch_{\bar{s}} = 0$, and then $G = 0$, impossible. Thus $a > 0$. The determinant of its symmetric matrix is $-(ax - ch)^2/4$, so $x = (c/a)h$ at every vertex. Choose

$$\alpha > \max_{v \neq 0} \frac{y_v^2}{4q(v)}, \quad \beta = \alpha c/a.$$

Again the determinant slack is strict. In both cases $q(v) > 0$ at every noncontact vertex follows from uniqueness, and finite cardinality gives a single parameter. Their first diagonal entries αh_v are positive, so the strict determinant slack implies PD membership. The contact itself has PSD matrix $e_2 e_2^T$. Thus a singular limit actually leads to an attaining set; it cannot obstruct closed-interval feasibility.

B.8 The explicit threshold counterexample

At $d = 0$, write a, b, c for the barycentric coordinates on the face opposite the contact, in the order $\bar{s}, v_2, v_3$. Substituting $b = 1 - a - c$ gives

$$q_0 = 2a^2 + \frac{11}{2}ac - \frac{5}{4}a + \frac{5}{4}c^2 - \frac{1}{2}c + \frac{1}{4}.$$

Its Hessian has determinant $-81/4 < 0$, so an interior stationary point cannot minimize it. On $c = 0$, the polynomial $2a^2 - 5a/4 + 1/4$ has minimum $7/128$ at $a = 5/16$. On $a = 0$ the minimum of $5c^2/4 - c/2 + 1/4$ is $1/5$. On $b = 0$ the expression is $1 + 9a(1 - a)/4$, with minimum one. This proves the face bound used in [proposition 4.11](#). The determinant of the ray matrix is $-3(3d + 4)/2$, so these instances also have full-dimensional optimal simplices.

It remains to exclude closed interval intersections. Put $\alpha = t^2$. The three vertex q values are $1 - d, 1/4 - d, 1 - d$. Let L_0, L_3 be the left endpoints of the intervals for $\bar{s}, v_3$, and R_2 the right endpoint for v_2 :

$$\begin{aligned} L_0(t) &= \frac{3t^2/2 + 1/2 - 2t\sqrt{1-d}}{1/4 - d}, \\ R_2(t) &= \frac{t^2/2 - 3/2 + 2t\sqrt{1/4-d}}{1-d}, \\ L_3(t) &= \frac{3t^2 - 1 - 2t\sqrt{1-d}}{4-d}. \end{aligned}$$

For $0 < d \leq 1/8$, $1 - \sqrt{1-d} \geq d/2$ and $\sqrt{1/4-d} \leq (1-2d)/2$. If $t \geq 1$, the numerator of L_0 is at least $(3t-1)(t-1)/2 + td \geq 0$. Hence

$$L_0 \geq 2(3t-1)(t-1) + 4td, \quad R_2 \leq \frac{4}{7}(t+3)(t-1),$$

where the second bound discards a negative term and uses $1/(1-d) \leq 8/7$. Thus

$$L_0 - R_2 \geq (t-1)(38t-26)/7 + 4dt > 0.$$

If $0 < t \leq 1$, split the numerator of L_3 as $(3t+1)(t-1) + 2t(1-\sqrt{1-d})$. The first summand is nonpositive and the second nonnegative, so

$$L_3 \geq \frac{8}{31}(3t+1)(t-1) + td/4.$$

The numerator of R_2 is at most $(t+3)(t-1)/2 - 2td \leq 0$, giving $R_2 \leq (t+3)(t-1)/2 - 2td$. Therefore

$$L_3 - R_2 \geq (1-t)(77-17t)/62 + (9/4)dt > 0.$$

In each case two closed intervals are disjoint. This proves the strict orbit gap throughout $0 < d < 7/128$. Taking d arbitrarily small proves the sharp cylinder threshold analytically; finite checks at three values are not needed for that conclusion.

B.9 Saved numerical calibration and its exact correction

The exact bounds above distinguish certification from numerical set optimization. On the fixed-ray family, normalized numerical optimization of the uncompleted family suggests a leading coefficient about 56.7 in $\sqrt{\varepsilon}/z_0$. The completed-family search found about 136.707. Only the rational interval $[136.7, 137]$ is used as a proved completed-family coefficient in the stated small- ε range. Similarly, the construction that lowers the third vertex proves a comparison with the best finite-height uncompleted set, but does not prove that all optimal completed sets arise this way. Each printed lower witness has its generating orbit set containing the vertex in its interior. On the first family, restricting a search to such generators therefore loses at most $137/(1367/10) - 1 < 0.22\%$ in the stated small- ε range. This estimate is specific to that family.

A retained near-boundary corner also illustrates the need to normalize before treating numerical infeasibility as evidence. Its initial original-coordinate optimization underestimated the orbit ratio. Rational primal and dual certificates instead give

$$\frac{763}{25000} = 0.03052 \leq \frac{z_A}{z_K} \leq \frac{191}{6250} = 0.03056.$$

The retained verification records fix the input as the exact binary values of the stored floating-point data. They report a rational primal check by PSD membership with the vertex PD. The reported dual check uses the trace identity in [appendix B.5](#), with PSD endpoint matrices and a PD vertex matrix. The objective normalization uses the exact corner bound of that specified input. The record reports an exact bracket, rather than a numerical solver bound. Its full rational primal and dual matrices were not archived; the companion retains the checking source and its successful exact-arithmetic receipt. This historical correction has that reproducibility limitation and is not used in any theorem above.

C Closure proofs and exact certificates

C.1 Parameters and coefficient certificates

Put

$$M_b = M(\bar{s}), \quad \Xi = F^T M_b, \quad \Sigma = \text{sym}(\Xi), \quad N_j = M_b^{-1} M_0(p_j), \quad m = M_b^{-1} e_1,$$

and $E = e_1 e_1^T$, $N_E = M_b^{-1} E$, $Z = \text{sym}(\Xi N_E)$. Then

$$A_F(\bar{s} + \mu p_j) = \Sigma + \mu \text{sym}(\Xi N_j), \quad v^T Z v = (v^T \Xi m) v_1.$$

A vector v is *kept* if $v^T Z v \geq 0$. Every s in the upward completion B_F satisfies $v^T A_F(s) v \geq 0$ for every kept v : this follows first for $s = c + t e_w$, $c \in C_F$, $t \geq 0$, and then by continuity. All certificate arguments below use this necessary condition.

Lemma C.1 (Transformed point-rule parameters). *The uncompleted point-rule sets under all determinant-preserving homogeneous automorphisms are exactly those with $\Xi = \Sigma \succ 0$ symmetric.*

Proof. In coordinates where the chosen point-rule direction is e_1 , the rule requires the image matrix to be symmetric with positive trace. For an automorphism $M \mapsto A M B^T$, $\det A \det B = 1$, the preimage of the positive-semidefinite symmetric-part cone is C_F with $F^T = B^{-1} A$. At the vertex, the rule gives $Y = A M_b B^T$ symmetric with positive trace. Since $\det Y = \det M_b > 0$, it is positive definite. Therefore $F^T M_b = B^{-1} Y B^{-T} \succ 0$ is symmetric. Conversely, given $\Sigma \succ 0$, set $F^T = \Sigma M_b^{-1}$. Its determinant is positive. Choose $B = t I$, $A = t F^T$, with $t^4 \det F = 1$; then $\det A \det B = 1$ and the image of the vertex is $t^2 \Sigma \succ 0$. Transposition fixes the base symmetric-part cone and gives no additional parameter family. $\square$

Lemma C.2 (A parameter half-space for completions). *For $\det F > 0$ and $\bar{s} \in \text{int } B_F$, set $u = (1, -\bar{y})^T$. Then $u^T \Sigma u > 0$.*

Proof. The affine image of $s \mapsto \text{sym}(F^T M(s))$ has rank three or two. In the rank-two case a nonzero kernel direction obeys $M_0(d) = tF^{-T}J$, where $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, so $(F^{-T}J)_{22} = 0$. Its image plane cannot contain the apex zero: $\text{sym}(F^T M(s)) = 0$ would imply $M(s) = t'F^{-T}J$, contradicting $M(s)_{22} = 1$. A plane meeting the positive-semidefinite cone but missing its apex cannot support that cone; hence it meets its positive definite interior. Thus $C_F \neq \emptyset$ implies $\text{int } C_F \neq \emptyset$. This also follows from the explicit completion classification in [proposition 3.5](#).

Interiors of convex sums now give a $\tau > 0$ with $s_\tau = \bar{s} - \tau e_w \in \text{int } C_F$. Let $\Xi_\tau = F^T M(s_\tau)$ and $q_\tau = q(\bar{s}) - \tau$. Then $\text{sym}(\Xi_\tau) \succ 0$, so $q_\tau > 0$, and

$$\Xi = \Xi_\tau + \frac{\tau}{q_\tau} \Xi_\tau u e_1^T.$$

As $e_1^T u = 1$, this gives $u^T \Sigma u = (1 + \tau/q_\tau) u^T \text{sym}(\Xi_\tau) u > 0$. $\square$

Lemma C.3 (A parameter-dependent kept vector). *For a point-rule parameter $\Sigma \succ 0$, let $v = J \Sigma m$ and $L_j(\Sigma) = m^T J N_j J \Sigma m$. Then*

$$v^T Z v = 0, \quad v^T \Sigma v = \det(\Sigma) m^T \Sigma m, \quad -v^T \Sigma N_j v = \det(\Sigma) L_j(\Sigma).$$

Consequently $a_j(B_F) \geq L_j(\Sigma)/(m^T \Sigma m)$ whenever that ratio is positive. Each L_j is linear in Σ .

Proof. The first identity is $(J \Sigma m)^T \Sigma m = 0$. The other two follow from $J^T \Sigma J = \text{adj}(\Sigma)$, $\Sigma \text{adj}(\Sigma) \Sigma = \det(\Sigma) \Sigma$, and $\Sigma J^T \Sigma = -\det(\Sigma) J$. Applying the kept inequality to $\bar{s} + \mu p_j \in B_F$ bounds its step by $m^T \Sigma m / L_j(\Sigma)$. $\square$

Lemma C.4 (Finite box certificate). *Let $\hat{\lambda} \geq 0$. Suppose a parameter map $\Xi(t)$ is affine on a box. For distinct selected rays, choose vectors v_j such that at every box vertex*

$$v_j^T \Xi N_E v_j \geq 0, \quad n_j = -v_j^T \Xi N_j v_j > 0.$$

Set $\mu_j = \max_{t \text{ vertex}} v_j^T \Xi(t) v_j / n_j(t)$ and assume $\mu_j > 0$. If $\sum_j \hat{\lambda}_j / \mu_j \geq 1$, then every completion in the box containing $\bar{s}$ satisfies $a^T \hat{\lambda} \geq 1$. For point-rule parameters, a positive lower bound from [lemma C.3](#) may replace $1/\mu_j$. Alternatively, a vector kept at all box vertices and satisfying $v^T \Xi(t) v < 0$ at all those vertices excludes every completion in that box that contains the vertex.

Proof. Keptness and positivity of n_j extend across the box because they are affine in its parameters. A ratio of affine functions with positive denominator lies between its values at the vertices: writing a point as a convex combination weights its vertex ratios by their positive denominators. Thus for $\mu > \mu_j$, $v_j^T A_F(\bar{s} + \mu p_j) v_j < 0$ throughout the box. The kept inequality gives $\alpha_j \leq \mu_j$ and then $a^T \hat{\lambda} \geq \sum_j \hat{\lambda}_j / \mu_j$. Unlisted rays contribute a nonnegative amount, and no ray is counted twice. The exclusion conclusion applies the kept inequality at $\bar{s}$. The parameter-dependent ratio is bounded in the same way using its linear numerator and positive linear denominator. $\square$

Lemma C.5 (A coefficient-simplex certificate). *Let $\Pi \subseteq \mathbb{R}_+^N$ be a polytope and let I_0 index coordinates zero on Π . Suppose symmetric positive-semidefinite matrices Y_j satisfy $Y_j = 0$ for $j \in I_0$, and*

$$R = - \sum_j N_j Y_j \quad \text{is symmetric}, \quad Q(a) = R - \sum_j a_j Y_j \succeq 0, \quad \text{tr } Q(a) > 0$$

at every vertex of Π . No orbit cut vector belongs to $\Pi + \text{cone}\{e_j : j \in I_0\}$.

Proof. The conditions on Q extend to Π by affinity and along its added coordinates because the corresponding Y_j vanish. For an orbit set containing the vertex in its interior, $\Sigma \succ 0$ and $a_j \Sigma + \text{sym}(\Xi N_j) \succeq 0$; when $a_j = 0$ this is the recession condition. Therefore

$$0 \leq \sum_j \langle a_j \Sigma + \text{sym}(\Xi N_j), Y_j \rangle = -\langle \Sigma, R - \sum_j a_j Y_j \rangle < 0,$$

where the equality uses symmetry of R and the strict inequality uses $\Sigma \succ 0$, $Q \succeq 0$, $Q \neq 0$. $\square$

To certify a closure point, subdivide $\{a \geq 0 : \hat{\lambda}^T a \leq 1\}$ into simplices and apply [lemma C.5](#) on each. If $\hat{\lambda}_j = 0$, that coordinate is an unbounded nonnegative direction and its multiplier must vanish. A simplex can also be excluded using $x \in X$ with $a^T x < 1$ at each vertex, provided $x_j = 0$ on every unbounded coordinate. These conditions ensure $a^T x < 1$ throughout that simplex and its added cone.

C.2 Portable finite certificates

The electronic supplement contains the exact rational subdivisions and witnesses in [table 5](#), together with separate exact-arithmetic checkers. These finite records establish the stated membership claims. No numerical search tolerance enters their acceptance conditions. The theorem statements specify the instance independently of the records, and the checkers reconstruct M_b , N_j , N_E , and the parameter charts from the rational instance data.

Table 5: Completed exact closure certificates. File names refer to the electronic supplement.

File	point or conclusion	cut	excluded	skipped
boxcert_thm14t_B.json	$(67/250, 41/100, 151/500) \in \mathcal{C}_B$	8070	2253	0
boxcert_thm14_B.json	$(3/11, 41/99, 10/33) \in \mathcal{C}_B$	2530	725	0
boxcert_thm14t_BP.json	$(51/350, 0, 17/150) \in \mathcal{C}_{B\mathcal{P}}$	420	0	6
boxcert_thm14_BP.json	$(1/5, 0, 1/6) \in \mathcal{C}_{B\mathcal{P}}$	33	0	5
boxcert_thm14w_B.json	$(2211, 8805, 2464)/10000 \in \mathcal{C}_B$	5518	1543	0
closure_cert_thm14_A.json	$(3/11, 41/99, 10/33) \in \mathcal{C}_A$	25	0	0
closure_cert_prop16_A.json	$(24/25, 1/100, 29/1000) \in \mathcal{C}_A$	92	0	0
closure_cert_wcorner_A_A.json	$(20, 20, 20, 0) \in \mathcal{C}_A$	1	0	0

For $\mathcal{B}$, [lemma C.2](#) permits normalization in the seven facets of $[-1, 1]^4$ with first coordinate nonnegative. More explicitly, $U = [u, e_2]$ and $U^T \Xi U = \begin{pmatrix} t_1 & t_2 \\ t_3 & t_4 \end{pmatrix}$, where $t_1 > 0$. Positive scaling puts one row entry at 1 or -1 , with the negative first-entry facet omitted. Each chart is therefore an axis-parallel box of dimension three. The records specify a binary subdivision along a longest side, choosing the lowest index on ties. Reconstructing the subdivision tree establishes coverage of each chart; each terminal box has a cut or exclusion witness from [lemma C.4](#). Charts may overlap, which does not affect coverage or the universal conclusion.

For $\mathcal{BP}$, normalize the symmetric parameter to trace one:

$$\Sigma = \begin{pmatrix} (1+u_1)/2 & u_2/2 \\ u_2/2 & (1-u_1)/2 \end{pmatrix}, \quad u_1^2 + u_2^2 < 1.$$

The saved certificates use a box containing this disk, with affine coordinates $u_1 = t_1 - t_2$, $u_2 = t_1 + t_2 - 1$ and $(t_1, t_2) \in [-1/2, 3/2]^2$. A skipped box must have exact minimum disk polynomial at least one.

This minimum is found by evaluating vertices, edge stationary points, and the interior stationary point. For the three-ray instance, the parameter-dependent forms are

$$m = (2/3, 0)^T, \quad m^T \Sigma m = 4a/9, \quad (L_1, L_2, L_3) = (-8b/3, 8b/3, 10b/9), \quad \Sigma = \begin{pmatrix} a & b \\ b & c \end{pmatrix}.$$

These explicit linear forms suffice for checking its boundary witnesses.

The coefficient-simplex records reconstruct subdivisions of $\text{conv}\{0, e_j/\hat{\lambda}_j : \hat{\lambda}_j > 0\}$ by bisecting a longest edge, with lexicographic tie-breaking. The exact squared lengths use the coordinates with positive $\hat{\lambda}_j$. Every listed piece must occur in the reconstructed tree and every terminal piece must be listed. All three current records use matrix witnesses, with no excluded pieces. The free-coordinate rule in [lemma C.5](#) covers the fourth coordinate in the four-ray certificate.

The point-rule lower bracket in [theorem 5.6](#) has a short explicit witness, independent of the subdivisions:

$$\Sigma = \begin{pmatrix} 1 & -392/621 \\ -392/621 & 47753/99290 \end{pmatrix}, \quad \mu = 2539/10000, \quad (\tau_1, \tau_2, \tau_3) = (0, 0, 227/64).$$

Its determinant is positive, and direct rational principal-minor checks give $\Sigma + \mu \text{sym}(\Sigma N_j) - \tau_j Z \succeq 0$ for each ray. Thus the completed set contains all three points at step μ , proving $z_{\mathcal{BP}}(\mathbf{1}) \geq \mu$. The orbit lower bracket 0.97538 comes from the rational orbit witness in [theorem 3.8](#).

C.3 A three-cut support-one proof

For [proposition 5.7](#), the exact ray matrices are

$$N_1 = \begin{pmatrix} -1 & 1/4 \\ 0 & -3/4 \end{pmatrix}, \quad N_2 = \begin{pmatrix} -47/32 & 1 \\ -19/32 & -3 \end{pmatrix}, \quad N_3 = \begin{pmatrix} -25/16 & 3/8 \\ -13/16 & -9/8 \end{pmatrix}.$$

Take $\Xi_i = \begin{pmatrix} a_i & b_i + k_i \\ b_i - k_i & c_i \end{pmatrix}$ with the three rows of [table 6](#). They are rows 15, 16, and 59, numbered from zero, of the retained sixty-set record. The independent proof below needs only these three rows.

Table 6: Rational orbit parameters for the support-one closure lower bound.

i	a_i	b_i	c_i	k_i
1	7753868/7754071	46303/9049648	203/7754071	-9009394/6985127
2	1381221/3672875	1434795/2962033	2291654/3672875	-4127451/3621740
3	3693704/9822089	3536863/7301600	6128385/9822089	-10687459/9385356

The certified steps are

$$(\mu_{ij}) = \begin{pmatrix} 5557766/5579491 & 564958/1365417 & 15591393/8400091 \\ 1658819/1667061 & 15022732/6553787 & 5144194/6131869 \\ 9857221/9905922 & 19123064/8369713 & 6479877/7726690 \end{pmatrix}.$$

For $\Sigma_i = \text{sym}(\Xi_i)$, direct rational arithmetic gives $(\Sigma_i)_{11} \geq 1/10$ and $\det \Sigma_i \geq 10^{-13}$. Every matrix $H_{ij} = \text{sym}(\Xi_i(I + \mu_{ij}N_j))$ has $(H_{ij})_{11} \geq 10^{-3}$ and $\det H_{ij} \geq 10^{-7}$, except H_{22} and H_{32} , whose determinants are at least 10^{-8} . All these inequalities follow by inserting the displayed fractions

and cross-multiplying positive denominators. Hence $\Sigma_i \succ 0$, $\det F_i > 0$ for $F_i^T = \Xi_i M_b^{-1}$, and $\alpha_j(C_{F_i}) \geq \mu_{ij}$.

Set $G_{ij} = 1/\mu_{ij}$. Its inequalities $G_i \lambda \geq 1$ are valid weakenings of the actual intersection cuts. Every point in the full closure satisfies them. For

$$y = (28554901, 56139688, 14841527)^T / 10^8,$$

the three components of $\mathbf{1} - G^T y$ are

$$\begin{aligned} d_1 &= \frac{219837048696181240067}{9087695197410765423400000000}, \\ d_2 &= \frac{268223412629374939}{12446434832390989600000000}, \\ d_3 &= \frac{3947513071580764783}{157490760258981370980000000}. \end{aligned}$$

They are positive. Therefore $y \geq 0$, $G^T y \leq \mathbf{1}$, and

$$\mathbf{1}^T \lambda \geq y^T G \lambda \geq \mathbf{1}^T y = 24884029/25000000 = 0.99536116 > 0.9953611.$$

This is a complete rational lower-bound proof. The full sixty-set LP record additionally has the exact value

$$\frac{396503562307491652037015395705021177207626376155404352}{398351441316353660966368086274075897893464946699521021} \simeq 0.995361184077.$$

It is the value of that finite relaxation, not an exact evaluation of the infinite-family closure. The upper certificate in [table 5](#) completes the strict nonexactness assertion.

C.4 Point-rule loss at the three-ray corner

At [theorem 5.6](#), $N_1 = \begin{pmatrix} -7 & 7/3 \\ -6 & 0 \end{pmatrix}$. For $\Sigma = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \succ 0$, the kept vector e_1 gives $a_1 \geq 7 + 6b/a$. [Lemma C.3](#) gives $a_1 \geq -6b/a$ whenever that bound is positive; nonnegativity covers the other values. Thus

$$a_1 \geq \max\{7 + 6b/a, -6b/a\} \geq 7/2.$$

The symmetric parameter

$$\Sigma_* = \begin{pmatrix} 1 & -7/12 \\ -7/12 & 7/18 \end{pmatrix}, \quad \det \Sigma_* = 7/144 > 0,$$

satisfies $\text{sym}(\Sigma_*(I + (2/7)N_1)) = 0$. Its uncompleted set contains the ray endpoint at step 2/7; its completion does too. The universal bound proves exact attainment of $\inf a_1 = 7/2$.

To bound the corner value away from zero when ray 1 becomes inexpensive, use the rational orbit parameter

$$\Xi = \begin{pmatrix} a & b+k \\ b-k & c \end{pmatrix}, \quad (a, b, c, k) = \left(\frac{658069}{904895}, -\frac{83794}{200103}, \frac{246826}{904895}, -\frac{385931}{893930} \right).$$

Direct principal-minor checks give

$$\text{sym}(\Xi) \succ 0, \quad \text{sym}(\Xi N_1) \succeq 0, \quad \text{sym}(\Xi) + (367/759) \text{sym}(\Xi N_j) \succeq 0 \quad (j = 2, 3).$$

Thus ray 1 is recessive, and both other steps are at least 367/759. The valid orbit cut implies $\lambda_2 + \lambda_3 \geq 367/759$ for every $\lambda \in X$. This proves all bounds in the first part of [theorem 5.8](#).

C.5 The four-ray corner

For the $\mathcal{A}$ claim use

$$Y_1 = \begin{pmatrix} 141 & -63 \\ -63 & 29 \end{pmatrix}, \quad Y_2 = \begin{pmatrix} 4 & 67 \\ 67 & 1322 \end{pmatrix}, \quad Y_3 = \begin{pmatrix} 31 & -132 \\ -132 & 565 \end{pmatrix}, \quad Y_4 = 0.$$

Their determinants are 120, 799, 91. The matrix $R = -\sum_j N_j Y_j = \begin{pmatrix} 14 & 18 \\ 18 & 85 \end{pmatrix}$ has determinant 866. The matrices $20R - Y_j$, $j = 1, 2, 3$, have first diagonal entries 139, 276, 249 and determinants 53340, 18479, 40551. All are positive definite. [Lemma C.5](#), applied to $\text{conv}\{0, e_1/20, e_2/20, e_3/20\} + \mathbb{R}_+ e_4$, shows that every orbit cut has $a_1 + a_2 + a_3 > 1/20$. Therefore $(20, 20, 20, 0) \in \mathcal{C}_\mathcal{A}$.

There is also a qualitative explanation of this orbit obstruction. No C_F can have both e_w and e_x as recession directions; the same holds for e_w and $-e_y$. To see this, let $K' = \{d : \text{sym}(F^T M_0(d)) \succeq 0\}$. S -freeness implies $d_x d_y \leq 0$ on K' by considering large steps from an interior point. Thus a small perturbation of $e_x + \sigma e_w$ in the positive e_y direction is outside K' , for every $\sigma > 0$; the two-dimensional cone generated by e_x, e_w would lie in its boundary. If its defining linear map is injective, K' is an invertible linear image of the 2×2 positive-semidefinite cone, whose boundary cannot contain a two-dimensional convex cone: two nonproportional rank-one positive-semidefinite matrices have a positive definite sum. If the map has a nonzero kernel vector ℓ , then ℓ is a lineality direction of K' . The condition $(e_x + t\ell)_x (e_x + t\ell)_y \leq 0$ for both signs of small t forces $\ell_y = 0$. For every interior point s , both directions along ℓ stay in C_F , so $q(s + t\ell) \geq 0$ for all real t . Its linear coefficient is $\ell_w - s_y \ell_x$. Variation of s_y over an open interval forces $\ell_x = \ell_w = 0$, a contradiction. The second ray pair follows by exchanging x and $-y$. The exact $\mathcal{B}$ completion below has all three positive recession directions; its parameter $J^T M_b$ is nonsymmetric and hence outside $\mathcal{BP}$.

For $\mathcal{B}$, direct multiplication gives

$$\text{sym}(J^T M(s)) = \begin{pmatrix} -y & (w-1)/2 \\ (w-1)/2 & x \end{pmatrix}.$$

Its upward completion is the set in the proof of [theorem 5.8](#). With $r = \sqrt{-xy}$, every point of it satisfies $q \geq 1 - 2r + r^2 = (1-r)^2 \geq 0$, and its interior has $q > 0$. The first three rays are recessive and the fourth leaves at step two, giving the exact cut vector $(0, 0, 0, 1/2)$.

It remains to prove the sharp $\mathcal{BP}$ inequality. For $\Sigma = \begin{pmatrix} a & b \\ b & c \end{pmatrix} \succ 0$, write $\Delta = ac - b^2$, $A = a + b$, and $B = b + c$. Here $m = (1, 1)^T/2$. The kept forms and the two ray forms are

$$v^T Z v = \frac{1}{2} v_1 \langle \mathbf{1}, v \rangle_\Sigma, \quad v^T \Sigma N_1 v = \frac{1}{2} v_2 \langle \mathbf{1}, v \rangle_\Sigma, \quad v^T \Sigma N_2 v = \frac{1}{2} v_1 \langle (1, -1), v \rangle_\Sigma.$$

The unrestricted maxima of the negative ray forms divided by $v^T \Sigma v$ are

$$G_1 = \frac{\sqrt{a(a+2b+c)/\Delta} - 1}{4}, \quad G_2 = \frac{\sqrt{c(a-2b+c)/\Delta} - 1}{4}. \quad (31)$$

These formulas follow from the two generalized eigenvalue equations: each has trace $1/2$, and its determinant is respectively $-(a+b)^2/(16\Delta)$ or $-(c-b)^2/(16\Delta)$. The uncompleted coefficients are G_j ; enlarging to the completion gives $a_j \leq G_j$. If a maximizing vector is kept, its necessary inequality gives the reverse bound and hence $a_j = G_j$.

An exact symmetry exchanges the two rays. Let

$$T(x, y, w) = (-y, -x, w), \quad H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad \Sigma' = H \Sigma^{-1} H, \quad L = \Sigma H/2.$$

Writing $A_\Sigma(s) = \text{sym}(\Sigma M_b^{-1} M(s))$, direct multiplication gives

$$A_\Sigma(Ts) = LA_{\Sigma'}(s)L^T. \quad (32)$$

This congruence preserves the sign of the kept form, maps maximizing directions to maximizing directions, and exchanges the completed coefficients: $a_1(\Sigma') = a_2(\Sigma)$ and $a_2(\Sigma') = a_1(\Sigma)$. Up to a positive factor,

$$\Sigma' = \begin{pmatrix} a+c-2b & c-a \\ c-a & a+c+2b \end{pmatrix}.$$

First suppose $c \leq b$. Then $b \geq c > 0$, $a > c$, and the positive maximizer for ray 1 is kept: for $v = (s, -1)$ its quotient is

$$f_1(s) = \frac{As - B}{2(as^2 - 2bs + c)},$$

whose positive numerator implies $s > B/A > 0$ and thus $v^T Z v = s(As - B)/2 > 0$. Moreover,

$$\frac{a(a+2b+c)}{\Delta} \geq \frac{a(a+3c)}{c(a-c)} \geq 9,$$

where the last inequality is $(a-3c)^2 \geq 0$. Thus $a_1 = G_1 \geq 1/2$. If instead $a+b \leq 0$, the transformed parameter satisfies $c' - b' = 2(a+b)/\Delta \leq 0$, so the same argument gives $a_2 \geq 1/2$. Both cases are strictly above the required sum bound.

We may therefore assume $a+b > 0$ and $c > b$. The transformed parameter has the same two strict inequalities. If both unrestricted maximizing directions are kept, then

$$4(a_1 + a_2) + 2 = \frac{\sqrt{a(a+2b+c)} + \sqrt{c(a-2b+c)}}{\sqrt{\Delta}} \geq 2\sqrt{2}.$$

Indeed, AM–GM and

$$ac((a+c)^2 - 4b^2) - 4\Delta^2 = ac(a-c)^2 + 4b^2\Delta \geq 0$$

prove the inequality. Equality requires $a = c$, $b = 0$.

If at least one maximizing direction is not kept, use [equation \(32\)](#) to make it ray 1. Its positive maximum occurs at

$$s_* = \frac{a(b+c) + \sqrt{a\Delta(a+2b+c)}}{a(a+b)}.$$

This formula follows by differentiating f_1 . At that maximum the numerator $As_* - B$ is positive, so failure of keptness means $s_* < 0$. Normalize $a = 1$ and put $\beta = -b$. Then $0 < \beta < 1$, and squaring the negative-numerator condition gives

$$\beta^2 < c < \frac{2\beta^2}{1+\beta}.$$

The equivalence uses the identity

$$(c - \beta^2)(1 - 2\beta + c) - (\beta - c)^2 = (1 - \beta)((1 + \beta)c - 2\beta^2).$$

The kept vector $(0, -1)$ gives $a_1 \geq (\beta - c)/(2c)$. The kept-boundary vector $J\Sigma\mathbf{1}$ gives $a_2 \geq (\beta - c)/(1 - 2\beta + c)$. Both denominators are positive. Each ratio decreases with c , so

$$a_1 + a_2 > \frac{1-\beta}{4\beta} + \beta = \beta + \frac{1}{4\beta} - \frac{1}{4} \geq \frac{3}{4}.$$

This proves the sharp lower bound in every case without a separate ray-2 admissibility calculation.

For $\Sigma = I$ both maximizing vectors are kept, and direct eigenvalue calculation on rays 1, 2, and 4 gives

$$a = \left((\sqrt{2} - 1)/4, (\sqrt{2} - 1)/4, 0, (\sqrt{2} + 1)/4 \right).$$

For ray 4 the endpoint kernel vector $(1, \sqrt{2} - 1)$ is kept, so upward completion does not extend its step. Ray 3 is recessive by construction. This proves attainment, uniqueness up to positive scaling, and the claimed characterization of $(t, t, 0, 0)$. At the vertex, the point-rule rotation has angle $\pi/4$ and satisfies $R_{\pi/4}M_b = \sqrt{2}I$, so this is also the untransformed point-rule set.

C.6 An exact local factor lower bound

At [theorem 5.6](#), take $\omega = (11/20, 10^{-6}, 9/20)$ and $z_0 = 649/2500$. Let $t_0 = \bar{s}$ and $t_j = \bar{s} + (z_0/\omega_j)p_j$. The tetrahedron they span is the image of $\{\lambda \geq 0 : \omega^T \lambda \leq z_0\}$. Each triangular facet has rank-two projection onto (x, y) : the four signed projection determinants, in vertex order 012, 013, 023, 123, are

$$11027808/5, \quad 5475613/562500, \quad -190382852/45, \quad -1139162725613/562500.$$

On such a facet the Hessian of $w - xy$ is indefinite, so a minimum cannot lie in its relative interior. A minimum cannot lie in the tetrahedron's interior either, since $\partial q/\partial w = 1$. It suffices to minimize on its six edges. Substituting $(1 - t)t_i + tt_j$ gives a univariate quadratic; its exact minimum on $[0, 1]$ is listed below.

ij	01	02	03
$\min q$	3/16	3/2	3/2
ij	12	13	23
$\min q$	6480592821/4840070400112	19101/15625	1361266990991/4320082399920

Every value is positive. Thus $z_K(\omega) > 649/2500$. The completed-family certificate in [table 5](#) gives $\hat{\lambda} = (2211, 8805, 2464)/10000 \in \mathcal{C}_B$ and

$$\omega^T \hat{\lambda} = 464971761/2000000000 < 0.2324859.$$

Consequently $\rho_A \geq \rho_B > (649/2500)/(464971761/2000000000) > 1.1166$.

C.7 Retained numerical closure records

[Table 7](#) gives the eleven-corner survey, all with unit costs and recorded corner bound one. The single-cut $\mathcal{B}$ search used restricted sampling and local refinement, so its values are heuristic lower bounds on the supremum. Closure cut generation uses finitely many valid cuts: its LP value is a lower bound on the full closure value. Dividing that value by the best pricing value found is a numerical correction, not a certified upper bound, because the pricing search is incomplete. Equal values in the table therefore indicate observed agreement only.

The larger orbit example is defined after [theorem 5.6](#). R1–R8 identify numerical data records in the electronic companion, which provides their vertices, ray matrices and source mapping. They are not additional analytic examples. The survey omits one near-boundary record because its recorded single-cut values disagree across implementations. Observed closure gain was at most about 0.084 of the corner bound and about 1.12 times the single-cut value. The point-rule families gave matching single and closure estimates in 17 of 18 comparisons: eleven for $\mathcal{P}$ and seven for $\mathcal{BP}$, with record R1 in $\mathcal{BP}$ the exception. These observations do not prove convexity of their coefficient up-sets.

Table 7: Recorded single-cut and closure values; arrows show the observed change. Only the separately stated rational certificates establish exact brackets.

Example or record	support	$\mathcal{A}$	$\mathcal{B}$	$\mathcal{P}$	$\mathcal{BP}$
Theorem 5.6	2	.97539 $\rightarrow$.97539	.97539 $\rightarrow$.97539	.2218	.2540 $\rightarrow$.2540
Larger orbit example	2	.83477 $\rightarrow$.83477	.83477 $\rightarrow$.83477	.3431	.3431 $\rightarrow$.3431
R1	2	.6224 $\rightarrow$.6354	.7078 $\rightarrow$.7921	.3532	.4238 $\rightarrow$.4479
R2	2	.4526 $\rightarrow$.5065	.4526 $\rightarrow$.5065	.1199	.1693 $\rightarrow$.1693
Proposition 5.7	1	.98385 $\rightarrow$.99536	.98385 $\rightarrow$.99536	.3986	.4272 $\rightarrow$.4272
R3	1	.99923 $\rightarrow$.99967	—	.4228	—
R4	1	.98765 $\rightarrow$.99574	—	.3358	—
R5	1	.99311 $\rightarrow$.99311	.99311 $\rightarrow$.99311	.7706	.7934 $\rightarrow$.7934
R6	1	.99525 $\rightarrow$.99525	—	.5800	—
R7	1	.99062 $\rightarrow$.99062	—	.9422	—
R8	1	.99829 $\rightarrow$.99873	.99832 $\rightarrow$.99874	.3742	.3910 $\rightarrow$.3910

For [theorem 5.6](#) in direction **1**, the recorded $\mathcal{BP}$ values are close to $16/63$, while the untransformed point-rule set gives about 0.0896. Equality with $16/63$ is not established here. A scan of 291 grid directions and 297 edge directions at this corner gave largest single-cut loss about 1.1283 near $(.55, 0, .45)$. At positive costs $(.55, 10^{-6}, .45)$, the recorded closure estimates for both $\mathcal{A}$ and $\mathcal{B}$ were about .2301778, versus corner value .2596998. Only the lower factor bound in [appendix C.6](#) is certified; a finite scan does not upper-bound the worst loss.

The six recorded reoptimization trajectories are

corner	successive LP bounds	cuts reaching one
Theorem 5.6	.97539, .98097, 1	3
Proposition 5.7	.98385, 1	2
R8	.99829, 1	2
larger orbit example	.83477, .87574, .94583, .98023, .98023, 1	6
R2	.45263, .47486, .47981, .71275, .71275, .72135, .72306, 1	8
R1	.6224, $\dots$, .91713	not in 12 rounds

Each round forms a new basis cone containing the current polyhedron and maps its orbit cut back to the original coordinates, so the resulting cuts remain valid for the original feasible set. Zero reduced costs can make a later corner bound zero; positive-cost theorems then do not apply. Five of the six trajectories reached the bound, which does not prove finite convergence of rebasing.

The unresolved extensions are whether a support-one restricted closure can be exact while no single cut is exact, whether the fixed-corner $\mathcal{A}$ and $\mathcal{B}$ factors at [theorem 5.6](#) are finite, and why the point-rule single and closure estimates so often agree. The support-one $\mathcal{B}$ closure upper certificate remains incomplete. No claim here depends on that incomplete record or on the stopped four-ray numerical search; the four-ray result has the analytic proof above.

D Minor geometry and rational certificates

D.1 Orbit, support and the tangent pencil

Proof of [proposition 6.3](#). Write $M = H + tJ$, with H symmetric. Then $\det M = \det H + t^2$. Thus $C_I = \{H + tJ : H \succeq 0, t \in \mathbb{R}\}$ is closed, convex and free, with interior $H \succ 0$. To prove maximality, take $N \notin C_I$ and diagonalize its symmetric part. It has a negative eigenvalue. Choose a positive

definite symmetric matrix H small enough in that eigendirection and large enough in the orthogonal direction that $\text{sym } N + H$ is indefinite. Set $Z = H - \frac{1}{2}(N - N^T) \in \text{int } C_I$. Then $(N + Z)/2$ is symmetric and has negative determinant. If a convex set contained both C_I and N , this midpoint would be an interior point: a ball around Z remains in C_I , and its midpoints with N form a ball. Its interior also contains I , with positive determinant. The segment between these interior points contains a singular matrix in the interior, contradicting freeness. This proves maximality without a classification theorem.

For $N = AMB^T$, positive semidefiniteness of $\text{sym}(F^T A^{-1} N B^{-T})$ is equivalent, by congruence, to that of $\text{sym}(B F^T A^{-1} N)$. This gives $F'^T = B F^T A^{-1}$. Taking $A = I$, $B = F^T$ shows that all $\det F > 0$ occur in the orbit of C_I ; maximality transfers under invertible transformations. The substitution $u = Fv$ in $v^T M F v \geq 0$ gives the transposition rule. Moreover, with the matrix trace inner product, $C_F^* = F \mathbb{S}_+^2$. Equality of two cones implies that $G = F^{-1} F'$ maps $\mathbb{S}_+^2$ onto itself by left multiplication. Since symmetric matrices are spanned by positive semidefinite matrices, G preserves the symmetric matrix space. Testing I, E_{11}, E_{22} , and $E_{12} + E_{21}$ gives $G = cI$; preservation of the positive cone gives $c > 0$. Hence the projective family has dimension three.

Let $A \bar{M} B^T = U' H'$ be a transformed polar decomposition. Pulling $C_{U'}$ back gives $F^T = B^{-1} U'^T A$ and $F^T \bar{M} = B^{-1} H' B^{-T} \succ 0$ symmetric. Conversely, if $F^T \bar{M} = H \succ 0$, apply the coordinate map $M \mapsto F^T M$; the transformed polar factor is I , so its pullback is C_F . This family is parametrized by $H \succ 0$, modulo a positive scalar, and has dimension two. The parameter transformation sends H to $B H B^T$, proving equivariance; for transposition it sends H to $F^{-T} H F^{-1}$. Finally $R^T \bar{M} \succ 0$ symmetric with $R \in SO(2)$ forces $R = U$ by polar uniqueness. The negative polar sign gives the opposite cone and is not a second point-rule member. $\square$

Proof of proposition 6.5. Positive objective weights make finite objective sublevels compact, so the minimum is attained. Choose a minimizer with smallest support. If its supported rays are dependent, choose a nonzero supported vector h with $Ph = 0$. Both small signs of h preserve nonnegativity and the same constraint point; optimality forces $\omega^T h = 0$. Move until a coordinate first vanishes, contradicting smallest support. Thus an independent-supported minimizer exists.

Fix any independent-supported minimizer with support J of size k . Put $W = \text{span}\{P_j : j \in J\}$ and $V = P_J\{h : \omega_J^T h = 0\}$, so $\dim V = k - 1$. The determinant is nonnegative on T_{z_K} : a negative value there would produce a lower-cost zero on the segment from the apex. Small directions in V remain in the supported objective-level face, so $B(M_*, V) = 0$ and the quadratic form is nonnegative on V . If the determinant gradient vanishes on W , the same argument for directions with decreasing objective, together with evenness of the quadratic term, makes the form nonnegative on all of W , so $k \leq 2$. Otherwise V is a nonnegative subspace of dimension $k - 1$ of a form with positive index two; hence $k \leq 3$.

If $k = 3$, the preceding exceptional case is impossible, and $M_* \neq 0$. The restriction of the form to $T = \{D_{\text{edge}} : B(M_*, D_{\text{edge}}) = 0\}$ has radical $\mathbb{R}M_*$ and signature $(1, 1)$ after quotienting by that radical. Indeed, choose Z with $B(M_*, Z) = 1$; $\text{span}\{M_*, Z\}$ has signature $(1, 1)$ and its orthogonal complement has signature $(1, 1)$. The two-dimensional nonnegative subspace $V \subseteq T$ must contain M_* , since otherwise its image in the quotient would be a two-dimensional nonnegative subspace of signature $(1, 1)$. The supported objective-level affine plane $M_* + V$ therefore contains zero. As it lies in $\bar{M} + W$, this implies $\bar{M} \in W$. For $k = 2$, local nonnegativity along its edge gives $B(M_*, D_{\text{edge}}) = 0$ and $\det D_{\text{edge}} \geq 0$, as asserted. $\square$

Proof of lemma 6.6. The two-dimensional identities

$$\text{adj}(ab^T) = (Jb)(Ja)^T, \quad \text{adj}(D_{\text{edge}}) = J^T D_{\text{edge}}^T J$$

give $G_1 a = 0$ and $(Jb)^T G_0 a = b^T \text{adj}(D_{\text{edge}}) a = \text{tr}(\text{adj}(M_*) D_{\text{edge}}) = 0$. Thus $G_0 a = c_0 b$; invertibility of G_0 gives $c_0 \neq 0$. Linearity of the adjugate and tangency give

$$\det(D_{\text{edge}} + tM_*) = \det D_{\text{edge}}, \quad J \text{adj}(D_{\text{edge}} + tM_*) = G_0 + tG_1, \quad \det(G_0 + tG_1) = \det D_{\text{edge}}.$$

At the contact point, $\text{sym}((F^T a)b^T) \succeq 0$ forces $F^T a = cb$ for $c > 0$: if the two vectors are not parallel the symmetric matrix is indefinite, and invertibility rules out $c = 0$. Write $Y = \text{sym}(F^T D_{\text{edge}})$ and $u = Jb$. The segment condition is $cbb^T + sY \succeq 0$ for both signs of s . It first gives $u^T Y u = 0$; positive semidefiniteness then gives $(cbb^T + sY)u = 0$, hence $Yu = 0$. Therefore $Y = \mu bb^T$, and with $t = -\mu/c$, $F^T(D_{\text{edge}} + tM_*)$ is skew-symmetric. It equals ηJ , so

$$F^T = \frac{\eta}{\det D_{\text{edge}}} J \text{adj}(D_{\text{edge}} + tM_*) = \theta(G_0 + tG_1).$$

The contact identity $F^T a = cb$ fixes $\theta c_0 > 0$. □

D.2 Complete data and proofs for A, B and S1

All entries below use the order (a, b, c, d) . Put $V_0 = \bar{M}$, $P_j = V_j - V_0$ and $\omega_j = 1$.

	A	B	S1
V_0	$(0, 5/2, -1/2, -1/2)$	$(-3, -5/2, 1/2, -2)$	$(-4, -1, -1/2, -2)$
V_1	$(-5, 3, 4, -4)$	$(1, 1, -7, -4)$	$(3, 3, -3, -3)$
V_2	$(-8, 12, -8, 8)$	$(-1/2, -1/2, -11/2, -7)$	$(95/24, 4, -4, -4)$
V_3	$(-3/2, -4, 2, -7/2)$	$(1/2, 1, -5/2, 0)$	$(17/3, 6, -3/2, -3/2)$
V_4	$(-7/2, -1/2, 3/2, -4)$	$(-7/2, 4, -7/2, 2)$	$(41/6, 3, -6, -2)$
M_*	$(-6, 6, 0, 0)$	$(0, 0, -6, -6)$	$(3, 3, -3, -3)$
λ_*	$(2/3, 1/3, 0, 0)$	$(1/3, 2/3, 0, 0)$	$(1, 0, 0, 0)$
$\det \bar{M}$	$5/4$	$29/4$	$15/2$
$\det[P_1 \cdots P_4]$	$-2775/4$	$1005/4$	$533/16$

Table 8: Rational full-dimensional minor corners.

Exact minimization on a simplex. For $V(\beta) = \sum_{i=0}^4 \beta_i V_i$, let $H_{ij} = B(V_i, V_j)$. Then $\det V(\beta) = \beta^T H \beta$ on $\beta \geq 0$, $\mathbf{1}^T \beta = 1$. On every nonempty face I , an interior stationary point solves

$$\begin{pmatrix} 2H_{II} & -\mathbf{1} \\ \mathbf{1}^T & 0 \end{pmatrix} \begin{pmatrix} \beta_I \\ \mu \end{pmatrix} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (33)$$

Only solutions with all barycentric coordinates positive need be retained. A singular system cannot hide a new minimum: when a stationary solution exists, a null vector (h, η) satisfies $2H_{II}h = \eta \mathbf{1}$, $\mathbf{1}^T h = 0$. Multiplying by the stationary barycentric vector gives $\eta = 0$. Thus the objective is constant along $\beta_I + th$, and its minimum also occurs on lower-dimensional faces. A nonconstant segment of minima would have two distinct boundary minimizers. This proves that face enumeration establishes both the value and uniqueness.

For A, B and S1, the respective polar matrices are

$$H_A = \begin{pmatrix} 5/4 & -3 & 15 & -25/8 & -9/8 \\ -3 & 8 & -16 & 67/4 & 63/4 \\ 15 & -16 & 32 & -20 & -9 \\ -25/8 & 67/4 & -20 & 53/4 & 101/8 \\ -9/8 & 63/4 & -9 & 101/8 & 59/4 \end{pmatrix}, \quad H_B = \begin{pmatrix} 29/4 & -4 & 17/4 & -31/8 & -39/8 \\ -4 & 3 & -3/2 & 15/4 & 95/4 \\ 17/4 & -3/2 & 3/4 & 3/8 & 175/8 \\ -31/8 & 15/4 & 3/8 & 5/2 & 29/4 \\ -39/8 & 95/4 & 175/8 & 29/4 & 7 \end{pmatrix},$$

$$H_{S1} = \begin{pmatrix} 15/2 & 9/4 & 73/24 & -23/12 & -61/12 \\ 9/4 & 0 & 1/16 & 1/2 & 1/4 \\ 73/24 & 1/16 & 1/6 & 67/96 & 3/8 \\ -23/12 & 1/2 & 67/96 & 1/2 & 227/24 \\ -61/12 & 1/4 & 3/8 & 227/24 & 13/3 \end{pmatrix}.$$

All 31 systems in each example are nonsingular. The following table lists every stationary value whose barycentric coordinates are positive; “–” means that no such stationary point occurs. A face is named by its vertex indices. The five vertex values are the corresponding diagonal entries of H .

Face	A	B	S1
01	4/61	23/73	–
03	435/1328	199/1120	11/1704
04	1099/1168	1727/1536	959/3168
12	0	0	–
13	57/4	105/32	–
14	2081/140	8689/600	–
23	96/341	111/160	3721/6720
24	1564/259	30289/2304	–
34	2307/176	561/80	50281/8112
012	36/37	27/70	–
013	6497/16997	1048/2361	2511/6704
014	–	204432/73729	162/265
023	88880/55709	7193/7848	33595/64082
024	–	573681/114824	1487/1829
034	–	2317/1929	575117/361346
123	864/619	–	–
134	30431/2165	–	–
0123	142884/67651	–	–
0234	–	–	15689521/9829632

Every unlisted face of size at least two has no positive-barycentric stationary point. Thus the unique zero is on face 12 with weights $(2/3, 1/3)$ for A and $(1/3, 2/3)$ for B, and at vertex 1 for S1. It follows that $z_K = 1$ with the stated unique minimizers.

Remaining assertions of [theorem 6.7](#). For A, $D_{\text{edge}} = V_1 - V_2 = (3, -9, 12, -12)$ has determinant 72 and $\nabla \det(M_*)^T D_{\text{edge}} = 0$. Its pencil is

$$G_0 = \begin{pmatrix} -12 & 3 \\ 12 & -9 \end{pmatrix}, \quad G_1 = \begin{pmatrix} 0 & -6 \\ 0 & 6 \end{pmatrix}.$$

The positive sign of the pencil scale is required at M_* . The intervals on its parameter imposed by the five vertices are

$$\begin{aligned} K(V_0) &= [6 - \sqrt{10}, 6 + \sqrt{10}], & K(V_1) &= (-\infty, 3], & K(V_2) &= [-3/2, \infty), \\ K(V_3) &= [-49/6 - 2\sqrt{106}/3, -49/6 + 2\sqrt{106}/3], \\ K(V_4) &= [-33/10 - 2\sqrt{118}/5, -33/10 + 2\sqrt{118}/5]. \end{aligned}$$

These follow by the two diagonal and determinant inequalities for $\text{sym}((G_0 + tG_1)V_i)$. Already $K(V_0)$ and $K(V_3)$ are disjoint. For B, $D_{\text{edge}} = (3/2, 3/2, -3/2, 3)$ has determinant 27/4 and its sign-normalized pencil is

$$G_0 = \begin{pmatrix} -3/2 & -3/2 \\ 3 & -3/2 \end{pmatrix}, \quad G_1 = \begin{pmatrix} -6 & 0 \\ -6 & 0 \end{pmatrix}.$$

The intervals are

$$\begin{aligned} K(V_0) &= [(11 - \sqrt{87})/2, (11 + \sqrt{87})/2], & K(V_1) &= (-\infty, 3/2], & K(V_2) &= [-3, \infty), \\ K(V_3) &= [-9/4 - \sqrt{30}/2, -9/4 + \sqrt{30}/2], \\ K(V_4) &= [-1/12 - \sqrt{21}/15, -1/12 + \sqrt{21}/15]. \end{aligned}$$

Again $K(V_0)$ and $K(V_3)$ are disjoint. The multipliers in $\omega = -\sigma P^T \nabla \det(M_*) + \nu$ are

	σ	ν	$\nabla \det(M_*)^T \bar{M}$
A	1/6	(0, 0, 3/2, 5/2)	6
B	1/3	(0, 0, 1, 15)	3
S1	2/9	(0, 1/36, 2/9, 1/9)	9/2

For S1, the normal products on the four incident edges to V_0, V_2, V_3, V_4 are respectively 9/2, 1/8, 1, 1/2, all positive. For A, the unique corner contact satisfies $\nabla \det(M_*)^T (\bar{M} - M_*) = 6 > 0$. The attainment criterion in [theorem 2.3](#), applied to $\{\det \leq 0\}$ and then [lemma 6.1](#), gives an unrestricted maximal free set attaining one.

For the strict supremum gaps, set $W_0 = V_0$ and $W_j = (1 - z)V_0 + zV_j$. Represent a symmetric certificate matrix by $[u, v, h] = \begin{pmatrix} u & v \\ v & h \end{pmatrix}$. The following complete rational certificates satisfy $\sum_{i=0}^4 W_i Y_i = 0$ and $Y_i \succ 0$. Both assertions are checked by matrix multiplication and by $u > 0$, $uh - v^2 > 0$ in each row.

A, $z = 9/20$				
i	u	v	h	
0	120111307/200000000	22844639/100000000	8690461/100000000	
1	2000019/2000000	28763/50000	330939/1000000	
2	26271/1000000	6631/200000	20937/500000	
3	43823/1000000	-25577/1000000	14943/1000000	
4	11/1000000	0	11/1000000	
B, $z = 461/500$				
0	56409442877/1189000000000	-400381181/3625000000	93214817/362500000	
1	58606291/82000000	-211351/250000	1	
2	15799/20000	-382359/500000	370147/500000	
3	1/1000000	0	1/500000	
4	4523/1000000	14917/1000000	49221/1000000	
S1, $z = 779/1000$				
0	68915327/21600000000	3999047/1800000000	11344801/7200000000	
1	407/21600000	-1/100000	1/50000	
2	11101/100000	-12781/100000	3679/25000	
3	1	-76729/100000	471/800	
4	3653/100000	-6377/100000	11139/100000	

[Proposition 6.4](#) now gives all three upper bounds.

For robustness, the linear map $\mathcal{L} : (Y_i) \mapsto \sum_i W_i Y_i$ from symmetric blocks to $\mathbb{R}^{2 \times 2}$ has rank four. Its adjoint kernel consists of F with $\text{sym}(F^T W_i) = 0$ for every i . The apex condition gives $F^T = tJV_0^{-1}$. If $t \neq 0$, the other conditions force every $V_0^{-1}W_i$ to be scalar, contradicting the independence of the four rays. Hence the kernel is zero. In a neighborhood of A, the map retains full row rank, so projecting the given positive definite tuple onto its continuously varying kernel keeps all blocks positive definite. The orbit upper bound 9/20 persists. The corner bound is continuous here. Lower semicontinuity follows from compact objective sublevels and continuity of the determinant.

For upper semicontinuity, the feasible point $M_* + \varepsilon P_1$ has negative determinant for small $\varepsilon > 0$, since $\nabla \det(M_*)^T P_1 = -6$. This sign persists under small perturbations, so the segment from the perturbed apex reaches a zero at cost arbitrarily close to one. The strict gap therefore persists. $\square$

The three coarse bounds are the portable certificates needed for the counterexamples. Narrower archived verification intervals are reported with the computational evidence; their numerical precision is not needed for the preceding proofs.

D.3 Portable comparisons and bilinear brackets

Rational feasible parameters, checked at the indicated levels, include

corner	z	F^T
A	449533/1000000	$\begin{pmatrix} -10188147307/2513 & -24509003591/8528 \\ 34718214589/9897 & -20333469614/8855 \end{pmatrix}$
B	921947/1000000	$\begin{pmatrix} -15696987146/8949 & -650114959/308 \\ 2464903799/1862 & -19805038891/9695 \end{pmatrix}$
S1	389053/500000	$\begin{pmatrix} 2721203962/1605 & -11697519049/801 \\ 47980238700/6149 & -39926061217/9010 \end{pmatrix}$

In each case $\det F > 0$, $\text{sym}(F^T V_0) \succ 0$, and $\text{sym}(F^T W_i) \succeq 0$ for $i = 1, \dots, 4$. Thus B's ratio exceeds 0.921947, while its coarse upper bound is 0.922.

For completeness, the three bilinear slices have the following data in the same minor coordinate order. Their objective weights are all one.

	T	T'	V
V_0	$(3/2, -9/2, 0, 1)$	$(1, -1/2, 1/2, 1)$	$(2, -2, 3, 1)$
V_1	$(18, -1, -6, 1)$	$(-1/2, 1/2, -4, 1)$	$(0, 0, 0, 1)$
V_2	$(-18, -5, 6, 1)$	$(24, -10, 3, 1)$	$(1/4, 6, -2, 1)$
V_3	$(5/2, 0, 5/2, 1)$	$(15/2, 9/2, 1/2, 1)$	$(1/2, 1, -5/2, 1)$
λ_*	$(1/2, 1/2, 0)$	$(6/7, 1/7, 0)$	$(1, 0, 0)$

They are the two tangent-edge and support-one bilinear examples, respectively. Their corner bounds are one by the corresponding bilinear proofs. The following lower parameters prove the lower ends of the brackets in [corollary 6.8](#) directly:

$$\begin{aligned}
F_T^T &= \begin{pmatrix} 3557730690827343/67108864 & 2666317851727117/16777216 \\ -22689768773827/93440775 & 1900883772391943/99891300 \end{pmatrix}, & z_T^{\text{lo}} &= 9753853/10^7, \\
F_{T'}^T &= \begin{pmatrix} 1151288814435793/80718478 & -1212462267789092/97079703 \\ -56281863190224/49263763 & 709421204462399/87313318 \end{pmatrix}, & z_{T'}^{\text{lo}} &= 8347663/10^7, \\
F_V^T &= \begin{pmatrix} 722028544872305/33554432 & -251095200246904/33842369 \\ 121179366244954/66838953 & 2714408638762620/85585871 \end{pmatrix}, & z_V^{\text{lo}} &= 1967693/2000000.
\end{aligned}$$

The tests are the same rational determinant and diagonal tests as above. At the respective upper levels $z = 1219233/1250000$, $2086943/2500000$, and $2459617/2500000$, the following $Y_i = [u, v, h]$ give the complete dual certificates:

T, $z = 1219233/1250000$				
i	u		v	h
0	284680208922551/1125000000000000	125759927996883/2500000000000000	2500000002793643/2500000000000000	
1	773793481/1800000000	15927467/100000000	5901223/100000000	
2	34295509/100000000	-8578191/100000000	2145641/100000000	
3	2439499/100000000	-509103/4000000	66403807/100000000	
T', $z = 2086943/2500000$				
0	443012111728267/3750000000000000	141798875388169/1250000000000000	68080425260229/6250000000000000	
1	156624919/750000000	2856141/6250000	1	
2	1033917/100000000	3925397/100000000	14903451/100000000	
3	185279/50000000	-1727869/100000000	4028567/50000000	
V, $z = 2459617/2500000$				
0	283184926112931/2000000000000000	-243552272820433/2000000000000000	209466408183833/2000000000000000	
1	1	41186779/400000000	1060221/100000000	
2	7891043/50000000	-5452943/50000000	753629/10000000	
3	287531/10000000	2685447/100000000	627031/25000000	

Each block is positive definite and $\sum_{i=0}^3 W_i Y_i = 0$. These data prove both ends of all three brackets. In particular, $779/1000 < 8347663/10^7$ and $8347772/10^7 < 921947/10^6 < 461/500 < 9753853/10^7$, which prove every comparison in [corollary 6.8](#) without relying on unprinted endpoint witnesses.

D.4 Scaling and other minor projections

Proof of [proposition 6.10](#). The determinant is

$$\delta[(1 - \lambda_2)(1 - \lambda_3) + \lambda_1 \lambda_4].$$

It cannot vanish at cost below one, and it vanishes at e_2, e_3 . The polar factor is I . Its ray steps are $(2\sqrt{\delta}, 1, 1, 2/\sqrt{\delta})$, by the determinant test on $\text{sym}(\bar{M} + tP_j)$. The point-rule parameter $F^T = \text{diag}(1, 1/\delta)$ has $F^T \bar{M} = I$ and steps $(2, 1, 1, 2)$.

For a rotation parameter $F^T = \begin{pmatrix} c & s \\ -s & c \end{pmatrix}$, positive definiteness at the apex forces $c > 0$ and $|u| < 2\sqrt{\delta}/(1 - \delta) \leq (8/3)\sqrt{\delta}$, where $u = s/c$. The first ray's positive semidefiniteness requires

$$(t - u(1 - \delta))^2 \leq 4(\delta - tu).$$

For $u \geq 0$ this implies $t < 4\sqrt{\delta}$. For $u < 0$, expansion gives $t^2 \leq 4\delta + 4t|u|$, so

$$t \leq 2|u| + 2\sqrt{u^2 + \delta} < \frac{16 + 2\sqrt{73}}{3}\sqrt{\delta}.$$

The cut bound is no greater than its first-ray step. Finally left multiplication by $\text{diag}(1, \delta)$ maps the stated unscaled corner to this corner and maps its exact C_I to the exact point-rule parameter; the newly computed polar-factor choice is still C_I . $\square$

Other minor projections. For a principal minor, M is symmetric and $\det M = ad - b^2$ has signature $(1, 2)$. The positive-determinant result in [proposition 6.11](#) concerns the full singular locus, including positive and negative rank-one matrices. At a negative-determinant apex, reversing the form gives signature $(2, 1)$; the general homogeneous signature- $(n, 1)$ orbit characterization applies to that different component. It does not turn the polar point rule into the full family. If the diagonal bounds are imposed, the forbidden projection is instead the rank-one positive semidefinite cone. In that setting, at an indefinite apex, the maximal outer-product-free sets containing that apex in their interiors are halfspaces $\langle A, M \rangle \geq 0$ with $A \preceq 0$ and $\langle A, \bar{M} \rangle > 0$ ([Bienstock et al., 2020](#)). Their intersection cuts are dominated on the corner cone by the valid inequality $\langle A, M \rangle \leq 0$. Indeed,

writing $-A = \mu_1 g_1 g_1^T + \mu_2 g_2 g_2^T$, $\mu_i \geq 0$, shows this is a nonnegative combination of at most two valid inequalities $g_i^T M g_i \geq 0$. A particular eigenvector separator need not select these two vectors.

With exactly one diagonal entry, say $a = x_i^2$, $b = x_i x_l$, $c = x_j x_i$, $d = x_j x_l$, the projection has closure $\{\det M = 0, a \geq 0\}$. For $a > 0$, choose $x_i = \sqrt{a}$, $x_l = b/\sqrt{a}$, $x_j = c/\sqrt{a}$; then $d = bc/a$ follows from the determinant equation. For $a = 0$, the determinant requires $bc = 0$. If $b \neq 0$, approach with $a > 0$, fixed b, d , and $c = ad/b$; the case $c \neq 0$ is symmetric. If $b = c = 0$, choose b, c tending to zero with $bc = ad$, including either sign of d . Thus every boundary point is a limit of projected outer products. This smaller forbidden set can admit larger free sets and a larger corner bound. The four-distinct-index orbit guarantees remain valid cuts, but their ratios to that larger bound can only decrease.

E Proofs for successive cuts

E.1 Ball inclusions for the bilinear rules

We give the complete set formulas used in [lemma 7.5](#). For the side $S_+ = \{w \leq xy\}$, define

$$\hat{x}_+(s) = \left(\frac{x-y}{2}, \frac{w+1}{2} \right), \quad \hat{y}_+(s) = \left(\frac{x+y}{2}, \frac{w-1}{2} \right).$$

For $S_- = \{w \geq xy\}$, define

$$\hat{x}_-(s) = \left(\frac{x+y}{2}, \frac{1-w}{2} \right), \quad \hat{y}_-(s) = \left(\frac{x-y}{2}, \frac{-1-w}{2} \right).$$

Then $\|\hat{x}_\sigma(s)\|^2 - \|\hat{y}_\sigma(s)\|^2$ equals the signed violation $w - xy$ or $xy - w$. Fix the violated side, suppress its subscript, and put

$$a = \|\hat{x}(\bar{s})\|, \quad b = \|\hat{y}(\bar{s})\|, \quad \lambda^S = \frac{\hat{x}(\bar{s})}{a}, \quad \ell = \lambda_2^S.$$

Here $a > b \geq 0$, since $a^2 - b^2 = q > 0$. The unit vector λ^S is distinct from the ray coordinates. The analytical default Case-4 set is

$$C_S = \{s : \phi_\ell(\hat{y}(s)) \leq (\lambda^S)^T \hat{x}(s)\}, \quad \phi_\ell(y) = \begin{cases} \|y\|, & y_2 \leq \ell \|y\|, \\ \sqrt{1-\ell^2} |y_1| + \ell y_2, & y_2 > \ell \|y\|. \end{cases} \quad (34)$$

Indeed, ϕ_ℓ is the support function of the compact convex set $\{z : \|z\| \leq 1, z_2 \leq \ell\}$: either the norm maximizer is permitted, or the maximizer lies on $z_2 = \ell$. Thus ϕ_ℓ is convex and $\phi_\ell(y) \leq \|y\|$. The set C_S is closed and convex and contains

$$C_0 = \{s : \|\hat{y}(s)\| \leq (\lambda^S)^T \hat{x}(s)\}.$$

For completeness, these formulas also show directly that C_S is free of the violated side. Consider side $+$ first. The default vector cannot have $\ell = -1$: that would require $\bar{x} = \bar{y}$ and $\bar{w} < -1$, contradicting $\bar{w} - \bar{x}\bar{y} > 0$. For $-1 < \ell < 1$, minimizing

$$\|(y_1, y_2 - t)\| + \ell t \quad \text{over } t \geq 0$$

gives exactly $\phi_\ell(y)$, and the minimum is attained. This is the condition for $s - 2te_w$ to belong to C_0 , so $C_S = C_0 + \mathbb{R}_+ e_w$. For $\ell = 1$ the minimum occurs at zero, and $C_S = C_0$, which is already

upward closed. Every point of C_0 satisfies $\|\hat{y}\| \leq (\lambda^S)^T \hat{x} \leq \|\hat{x}\|$, and hence $w - xy \geq 0$. Moving upward preserves this inequality. Since its zero set has empty interior and $\partial(w - xy)/\partial w = 1$, an interior point of C_S has $w - xy > 0$. For side $-$, the substitution $(x, y, w) \mapsto (x, -y, -w)$ reduces the assertion to side $+$.

Both affine coordinate maps have linear-part norm at most $1/\sqrt{2}$; for example,

$$\|\Delta \hat{x}_+\|^2 = \frac{(\Delta x - \Delta y)^2 + \Delta w^2}{4} \leq \frac{\|\Delta s\|^2}{2}.$$

For $\|s - \bar{s}\| \leq t$, it follows that

$$(\lambda^S)^T \hat{x}(s) \geq a - t/\sqrt{2}, \quad \|\hat{y}(s)\| \leq b + t/\sqrt{2}.$$

Consequently C_0 , and hence C_S , contains the closed ball with radius

$$\rho_S = \frac{a - b}{\sqrt{2}} = \frac{q}{\sqrt{2}(a + b)}. \quad (35)$$

On $[-B, B]^3$, each of a and b is at most $L_B = (B^2 + (B + 1)^2/4)^{1/2}$. This proves lemma 7.5(a), including the interior condition at the vertex.

For the orbit inclusion, set $M = M_\sigma$ as in lemma 7.5 and choose $F^T = M(\bar{s})^{-1}$. Its determinant is $1/q > 0$. The linear part satisfies

$$M_{0,+}(\Delta s) = \begin{pmatrix} \Delta w & \Delta x \\ \Delta y & 0 \end{pmatrix}, \quad M_{0,-}(\Delta s) = \begin{pmatrix} \Delta x & \Delta w \\ 0 & \Delta y \end{pmatrix}, \quad \|M_{0,\sigma}(\Delta s)\|_F = \|\Delta s\|.$$

If $\|\Delta s\| \leq \sigma_{\min}(M(\bar{s}))$, then

$$\|M(\bar{s})^{-1} M_{0,\sigma}(\Delta s)\|_2 \leq 1,$$

and therefore

$$\text{sym}(F^T M(\bar{s} + \Delta s)) = I + \text{sym}(M(\bar{s})^{-1} M_{0,\sigma}(\Delta s)) \succeq 0.$$

Thus the orbit set contains the ball of radius $\sigma_{\min}(M(\bar{s}))$. The determinant and singular-value identities give

$$\sigma_{\min}(M(\bar{s})) = \frac{q}{\sigma_{\max}(M(\bar{s}))} \geq \frac{q}{\|M(\bar{s})\|_F} \geq \frac{q}{\sqrt{3B^2 + 1}}. \quad (36)$$

There is no need to assume this orbit set is maximal. To check that it is free of the violated side, an interior point must have $\text{sym}(F^T M(s)) \succ 0$: a zero quadratic form at a singular positive semidefinite matrix would give a supporting affine inequality that is strict at $\bar{s}$. For a real 2×2 matrix A , writing its symmetric and skew parts shows $\det A = \det(\text{sym } A) + h^2$ for some real h . Hence a positive definite symmetric part implies $\det(F^T M(s)) > 0$, and so $\det M(s) > 0$. This proves lemma 7.5(b).

Finally the reference set in (b) has $\alpha_j \|r_j^r\| \geq \rho_A = q/T_B$. For geometric weights its objective is at least ρ_A , so a θ -approximate maximizer has $\delta \geq \theta \rho_A$. For perturbed weights its objective is at least $\tau \rho_A$, and each finite returned step satisfies

$$\alpha_j \|r_j^r\| = \frac{u_j \alpha_j}{\hat{\omega}_j + \tau} \geq \frac{\theta \tau \rho_A}{1 + \tau}.$$

This proves lemma 7.5(c).

E.2 Proof of the suboptimal-limit construction

We prove [theorem 7.8](#) without computational verification. Since $xy = w \geq 1$, AM–GM gives

$$\varepsilon x + y + w \geq 2\sqrt{\varepsilon xy} + w = 2\sqrt{\varepsilon w} + w \geq 1 + 2\sqrt{\varepsilon}.$$

The allowed point $(1/\sqrt{\varepsilon}, \sqrt{\varepsilon}, 1)$ attains this bound. With the one-sided constraint $w \leq xy$, the same lower bound holds because $xy \geq w \geq 1$.

Each C_a is a closed, full-dimensional convex set, and

$$\frac{(ax + y/a)^2}{4} - xy = \frac{(ax - y/a)^2}{4} \geq 0$$

proves that its interior avoids $\{w \leq xy\}$. The invertible map $T_a(x, y, w) = (x/a, ay, w)$ preserves xy and maps C_1 onto C_a . To prove maximality it therefore suffices to consider $a = 1$. In coordinates $u = (x + y)/2$, $v = (x - y)/2$, we have

$$C_1 = \{w \geq u^2\}, \quad S = \{w \leq u^2 - v^2\}.$$

Let $p \notin C_1$, choose $0 < \tau < 1$, and take

$$u_c = u_p, \quad v_c = -\frac{\tau v_p}{1 - \tau}, \quad w_c = u_p^2 + \eta$$

with $\eta > 0$ sufficiently small. Then $c \in \text{int } C_1$, and $z = (1 - \tau)c + \tau p$ has $u_z = u_p$, $v_z = 0$, and

$$u_z^2 - v_z^2 - w_z = \tau(u_p^2 - w_p) - (1 - \tau)\eta > 0.$$

It is in $\text{int } S$. A ball around c inside C_1 scales to a ball around z inside $\text{conv}(C_1 \cup \{p\})$, so z is also an interior point of that convex hull. Every strict convex enlargement of C_1 therefore fails to be S -free.

At $(0, 0, 1)$, the default vector of [equation \(34\)](#) is $\lambda^S = (0, 1)$. Its set is

$$\sqrt{(x + y)^2 + (w - 1)^2} \leq w + 1,$$

which is equivalent to $w \geq (x + y)^2/4$. Thus it is C_1 , and the maps T_a give the asserted images.

At round zero the box has the unique optimum $(0, 0, 1)$ and basis rays e_x, e_y, e_w . With $a = 2/\xi_1$, the steps of C_a are $\xi_1, 4/\xi_1, \infty$, so the first cut is $x/\xi_1 + \xi_1 y/4 \geq 1$. Suppose inductively that P_r is the box intersected with

$$\frac{x}{\xi_k} + \frac{\xi_k y}{4} \geq 1, \quad 1 \leq k \leq r. \quad (37)$$

For $r \geq 1$, put $\xi = \xi_r$. The point $(\xi, 0, 1)$ satisfies every row, with every earlier cut and every box row other than $y \geq 0$ and $w \geq 1$ slack. The three tight rows define the basis rays

$$e_x, \quad p_\xi = (-\xi^2/4, 1, 0), \quad e_w. \quad (38)$$

The objective's directional costs are

$$\varepsilon, \quad 1 - \varepsilon \xi^2/4, \quad 1,$$

all strictly positive because $\xi < 2/\sqrt{\varepsilon}$. The polytope lies in this cone, so $(\xi, 0, 1)$ is its unique optimum.

Write $\xi' = \xi_{r+1}$ and $a = 2/\xi'$. The selected set contains the vertex in its interior because $(a\xi)^2/4 = (\xi/\xi')^2 < 1$. The step on e_x is $\xi' - \xi$, and that on e_w is infinite. Along p_ξ , the expression inside the square is

$$ax(t) + y(t)/a = \frac{2\xi}{\xi'} + \frac{\xi'^2 - \xi^2}{2\xi'}t.$$

It increases from a value in $[0, 2)$, and reaches two first at $t = 4/(\xi + \xi')$. In cone coordinates

$$\lambda_x = x - \xi + \xi^2 y/4, \quad \lambda_p = y, \quad \lambda_w = w - 1,$$

the new cut is

$$\frac{\lambda_x}{\xi' - \xi} + \frac{\xi + \xi'}{4}\lambda_p \geq 1.$$

Multiplication by $\xi' - \xi$ simplifies it to $x/\xi' + \xi'y/4 \geq 1$. This also covers round zero by setting $\xi = 0$. The next point $(\xi', 0, 1)$ satisfies all old rows and the new cut. Its tight basis has the same form, with ξ' in place of ξ , and the same strictly positive directional-cost argument proves unique optimality. This completes the induction. Every point has $w - xy = 1$, and its value converges to $1 + \varepsilon\xi_\infty < 1 + 2\sqrt{\varepsilon}$.

It remains to check uniform pointedness. Set $T = \xi_\infty^2/4$ and $h = (1, T+1, 1)$. For the normalized rays in [equation \(38\)](#), write $t = \xi^2/4 \in [0, T]$. Their inner products with h are

$$1, \quad \frac{T+1-t}{\sqrt{1+t^2}}, \quad 1,$$

each at least $1/\sqrt{1+T^2}$. Every convex combination of the normalized rays consequently has norm at least

$$\gamma_r \geq \frac{1}{\sqrt{1+T^2}\sqrt{2+(T+1)^2}} > 0. \quad (39)$$

The argument includes the initial orthant. On the other hand the new vertex is in the retained part of the old cone, so

$$d_r \leq \|v^{r+1} - v^r\| = \xi_{r+1} - \xi_r \rightarrow 0.$$

The accumulation point $(\xi_\infty, 0, 1)$ is infeasible. Hence the uniform depth condition fails while uniform pointedness holds.

F Experimental definitions, corrected records, and reproducibility

F.1 Populations and numerical conventions

The numerical evidence uses several distinct populations. The one-cut McCormick experiment measures the gap to a problem enforcing one selected bilinear term. The external multiround loop measures the gap to all bilinear equations. The inside-SCIP root experiment uses recorded reference objective values. The MINLPLib fidelity sample is drawn from attempted cuts in an instrumented search. It is not a random sample of independent LP vertices, and its per-corner and equal-instance-weighted summaries answer different questions.

In the fidelity campaign, 64 instances were sampled from 394 eligible MINLPLib entries using seed 20261001. Sixteen never called the handler; 48 supplied dumps. Thirty solves finished and 34 reached a limit. A first attempt sample, seed 12345, takes at most 50 attempts per active instance. It has 2,101 ray-bearing and 59 rayless records. A disjoint second sample, seed 54321, takes at most 150 additional attempts on the 26 instances selected by the positive numerical corner class. It has

2,477 ray-bearing and 148 rayless records. Ratio frequencies consequently come from an adaptive, instance-stratified sample. The generator campaign uses seeds 11 and 12 and generates 150 and 200 models, with cut attempts on 47 and 73 models.

Table 9: Retained numerical conventions. Floors pertain to implemented criteria, not to the exact optimization statements.

Population	Convention
Fidelity	Zero-rate class $\omega_j \leq 10^{-9} \max_i \omega_i$; ratio costs floored at this threshold. Matching ray steps have relative difference at most 10^{-6} , or both are infinite.
External multiround	Orbit weights $\max\{\omega_j, 10^{-9} \max(1, \max_i \omega_i)\}$. Tie proxy excludes normalized costs below 10^{-4} and uses relative tolerance 10^{-3} .
Inside-SCIP selection	Corner surrogate weights $\max\{\omega_j, 10^{-6} \max(\max_i \omega_i, 10^{-9})\}$; the floor is positive even if all raw weights vanish.
Reference screen	Objective comparison tolerance $10^{-6} \max(1, z_{\text{ref}})$; this does not certify primal feasibility or exact optimality.

F.2 The ten corner corrections

For a pair of scaled rays, the old oracle parameterized their direction by $\nu(\theta) = (\theta/\omega_i, (1 - \theta)/\omega_j)$ and used a quadratic root formula. If $P\nu(\theta) = 0$, rounding residuals in the coefficients can return a spurious finite objective. In all ten affected corners, the selected pair is antiparallel and the returned point has $q > 0$, hence is infeasible. The corrected oracle rechecks the boundary value at every candidate. Antiparallel pairs occur in 38 of the 47 small and 66 of the 73 larger corners; this is a realistic boundary case rather than an isolated input construction.

Table 10: Old infeasible corner values and conservative outward-rounded displays of the retained rational lower/upper checks.

Seed	Trial	Old value	Lower display	Upper display
11	3	$8.15 \cdot 10^{-9}$	0.048801	0.048851
11	110	0.103516	0.133667	0.133802
11	122	0.015625	0.032941	0.032975
12	14	0.062500	0.198744	0.198944
12	28	0.125000	0.137107	0.137246
12	71	0.031250	0.049226	0.049276
12	115	0.003906	0.027682	0.027711
12	140	0.003906	0.040754	0.040796
12	142	0.125000	0.196624	0.196821
12	174	0.007812	0.071949	0.072022

The original one-cut summaries remain available to identify the correction; the corrected table uses the affected-corner rechecks. The unchanged SCIP-cut LP means are 0.8952360 and 0.8700128. Corrected SCIP corner means are 0.7239198 and 0.6159066, orbit/corner means 0.7826349 and 0.6745576, and orbit-cut LP means 0.9311004 and 0.9099002. These are numerical aggregate results, distinct from the ten rational checks.

The original root-loop cohorts have 12 small and ten larger instances. Replacing only the original corner oracle yields larger-cohort round-10 fractions 0.930669 for the SCIP model, about 0.901 for orbit, and 0.904 for its completion. A separate fast runner on the same generated cohort gives 0.9036877 for orbit and 0.8920107 for its completion; its bug-affected orbit variant gives 0.8832702. These are different numerical trajectories, not duplicate estimates of a uniquely specified run. The corrected objective-parallel rule stalls near 0.854 on the small cohort and 0.613 on the larger cohort.

F.3 Fidelity and corner-oracle detail

The fidelity comparison has 7,533 generator rays in the first cohort, 36,263 in the second, and 566,397 MINLPLib rays. There are respectively one, zero, and 1,210 discrepancies. Rays after an attempt’s failing ray are excluded because the implementation supplied no step. Finite nonmonoidal coefficients equal $1/\alpha_j$ in all 7,619 records with coefficient data; monoidal coefficients were checked for presence rather than independently reconstructed.

The 307 bisection-class discrepancies split into 200 exactly reproduced numerical bisections and 107 unreproduced interval-rounding/bisection diagnoses. The remaining rounding class splits into 21 huge-step and 33 root-formula cases. The largest observed shortening, $5.9 \cdot 10^{-4}$, belongs only to this sample. A separate recorded inside-SCIP example halves a step; the smaller sample maximum is not a universal safeguard. The old-model comparison additionally differs in cutoff (10^7 versus 10^{12}), so it is not an unconditional formula-error frequency.

Table 11: Numerical methods used in the 1,067 positive reduced-space ratio records. No row certifies the whole cohort.

Method/class	Records	Limitation
Support at most two	602	Floating-point boundary calculations
Generic three-ray KKT enumeration	233	Degenerate minimizers may be missed
Solver-closed estimates	176	Solver status is not an exact certificate
Solver brackets	46	Saved lower/upper numerical bounds
Upper-only estimates	10	No retained global lower certificate

The higher-dimensional solver records have 175 optimal, 47 limit, and ten infeasible reports. Some saved full-space solver objectives lie above known feasible values. An exact feasible-point check for one **waterund25** record supplies an upper bound near 0.156185, not global optimality. The ten upper-only records can increase the reported ratios; they can reduce the fractions below 0.5 or 0.9 by at most 10/1067, or 0.94 percentage points. In the low-ratio class the determining rays split into 187 numerical zero-rate, 99 cheap positive, and 85 other rays.

Of 38,105 dynamism aborts, 34,834 occur in the first restriction piece and 3,271 in the second Case-4 piece. The sampled counts are 1,799 and 175. Tiny *A* alone occurs in 1,270 sampled and 28,071 full first-piece cases; 1,209 of those 1,270 exceed the specified $64u$ rounding threshold, where $u = 2^{-53}$. A separate rounding diagnostic finds 36 of 138 cases consistent with zero at its tolerance. None of these partitions proves that rounding noise explains most aborts.

After generation, 1,265 cuts are rejected by cleanup and 2,889 by range/efficacy checks. Later LP observations contain 296 of 767 checkable first-generator cuts, 595 of 2,155 second-generator cuts, and 3,152 of 21,886 MINLPLib cuts. The uncheckable counts are 130, 308, 2,982. The source audit also identifies two option descriptions whose Boolean effects are reversed, and a generated-cut limit per expression that resets at restarts. These implementation facts affect interpretation of the attempts; they are not evidence of solver gains.

F.4 Paired trajectories and policy sensitivity

The four generated caches each contain 60 instances, with seeds 1001–1004; the larger two experiments use their first 50 entries. Generation retains 240 accepted instances from 751 trials, with 511 small-gap exclusions. The main and additional-rule records contain 44 files/3,840 trajectories and 30 files/1,320 trajectories. Pairing uses size and cached instance index together. All 220 baseline denominators match the cached initial LP and global-bilinear reference values. These records have no duplicate conflicting trajectories or non-success statuses. Global references and trajectory values remain numerical.

Root-gap fractions are clipped for numerical overshoots. Across 52,347 stored round states, 3,728 exceed the cached reference; the largest excess is $6.93 \cdot 10^{-7}$ relative to $\max(1, |z_{\text{bil}}|)$, or $2.09 \cdot 10^{-4}$ of the root gap. Clipping is not a feasibility proof. After 20 rounds, the SCIP-model rule closes more than 0.999 of the gap on 48/60, 42/60, 40/50, 29/50 instances. Later-round means include those already-closed instances.

The pooled orbit-minus-SCIP paired differences are approximately -0.0013401 at round 1, -0.0117545 at round 10, and -0.0115808 at round 20. Comparisons across explored complete-cohort rules use 14 rule-versus-SCIP tests on the 220 paired instances, adjusted by Holm separately for each endpoint. AUC is the mean fraction at rounds 1–10. For the efficacy policy, adjusted p is 0.0247 for AUC over rounds 1–10 and 0.3343 at round 10; the second efficacy variant has 0.0645 and 0.7789. Failure to reject a difference is not equivalence.

Let $h_T(r)$ be the fraction closed at round r by a schedule that uses orbit for its first T rounds and then the SCIP-model rule, and let $h_O(r)$ use orbit throughout. The recovery contrast over rounds a –20 is

$$[h_T(20) - h_O(20)] - [h_T(a) - h_O(a)].$$

For $T = a = 3$, pooled over the $60 \times 6 \times 8$ and $50 \times 10 \times 20$ instances, its mean is approximately $+0.0060$ $[0.0020, 0.0100]$. Its raw paired- t p is 0.00369; Holm adjustment gives 0.0443 over 12 contrasts: four switching schedules, each evaluated on either size and on their 110-instance pool. The adjustment is 0.1328 over 36 tests spanning starts $a = 3, 5, 10$, and 0.1218 over 33 distinct tests after removing three duplicate contrasts. The 12-test Wilcoxon adjustment is about 0.056 for the pooled contrast, while its 10×20 contrast gives 0.035. The policy family, time window, and test determine which comparisons cross a conventional threshold. These are exploratory recovery results, not a validated replacement.

The majority-ray intervention chooses orbit on shares 0.219, 0.273, 0.287, 0.321 across sizes; random one-third selection has shares 0.337, 0.333, 0.335, 0.329. The majority rule uses the orbit set for a term only when its step is at least the SCIP step on strictly more than half the rays where at least one step is finite; the control selects orbit with probability one third. The pooled contrast is inconclusive, $+0.003$ $[-0.002, +0.007]$, and its signs are positive in two sizes, approximately zero in one, and negative in one. This intervention does not isolate step geometry at fixed exposure. Likewise, within-rule step/tie/degeneracy correlations do not establish causality. The efficacy pool contains the SCIP cut, orbit optimized for the current costs, orbit optimized for normalized costs plus 0.1 or 1, and orbit optimized for ray-norm weights. Efficacy is distance from the LP vertex to the cut hyperplane in the original LP coordinates. The two-candidate variant uses only SCIP and orbit; its tested gain is not established. The observed cone-pointedness proxy can be about 0.001; finitely many small values neither verify nor refute a uniform-depth hypothesis.

The one-orbit-root-round switching policy is entirely external to SCIP. The inside-SCIP implementation did not include the full-orbit LMI candidate. Its survival through SCIP’s numerical checks and cut selection is an untested integration question.

F.5 Minor and restricted-family diagnostics

The random minor data fix signature $(2, 2)$. There are 605 stored four-ray records, of which 600 have finite corner value, and 300 finite eight-ray records. Support-one/two counts are 499/101 and 268/32. Numerical orbit attainment occurs on 593/600 and 295/300, point-rule attainment on 521/600 and 274/300, rotation-family attainment on 435/600 and 240/300, and SCIP-set attainment on 5/600 and 6/300. Support-two tangency residuals have maximum absolute cosines $4.34 \cdot 10^{-12}$ and $3.01 \cdot 10^{-14}$. These observations support a geometric diagnostic; they do not prove a generic statement.

Both LP selection policies use coordinates $M \mapsto \bar{M}^{-1}M$, so the apex is I . At each tested bound z , the maximum-margin choice maximizes a common PSD margin $t \leq 1$ subject to $\text{sym}(G) \succeq tI$ and $\text{sym}(GV_j(z)) \succeq tI$, where $G = F^T$ and $V_j(z) = I + (z/\omega_j)\bar{M}^{-1}P_j$. The nearest-SCIP choice minimizes Frobenius distance to the transformed SCIP parameter, scaled to have minimum apex eigenvalue one, among parameters with the same normalization that contain T_z . Its target z is $(1 - 10^{-6})$ times the computed orbit bound.

The targeted minor implementation comparison matches 258 applied cuts in 273 separation rounds; 21 generated models have no recorded separation. The maximum normalized coefficient difference is $4.4964 \cdot 10^{-15}$ and the RHS difference $7.9936 \cdot 10^{-15}$. The standalone build without Ipopt does not enter this separator; the PySCIPOpt 6.2.1 build uses SCIP 10.0.3 with Ipopt 3.14.19. The saved fidelity logs contain errors and minor IDs rather than full coefficient vectors. Random records retain seeds, indices, supports and ratios rather than full apex/ray arrays; LP records retain trials, metrics and statuses rather than complete bases.

The following tight brackets are retained exact-check outputs.

Instance	Family	Archived bracket
A	Orbit	[0.4495347, 0.4495363]
B	Orbit	[0.9219403, 0.9219593]
S1	Orbit	[0.7780998, 0.7781193]
A	Point rule	[0.1975640, 0.1975643]
B	Point rule	[0.3634909, 0.3634912]
S1	Point rule	[0.4542877, 0.4542880]
A	Rotation	[0.3172917, 0.3172929]
B	Rotation	[0.6528369, 0.6528387]
S1	Rotation	[0.4855239, 0.4855241]

Complete endpoint witnesses for A and S1 were not saved; the explicit portable certificates in [Appendix D.2](#) use coarser caps. The fine B and embedded bilinear witnesses are retained.

Twenty-four rounded corners have successful one-sided exact upper checks and numerical lower family values. Their logs do not retain all upper dual witnesses; the random subset also omits rounded coordinates and rational endpoints. They cannot supply 24 exact two-sided comparisons. Rational tangent searches retain 1,487 constructed corners and 354 grid-obstruction candidates; a support-one search has 717 eligible corners and 89 misses. These are capped searches. The superseded generic-minor solver records are excluded from results.

The related bilinear checks include 888 random finite corners with support counts 749/139/0, 24 finite two-variable orbit checks, and a 40-trial point-rule test with four infinite corners excluded. On its 36 finite corners the plain point rule misses the corner bound on 13, while the full orbit reaches it numerically; the enlarged point-rule search still has 13 misses. Small support-two and adversarial searches motivate exact examples rather than estimate universal failure rates. Earlier global-solver comparisons are numerical validations and did not detect every antiparallel-ray failure.

The depth-family calibration also retains the following numerical progression. The A column uses normalized SDP bisection and a checked returned set; it does not give exact two-sided endpoint certificates. The B column records a heuristic value found, not a global optimum.

ε	D	A ratio	Best B found	SCIP point rule
10^{-1}	4.9	1	1	0.579
10^{-2}	14.3	1	1	0.198
10^{-3}	44.8	1	1	0.063
10^{-4}	141	0.9805	0.9805	0.020
10^{-5}	447	0.4207	0.4323	0.0063
10^{-6}	1,414	0.0994	0.1367	0.0020
10^{-7}	4,472	0.0247	0.0432	0.00063
10^{-8}	14,142	0.0068	0.0137	0.00020

The extended search supplies the $\varepsilon = 10^{-5}$ B entry; the shorter search had 0.4310. Numerical normalized-frame calibration suggests an A asymptotic coefficient near 56.7, whereas an explicit completion attains the certified coefficient 136.7 and the box bound is 137. Equality of the best-height A and B values is numerical. One severely scaled SDP result under-reports that calibration and is treated as a numerical failure. The fixed nonsingular three-ray family has numerical condition number about 12.0136; constancy and nonsingularity are mathematical properties rather than consequences of that decimal.

Separate fixed-point-rule sampling retains 15,731 corners in four location regimes. Under $D \leq 2$ alone the observed minimum ratio is about 0.0140, at ray condition number 434. With the additional $\kappa \leq 10$ restriction, minima are 0.2397 for the completion and 0.1823 for the uncompleted set. These are sample minima, not certified worst-case constants. A 150-corner support-one comparison agrees with the interval exactness criterion apart from two borderline SDP under-reports; 138 satisfy the reported cylinder condition and all are numerically exact. The proof of the criterion remains separate.

F.6 Inside-SCIP cohorts, validity diagnostics, and timing limits

The root campaign has 2,429 runs on 347 instances at seed 0 with seven settings, and 1,220 runs on 305 instances at each of seeds 1 and 2 with four settings. Complete cross-setting cohorts have 305, 305, 304 instances; referenced complete cohorts have 302, 302, 301; defined comparable root-gap fractions have 251, 250, 248. All 4,869 retained root runs return zero. An efficacy root bound on `tricp` exceeds its zero reference by about $1.7 \cdot 10^{-5}$ at a large objective scale. A separate debug rerun reproduces it without a row exclusion but has terminal node-cutoff diagnostics. It is not a clean validity certificate.

The strengthened variants at seed 0 change mean root-gap fraction by about +0.006 for the point rule, -0.008 for corner minus strengthened point rule, and approximately zero for efficacy minus strengthened point rule. Root CPU cost increases in that within-batch comparison. Search changes directions in 76.2% of corner searches and 78.9% of efficacy searches in the all-root population. Median per-instance surrogate gains are 1.428 and 1.044 on the 267 instances with a change. These populations differ from the 251 seed-0 root-gap comparisons. Offline attainment counts based on the ten incorrect corner values are excluded; their LP re-solve values are independent of that denominator error.

The full numerical-degeneracy scan retains 305 summaries totaling 109,389 corners: 89,307 (81.6415%) attain the floored criterion on a zero-rate ray, and 10,040 (9.1782%) have all costs in

the zero class. Individual dumps, solver output, and per-instance exit codes for that scan were not retained. This test concerns the criterion’s minimizer, whereas the fidelity test concerns feasibility of a zero-rate face.

The ordinary full grid has 480 runs, four settings on 60 instances and seeds 1 and 2, at a 300 CPU-second limit. It reports 302 optimal outcomes, 177 limits, and one baseline numerical failure. Solved counts are 75 with cuts disabled, 75 with patched point rule, 76 with corner selection, and 76 with efficacy selection. The common-solved cohort has 73 instance–seed pairs. All cut-on settings capture rows and can change paths. The 120 stock reruns report 77 optimal outcomes and 43 limits, but were run in a different batch. An additional completion is sensitive to batch speed; matching paths do not remove that confound. The stock timing contrast is unusable for a conclusion about enabling cuts.

The debug grid instead uses five cut-on settings on 60 instances, seed 0, 30 CPU seconds, and disabled heuristics and symmetry. It reports 170 optimal outcomes, 129 limits, and one failure. Among 299 available final and 286 root bounds, none excludes the recorded feasible reference at the stated tolerance. Nevertheless 45 runs contain ERROR diagnostics and 215 unknown-variable warnings.

All 73 intersection-row messages concern `st_glmfp2`. The supplied coordinates require nonlinear objective helper value 7.6275, but the loaded helper is zero. Correcting it satisfies all 73 rows with minimum slack 0.0175438. Nine other-row messages are local secants whose domains exclude the supplied point. These particular false alarms are resolved; other bound, implication, and cutoff diagnostics remain unresolved. The 114,699 objective-bound warnings on 25 runs use incomplete loaded objectives and do not count invalid original-objective bounds. Three small downward discrepancies in reported `kall` debug optima were not individually primal-checked; a saved primal check on an ordinary full-solve run is a different case. The earlier symmetry-enabled attempt contains 43 logs and is an incomplete population.

F.7 Companion contents and replay limits

The companion preserves original file bytes, relative paths, SHA-256 digests, sizes, and record counts. Its checker validates that inventory without importing experiment modules or launching solvers. Repository-owned code, compact records, generated model caches, saved certificate data, and verification receipts are included. Third-party literature, benchmark models and solutions, solver binaries, environments, and bulky raw traces are excluded from the default companion. The companion does not supply a redistribution license.

Full saved exact data include the ratio-bound box trees, five completed closure box certificates, three current A closure certificates, and 60 rational cuts giving the support-one closure lower bound. Ratio box line counts include facet headers and completion markers and therefore exceed leaf counts. The strongest tangent certificates have 11,132, 1,882, 10,705, 11,943 leaves. The standard upper-constant certificate has 28,699 leaves; its current-header version has identical certificate data to the historical version. False attempted upper bounds that reached maximum search depth remain failed searches, not certificates.

The closure box leaf counts are 3,255, 38, 10,323, 426, 7,061; the A piece counts are 25, 92, 1. Current verifiers include ray-uniqueness, sign, coordinate-count, and free-coordinate pruning checks. Historical verifier flaws did not affect those saved certificates, but old verifier versions must not be used. Finite mutation tests do not prove general verifier soundness.

The near-boundary ratio bracket [0.03052, 0.03056] is a retained exact-check result for the binary-float data model. Its upper-check receipt does not save the rational dual matrices; its generation script solves an optimization problem. It must not be advertised as a complete saved-data replay.

The same distinction applies to missing fine minor endpoint witnesses and several rounded-corner checks.

The support-one B closure checkpoint remains incomplete, with 37,936 leaves and 171,528 pending boxes. Its value near 0.9953622 is numerical. The stopped diagnostic for $\mathcal{BP}$ at the four-ray corner of [theorem 5.8](#) is also incomplete; the analytic argument settles its intended claim. The closure survey evaluates 11 distinct corners for the orbit and uncompleted point-rule families after excluding the unreliable near-boundary row; seven also have completion and completed point-rule results. Among 18 point-rule single/closure comparisons on 11 distinct corners (11 uncompleted and seven completed), 17 match numerically. Re-basing reaches the original numerical bound in five of six examples. In the three-ray tangent example the objective reaches 1 after three cuts while the vertex still has $q \approx 12$; the recorded feasible vertex occurs only after four cuts. These are finite diagnostics, not convergence guarantees.

Saved run headers identify SCIP 10.0.3, SoPlex 8.0.3, and the relevant build flags. The recorded Python environment uses Python 3.13, NumPy 2.5.1, SciPy 1.18.0, CVXPY 1.9.3, Clarabel 0.11.1, SCS 3.3.1, SymPy 1.14.0, highspy, and PySCIPOpt 6.2.1. Some solver-bound analyses require Gurobi. The companion records current file hashes; run-specific source digests were not retained. Reproducing a numerical experiment and checking a saved exact certificate remain separate operations.

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